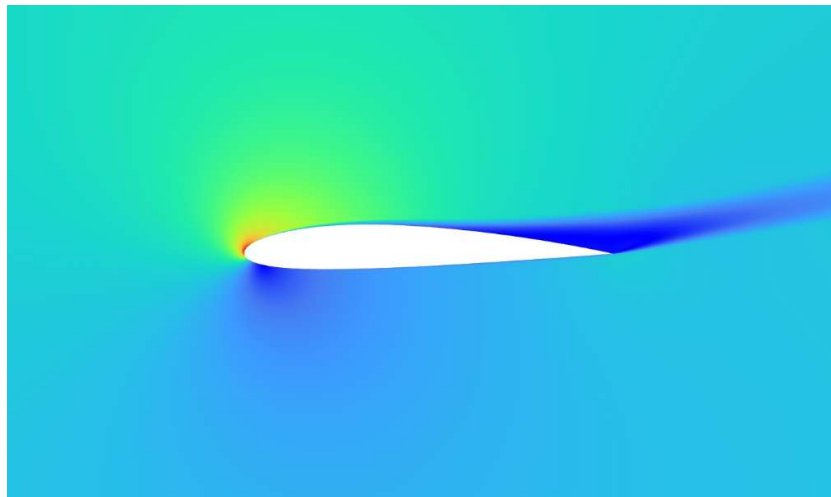


Aerodynamic Analysis of the NACA 2412 Airfoil at Varying Angles of Attack

A Computational Investigation Using ANSYS Fluent



Velocity contour of NACA 2412 at $\alpha = 17^\circ$

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Software used: ANSYS Fluent 2026 R1

Reynolds number: $Re = 3.1 \times 10^6$

Abstract

This investigation presents a two-dimensional steady-state computational fluid dynamics (CFD) analysis of the NACA 2412 airfoil at a Reynolds number of 3.1×10^6 , conducted using ANSYS Fluent 2026 R1. Simulations were performed across a range of angles of attack from -8° to 20° using the Shear Stress Transport (SST) $k-\omega$ turbulence model with a C-type structured mesh of 400,020 cells. A mesh independence study was conducted at 14° angle of attack using Richardson extrapolation with a refinement ratio of $\sqrt{2}$ across three mesh levels, confirming monotonic convergence of both lift and drag coefficients. Validation against experimental data from NACA Technical Report 824 (Abbott & Von Doenhoff, 1945) at 14° yielded a lift coefficient deviation of approximately 1.69% and drag coefficient deviation of around 12.2%. The drag coefficient overprediction is attributed to the SST $k-\omega$ turbulence model assuming the boundary layer is fully turbulent across the entire airfoil surface. In reality, the boundary layer begins as laminar flow near the leading edge before transitioning to turbulent flow further downstream. Since laminar flow produces significantly less skin friction drag than turbulent flow, the model overestimates drag by treating the laminar region as turbulent. Results indicate a maximum lift coefficient of 1.6428 occurring at a stall angle of 17° , with maximum aerodynamic efficiency achieved at 8° angle of attack, corresponding to a lift-to-drag ratio of 73.28. Post-stall behaviour was characterised by oscillatory residuals consistent with massively separated unsteady flow, with converged values obtained through iteration averaging. The findings demonstrate good agreement with experimental data across the pre-stall regime and highlight the limitations of steady-state RANS modelling at high angles of attack.

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1. Introduction and Theory

1.1 Introduction

Aerodynamic analysis of airfoils is an integral part of design of aircraft, wind turbines etc. For the aerodynamic analysis to be effective, it must be able to analyze the airfoil's lift and drag characteristics at different operating conditions. Computational Fluid Dynamics is an increasingly popular way of carrying out such analysis, as an alternative to expensive physical experiments in a wind tunnel which give detailed information about the dynamics of the flow, but not in real time.

The NACA 2412 is a cambered and asymmetric airfoil with maximum thickness of 12% chord and maximum camber of 2% at 40% chord location. It is one of the airfoils which have been widely studied under the scope of aerodynamic analysis and the results of which can be found in NACA Technical Report 824 (Abbott & Von Doenhoff, 1945).

This work presents a 2D steady state computational fluid dynamics study on the NACA 2412 airfoil at 3.1×10^6 Reynolds number. The air flow freestream velocity at 15°C International Standard Atmosphere (ISA) conditions was 45.28 ms^{-1} . The aerodynamics of the airfoil was computed at angles of attack ranging from -8° to 20° . This includes all aspects of the aerodynamics of the airfoil up to stall and separation. The SST $k-\omega$ turbulence model was chosen for its proven ability to predict adverse pressure gradient flows and boundary layer separation on the airfoil surface.

The primary objectives of this study are:

- To design and construct a structured C-type computational mesh suitable for 2D external airfoil flow analysis

- To conduct a mesh independence study using the Grid Convergence Index (GCI) method to ensure solution reliability
- To validate CFD results against NACA TR-824 experimental data and analyse lift, drag, and aerodynamic efficiency characteristics across the full angle of attack range
- To visualise flow field development including pressure distribution, velocity contours, and boundary layer behaviour at key angles of attack

1.2 Theory

The Reynolds number is a dimensionless parameter that characterises the ratio of inertial forces to viscous forces in a fluid flow. It is defined as:

$$Re = \frac{\rho V L}{\mu} = \frac{V L}{\nu}$$

Where ρ is the fluid density (kgm^{-3}), V is the freestream velocity (ms^{-1}), L is the characteristic length — in this case the airfoil chord length (m), μ is the dynamic viscosity ($\text{Pa}\cdot\text{s}$), and ν is the kinematic viscosity (m^2s^{-1}). The Reynolds number governs the nature of the flow and the behaviour of the boundary layer, directly influencing lift and drag characteristics. In this study $Re = 3.1 \times 10^6$, corresponding to a freestream velocity of 45.28 ms^{-1} at ISA standard conditions.

The lift coefficient is a dimensionless measure of the lift generated by the airfoil:

$$C_l = \frac{L}{\frac{1}{2} \rho V^2 A}$$

Where L is the lift force (N) and A is the reference area (m^2), taken as the chord length multiplied by the span for a 2D analysis.

The drag coefficient is a dimensionless measure of the aerodynamic drag acting on the airfoil:

$$C_d = \frac{D}{\frac{1}{2}\rho V^2 A}$$

Where D is the drag force (N).

The pressure coefficient describes the local static pressure relative to the freestream dynamic pressure:

$$C_p = \frac{P - P_\infty}{\frac{1}{2}\rho V^2}$$

Where P is the local static pressure (Pa) and P_∞ is the freestream static pressure (Pa). A value of $C_p = 1$ corresponds to the stagnation point, while negative values indicate regions of flow acceleration and suction.

The lift-to-drag ratio is a measure of aerodynamic efficiency, expressing the amount of lift generated per unit of drag:

$$\frac{C_l}{C_d} = \frac{L}{D}$$

The angle of attack at which this ratio is maximised represents the most aerodynamically efficient operating condition.

The dimensionless wall distance y^+ describes how close the first mesh cell is to the airfoil surface, relative to the local flow conditions at that surface:

$$y^+ = \frac{\rho u_\tau y}{\mu}$$

Where y is the physical distance from the wall to the first cell centre (m) and u_τ is the friction velocity:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

Where τ_w is the wall shear stress (Pa). For the SST $k-\omega$ model to accurately resolve the viscous sublayer without wall functions, a y^+ value below 1 is required. In this study a maximum y^+ of 1.24 was achieved across the airfoil surface, confirming adequate boundary layer resolution.

A mesh independence study was conducted using the Grid Convergence Index (GCI) method based on Richardson extrapolation (Roache, 1994). This method provides a quantitative estimate of the discretisation error in the numerical solution. The refinement ratio between successive mesh levels is defined as:

$$r = \frac{h_{coarse}}{h_{fine}}$$

Where h is the representative mesh size. A refinement ratio of $r = \sqrt{2}$ was used, resulting in approximately double the number of cells between each mesh level.

The observed order of convergence is a number that describes how quickly your solution improves as you refine the mesh and is calculated as:

$$p = \frac{1}{\ln(r)} \left| \ln \left(\frac{\epsilon_{32}}{\epsilon_{21}} \right) \right|$$

Where $\epsilon_{32} = \phi_3 - \phi_2$ and $\epsilon_{21} = \phi_2 - \phi_1$ are the differences between coarse-medium and medium-fine mesh solutions respectively. If $p = 2$, every time you halve your mesh size your error reduces by a factor of 4 (2^{p-2}). If $p = 1$, your error only halves. A higher p means your solution is converging faster and more efficiently with each refinement.

The convergence ratio determines the type of convergence behaviour:

$$R = \frac{\varepsilon_{21}}{\varepsilon_{32}}$$

A value of $0 < R < 1$ indicates monotonic convergence, which is the ideal behaviour for a mesh independence study.

Richardson extrapolation estimates the grid-independent solution at infinite mesh refinement and is known as the extrapolated solution (ϕ_{ext}):

$$\phi_{ext}^{21} = \frac{r^p \cdot \phi_1 - \phi_2}{r^p - 1}$$

Approximate Relative Error (e_a^{21}) is the percentage difference between your two successive mesh results and your medium mesh result. In the case below it is percentage difference between my fine mesh result (ϕ_1) and medium mesh result (ϕ_2).

$$e_a^{21} = \left| \frac{\phi_1 - \phi_2}{\phi_1} \right| \times 100\%$$

Extrapolated Relative Error (e_{ext}^{21}) is the percentage difference between your fine mesh result (ϕ_1) and the mathematically extrapolated exact solution (ϕ_{ext}^{21}).

$$e_{ext}^{21} = \left| \frac{\phi_{ext}^{21} - \phi_1}{\phi_{ext}^{21}} \right| \times 100\%$$

Grid Convergence Index (GCI) is a safety-factor adjusted error estimate that quantifies the uncertainty in your fine mesh solution due to discretisation.

$$GCI_{fine}^{21} = \frac{1.25 \cdot e_a^{21}}{r^p - 1}$$

A GCI value below 1% indicates that the solution is effectively mesh independent and further refinement would produce negligible change in results.

2. Methodology

The NACA 2412 airfoil was selected for this study due to the availability of validated experimental data in NACA Technical Report 824 (Abbott & Von Doenhoff, 1945). The airfoil coordinates were obtained from Airfoil Tools (airfoiltools.com) and imported into ANSYS DesignModeler. A chord length of 1.0 m was used, providing a convenient reference length for Reynolds number calculations and non-dimensionalisation of results.

Table 1: Geometric properties of the NACA 2412 Airfoil.

Parameter	Value
Maximum camber	2% chord
Location of maximum camber	40% chord
Maximum thickness	12% chord

A C-type domain was constructed around the airfoil in ANSYS DesignModeler. The C-type topology was selected as it is well suited for external airfoil flows, providing smooth mesh distribution around the leading edge and along the upper and lower surfaces.

Table 2: Domain dimensions of mesh relative to the chord length ($c = 1.0$ m).

Boundary	Distance from airfoil
Upstream (C-curve inlet)	$7.5c$
Downstream (outlet)	$14c$
Top and Bottom	$7.5c$

The domain boundaries were positioned sufficiently far from the airfoil to minimise boundary interference effects. A blockage ratio analysis confirmed that the domain produces a maximum blockage ratio of 1.73% at 15° angle of attack, within the accepted limit of 3% for external flow simulations.

The computational mesh was generated in ANSYS Meshing using edge sizing and face meshing controls to produce a structured C-type mesh. No automated inflation layer tool was used, instead, near-wall mesh refinement was achieved through manual edge sizing on the airfoil surface, producing a mesh with cell layers sufficiently thin to resolve the viscous sublayer.

Edge sizing with bias was applied for two reasons. To cluster cells near the airfoil surface, ensuring adequate resolution of the boundary layer along the entire chord and to cluster cells behind the trailing edge, capturing the developing wake and flow separation at high angles of attack. Different bias factors were applied to each edge sizing to control the degree of cell clustering. A higher bias factor produces a greater concentration of cells at the target location, transitioning to larger cells further away. This approach ensures mesh refinement is focused where the flow gradients are largest — near the airfoil surface and in the wake — without unnecessarily increasing the total cell count across the entire domain.

The fine mesh used for all production simulations consisted of **400,020 cells**. A wall y^+ analysis confirmed a maximum value of 1.24 across the airfoil surface at 18° angle of attack. This value remains well within the acceptable limit of $y^+ < 5$ for the SST $k-\omega$ turbulence model, confirming adequate boundary layer resolution across all simulated angles of attack.

The following boundary conditions were applied in ANSYS Fluent:

Table 3: Simulation boundary conditions.

Boundary	Type	Value
Inlet (C-curve, top, bottom)	Velocity inlet	$V = 45.28 \text{ ms}^{-1}$, direction via $\cos(\text{AoA})$ and $\sin(\text{AoA})$ components
Outlet	Pressure outlet	Gauge pressure = 0 Pa
Airfoil surface	No-slip wall	Stationary wall

The angle of attack was varied by adjusting the X and Y components of the inlet flow direction:

$$X\text{-component} = \cos(\alpha) \quad Y\text{-component} = \sin(\alpha)$$

Where α is the angle of attack in degrees. This approach keeps the airfoil geometry fixed while rotating the incoming flow direction, eliminating the need to remesh at each angle of attack. The default turbulence boundary conditions were used. Air was modelled as an incompressible fluid with constant properties corresponding to ISA standard conditions at 15°C:

Table 4: Airflow properties.

Property	Value
Density (ρ)	1.225 kgm ⁻³
Dynamic viscosity (μ)	1.7894×10^{-5} Pa·s
Freestream velocity (V)	45.28 ms ⁻¹
Reynolds number (Re)	3.1×10^6

The incompressible assumption is valid at this freestream velocity, corresponding to a Mach number of approximately 0.13 which is well below the compressibility threshold of Mach 0.3. (Note: above Mach 0.3 air starts to compress noticeably under pressure, meaning density changes become significant and must be accounted for in the equations.) All simulations were conducted in ANSYS Fluent 2026 R1 using the following settings:

Table 5: Simulation settings.

Setting	Value
Solver	Pressure-based, steady-state
Turbulence model	SST k- ω
Pressure-velocity coupling	Coupled
Gradient discretisation	Least Squares Cell Based
Pressure discretisation	Second Order
Momentum discretisation	Second Order Upwind
Turbulent kinetic energy	Second Order Upwind
Specific dissipation rate	Second Order Upwind

Convergence was monitored using both scaled residuals and lift and drag coefficient monitors. Residual convergence targets were set to 1×10^{-6} for all equations. A solution was considered converged when both the residuals dropped below target levels and the lift and drag monitors stabilised to a constant value. At post-stall angles of attack where steady-state convergence was not fully achievable due to unsteady separated flow, coefficient values were obtained by averaging over the final 50 iterations.

3. Mesh Independence Study

A mesh independence study was conducted to ensure that the CFD results are independent of the mesh resolution and that further refinement would produce negligible change in the solution. The Grid Convergence Index (GCI) method based on Richardson extrapolation (Roache, 1994) was employed to quantify the discretisation error across three mesh levels. The study was performed at an angle of attack of 14° and a Reynolds number of 3.1×10^6 , representing a near-stall condition where the flow field is most demanding on the mesh. The lift coefficient (Cl) and drag coefficient (Cd) were selected as the monitored quantities.

Three mesh levels were generated with a consistent refinement ratio of $r = \sqrt{2} \approx 1.414$, resulting in approximately double the number of cells between each successive level:

Table 6: Cell count of each mesh.

Mesh Level	Cell Count
Coarse (ϕ_3)	100,182
Medium (ϕ_2)	200,000
Fine (ϕ_1)	400,020

All mesh levels were generated using identical topology, boundary conditions, solver settings and fluid properties — only the mesh density was varied between levels.

The lift and drag coefficients obtained from each mesh level are presented in Table 7.

Table 7: Lift and drag coefficients calculated for each mesh.

Mesh Level	Cell Count	Cl	Cd
Coarse (ϕ_3)	100,182	1.5120	0.028638
Medium (ϕ_2)	200,000	1.5373	0.026840
Fine (ϕ_1)	400,020	1.5436	0.026340
Extrapolated (ϕ_{ext})	∞	1.54571	0.026143
Experimental (TR-824)	—	1.52	0.0233

Both Cl and Cd show a clear trend of converging toward the extrapolated value as mesh density increases, confirming systematic mesh refinement behaviour.

Table 8: The full GCI analysis results.

Parameter	Cl	Cd
Refinement ratio (r)	$\sqrt{2}$	$\sqrt{2}$
Convergence ratio (R)	0.249	0.278
Convergence type	Monotonic	Monotonic
Order of convergence (p)	3.990	3.663
Extrapolated value (ϕ_{ext21})	1.54571	0.026143
Approximate relative error (ea21)	0.41%	1.92%
Extrapolated relative error (eext21)	0.14%	0.75%
GCI fine (GCI21)	0.172%	0.936%

Both Cl and Cd exhibit monotonic convergence with convergence ratios of $R = 0.249$ and $R = 0.278$ respectively, confirming that the solution is converging systematically with mesh refinement. The observed order of convergence of $p \approx 3.99$ for Cl and $p \approx 3.66$ for Cd exceeds the theoretical second-order accuracy of the solver, indicating that the mesh refinement is operating in the asymptotic convergence regime.

The GCI fine values of 0.172% for Cl and 0.936% for Cd confirm that the fine mesh solution is within 1% of the grid-independent solution for both monitored quantities. This satisfies the accepted threshold of $GCI < 1\%$ for mesh independence, confirming that the 400,020 cell mesh is adequate for the production simulations.

Figure 1 and 2 presents the lift and drag coefficients plotted against the inverse of the cell count ($1/N$), respectively, following the Richardson extrapolation convention. As $1/N$ approaches zero, corresponding to infinite mesh refinement, the solution converges to the extrapolated value. The experimental TR-824 data point is included for reference, showing close agreement with the extrapolated CFD solution for Cl and a larger but expected deviation for Cd, attributed to the fully turbulent assumption of the SST $k-\omega$ model.

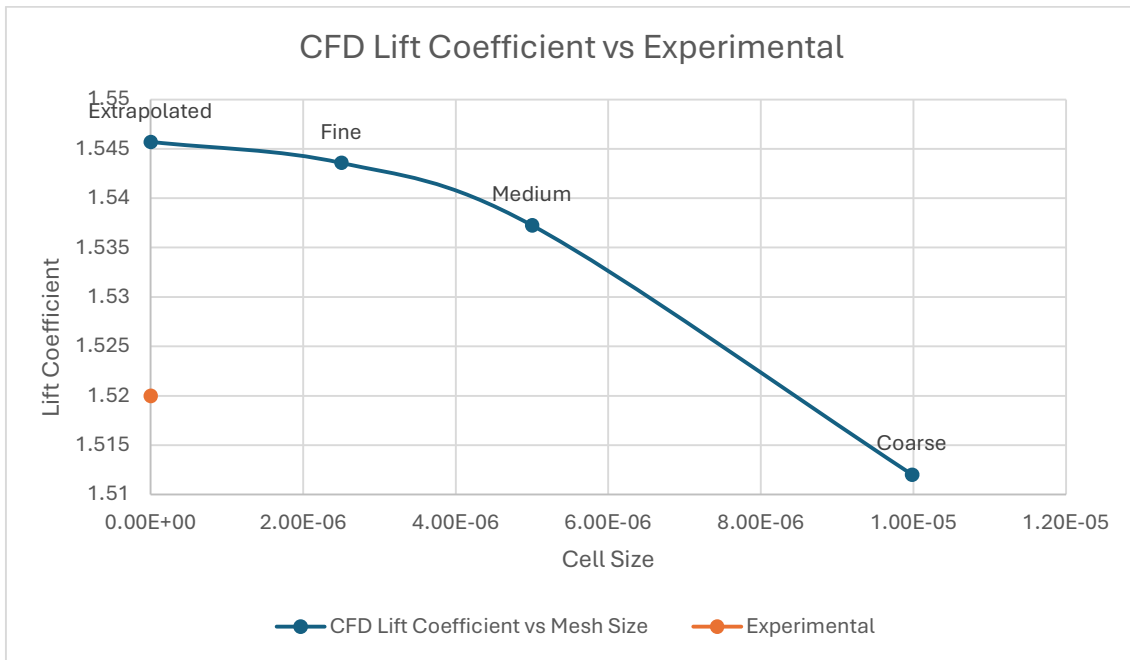


Figure 1: Lift coefficient for each mesh compared to the experimental coefficient.

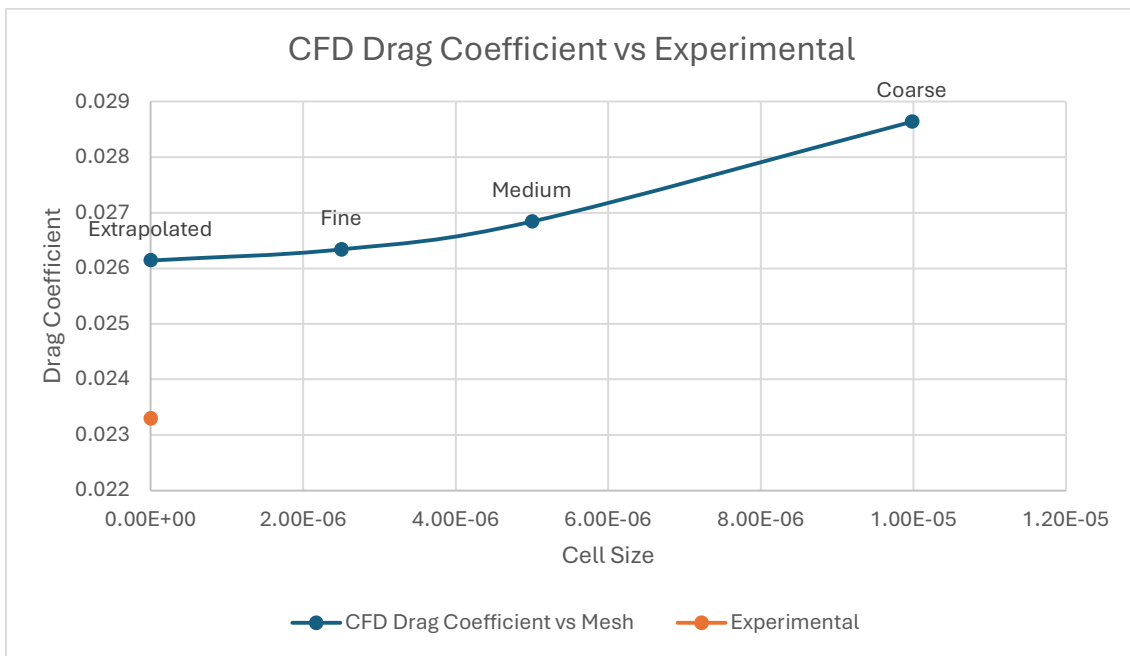


Figure 2: Drag coefficient for each mesh compared to the experimental coefficient.

Based on the GCI analysis, the 400,020 cell fine mesh was selected for all production simulations. This mesh satisfies the $GCI < 1\%$ threshold for both lift and drag coefficients,

confirming that results are effectively mesh independent. All subsequent angle of attack simulations were conducted using this mesh with identical boundary conditions and solver settings.

4. Validation

Validation is the process of demonstrating that the CFD simulation accurately represents physical reality by comparing results against established experimental data. In this study, the Richardson-extrapolated lift and drag coefficients at 14° angle of attack are compared against experimental data extracted from NACA Technical Report 824 (Abbott & Von Doenhoff, 1945) at $Re = 3.1 \times 10^6$. An angle of attack of 14° was selected as the validation point as it represents a near-stall condition where the flow field is most demanding on the turbulence model, providing a rigorous test of simulation accuracy.

Table 9: The CFD results are compared against TR-824.

Parameter	Experimental (TR-824)	CFD (Extrapolated)	% Deviation
C_l	1.52	1.5457	1.69%
C_d	0.0233	0.02614	12.2%

The lift coefficient shows excellent agreement with the experimental data, with a deviation of 1.69%. This confirms that the simulation accurately captures the pressure field around the airfoil and the resulting lift generation. The small overprediction of C_l is consistent with the fully turbulent assumption of the SST $k-\omega$ model, which slightly overestimates the momentum of the boundary layer, delaying separation and producing marginally higher lift than observed experimentally.

The drag coefficient shows a larger deviation of 12.2% from the experimental value. This is a well-documented limitation of fully turbulent RANS simulations at these Reynolds numbers. In the TR-824 wind tunnel experiments, the boundary layer undergoes natural laminar-

turbulent transition, beginning as laminar flow near the leading edge before transitioning to turbulent flow further downstream. Since laminar flow produces significantly less skin friction drag than turbulent flow, the fully turbulent SST $k-\omega$ model overestimates drag by treating the entire boundary layer as turbulent from the leading edge. This limitation could be addressed in future work by employing a transition-sensitive turbulence model such as the γ - $Re\theta$ transitional SST model.

The close agreement in lift coefficient confirms that the simulation is validated for use in the full angle of attack sweep. The drag coefficient deviation of 12.2% is within acceptable limits for fully turbulent RANS simulations and is consistent with findings reported in published CFD literature for airfoil studies at similar Reynolds numbers. The fine mesh simulation is therefore considered validated and suitable for the production AoA sweep.

5. Results and Discussion

This section presents the aerodynamic characteristics of the NACA 2412 airfoil at $Re = 3.1 \times 10^6$ across angles of attack ranging from -8° to 20° . Results are discussed in terms of lift coefficient, drag coefficient, aerodynamic efficiency, pressure distribution, and flow field visualisation. Experimental data from NACA TR-824 (Abbott & Von Doenhoff, 1945) is included for comparison where available.

Figure 3 presents the lift coefficient as a function of angle of attack for both CFD results and TR-824 experimental data.

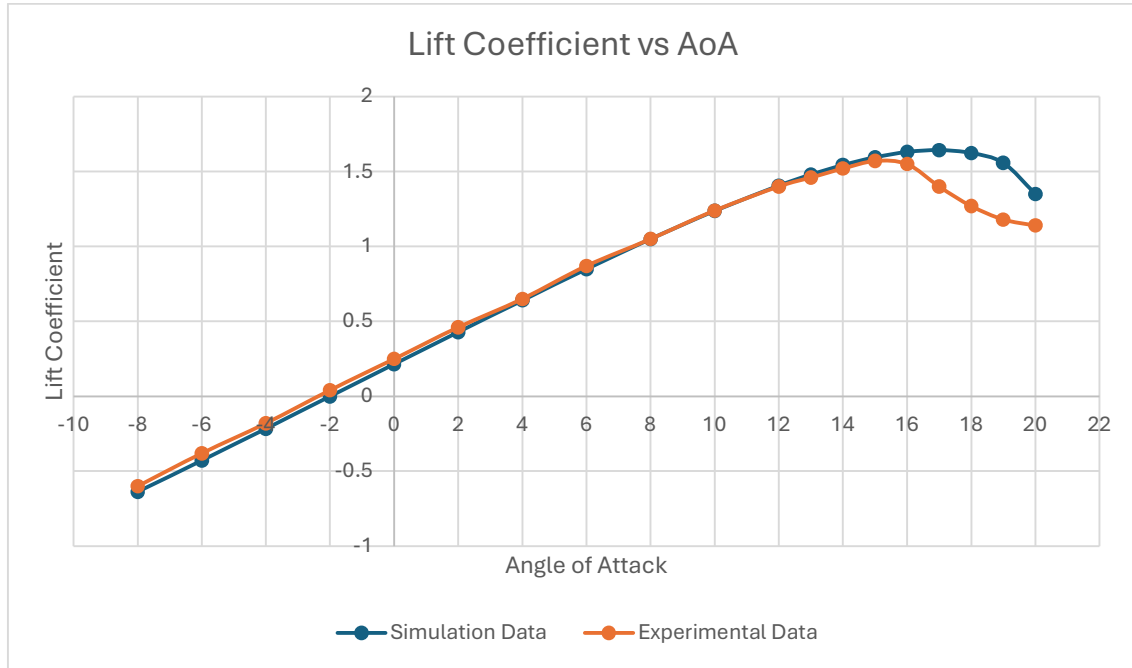


Figure 3: CFD lift coefficient and experimental lift coefficient for different angles of attack.

The lift coefficient increases linearly with angle of attack from -8° to approximately 12° , consistent with thin airfoil theory which predicts a lift curve slope of 0.110 per degree (2π per radian) for an idealised thin airfoil in inviscid flow. The CFD lift curve slope of 0.102 per degree shows good agreement with the theoretical value, with the small deviation attributed to viscous effects not accounted for in thin airfoil theory. The NACA 2412 generates a positive lift coefficient of 0.2139 at 0° angle of attack due to its cambered geometry, the asymmetric profile accelerates flow more rapidly over the upper surface than the lower surface even at zero incidence, producing a net upward pressure difference.

The CFD results show good agreement with TR-824 experimental data throughout the pre-stall linear region, with deviations consistently below 5%. The maximum lift coefficient, $C_{l_{\max}} = 1.6428$ is achieved at a stall angle of 17° , beyond which C_l drops sharply as flow separates from the upper surface. The experimental data shows stall occurring slightly earlier at approximately 16° , with a lower $C_{l_{\max}}$ of approximately 1.58. This difference is attributed to

the fully turbulent SST $k-\omega$ assumption — by treating the entire boundary layer as turbulent the model overestimates boundary layer momentum, delaying separation and producing a slightly higher and later stall than observed experimentally.

Post-stall behaviour beyond 17° shows a progressive decrease in C_l as the separated flow region grows with increasing angle of attack. At 19° and 20° the solution exhibited oscillatory convergence behaviour characteristic of unsteady massively separated flow — values at these angles were obtained by averaging over the final 50 iterations as discussed in Section 3.6.

Figure 4 presents the drag coefficient as a function of angle of attack.

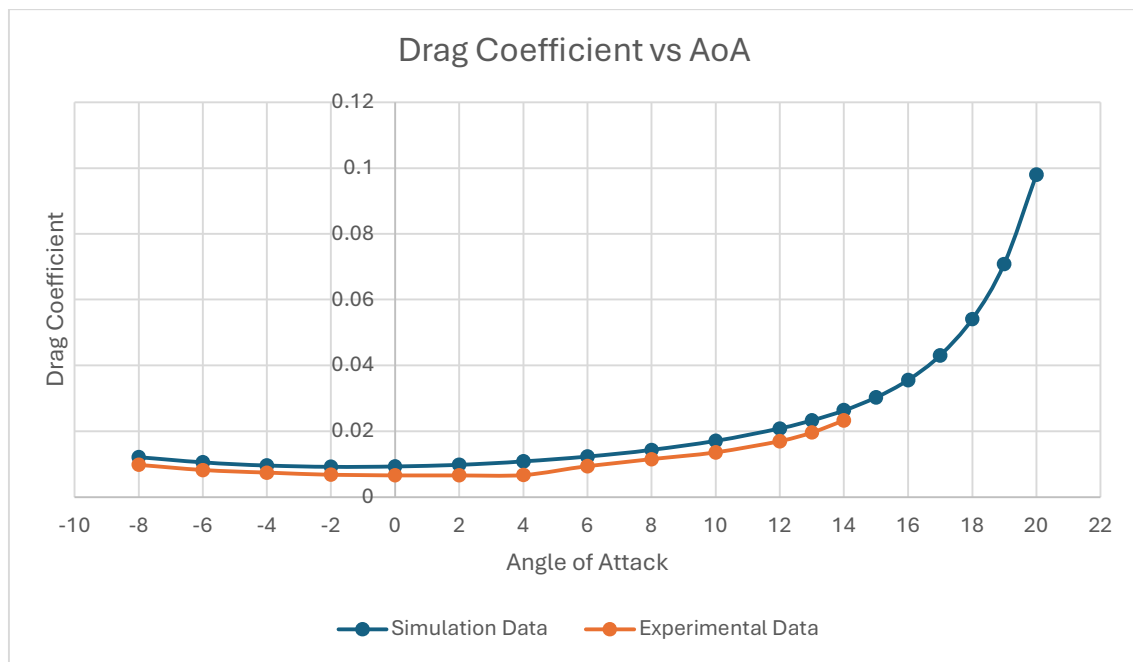


Figure 4: CFD drag coefficient and experimental drag coefficient for different angles of attack.

The drag coefficient follows a characteristic U-shaped curve in the pre-stall regime, with a minimum value of $C_{d_{\min}} = 0.009193$ occurring at -2° angle of attack. This minimum corresponds to the angle at which the pressure drag, and skin friction drag combined are lowest i.e. the airfoil is operating at its most streamlined condition relative to the incoming flow.

Beyond -2° , drag increases progressively as the adverse pressure gradient on the upper surface strengthens with increasing angle of attack, thickening the boundary layer and increasing both pressure drag and skin friction drag. The rate of drag increase accelerates significantly beyond 14° as the flow approaches separation, with drag rising steeply through the stall region.

Post-stall drag continues to increase despite the reduction in lift, as pressure drag becomes the dominant mechanism. The massive flow separation creates a large low-pressure wake behind the airfoil, generating significant form drag. At 20° the drag coefficient reaches 0.098 — more than ten times the minimum value — demonstrating the dramatic penalty of operating beyond the stall angle.

The CFD consistently overpredicts drag relative to TR-824 experimental data across all angles of attack, as discussed in Section 5. The overprediction is most pronounced at low angles of attack where the laminar boundary layer region is largest and reduces at higher angles of attack where the boundary layer is predominantly turbulent.

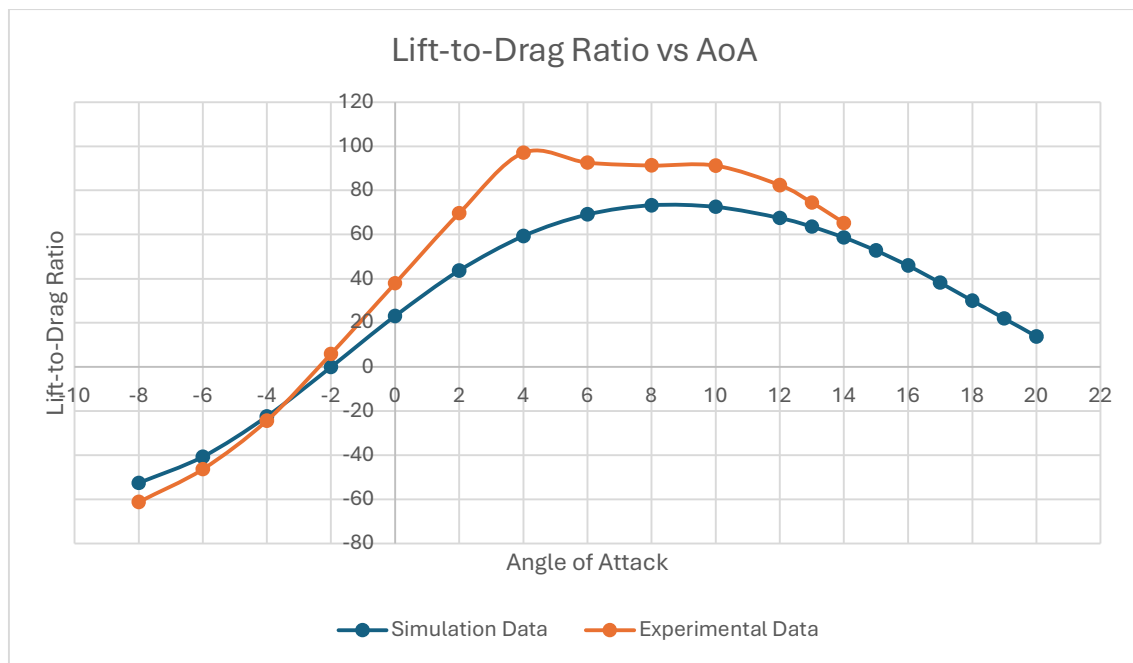


Figure 5: Lift-to-drag ratio for different angles of attack.

Figure 5 presents the lift-to-drag ratio (Cl/Cd) as a function of angle of attack, representing the aerodynamic efficiency of the airfoil at each operating condition. The maximum aerodynamic efficiency of $Cl/Cd = 73.28$ is achieved at 8° angle of attack, representing the optimal operating condition for the NACA 2412 airfoil at $Re = 3.1 \times 10^6$. At this angle the airfoil generates the greatest amount of lift per unit of drag — making it the ideal cruise condition for applications such as light aircraft and gliders where fuel efficiency or glide ratio is the primary design objective. Beyond 8° the lift-to-drag ratio decreases monotonically as drag increases faster than lift. The efficiency collapses rapidly beyond the stall angle of 17° , falling to 13.78 at 20° — less than one fifth of the peak value. This dramatic reduction highlights the severe aerodynamic penalty of operating in the post-stall regime.

At negative angles of attack the Cl/Cd ratio is negative, indicating the airfoil is generating downforce rather than lift — relevant for applications such as racing car wings where negative lift is desirable.

Figure 6-11 presents the pressure coefficient (C_p) distributions along the airfoil chord at selected angles of attack.

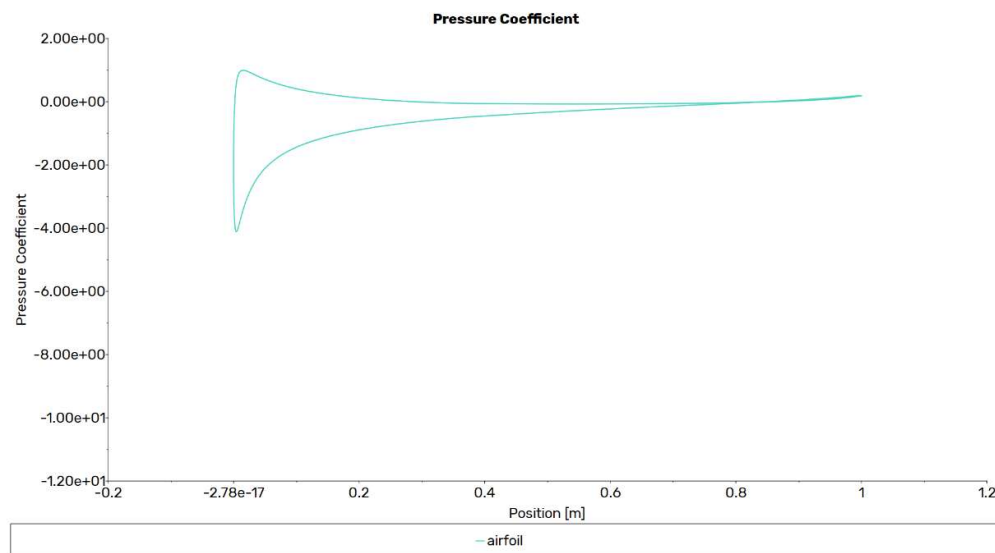


Figure 6: Pressure coefficient plot at -8° angle of attack.

At -8° , the C_p distribution retains the same fundamental form as positive AoA cases — the upper surface remains the suction side with a peak of approximately $C_p = -4.2$, while the lower surface sits at approximately $C_p = -1.7$. However, the enclosed area between the two curves is significantly smaller than at 8° , and crucially the lower surface C_p is now negative throughout the chord rather than positive as seen at 8° . This means both surfaces are at sub-atmospheric pressure, but the upper surface remains at lower pressure than the lower surface — generating a small net upward suction force that is outweighed by the downward pressure components at this extreme negative incidence, resulting in the net negative lift coefficient of -0.6381 .

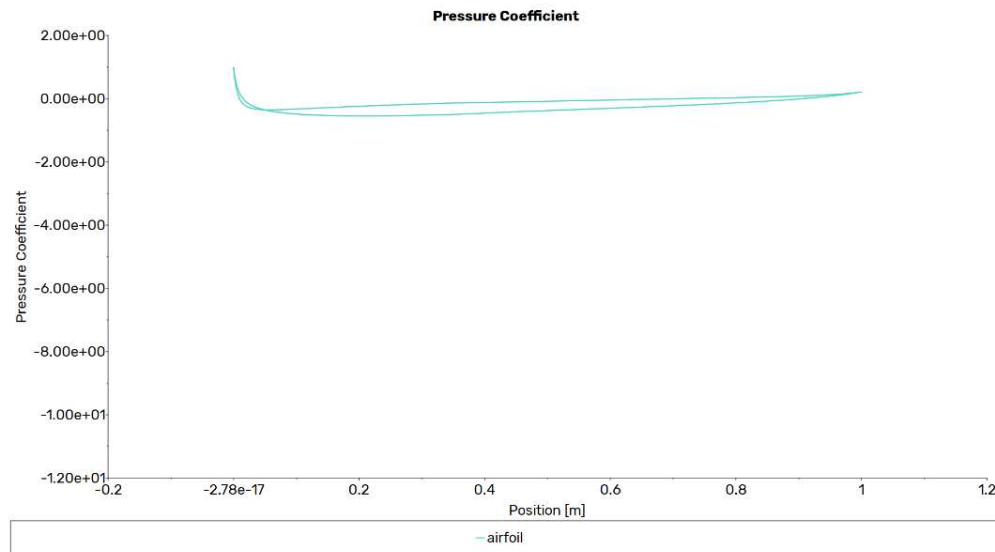


Figure 7: Pressure coefficient plot at 0° angle of attack.

At 0° , the C_p distribution shows a small but clear separation between the upper and lower surface curves. The stagnation point at the leading edge produces a C_p peak of approximately $+1.2$, immediately followed by a sharp suction spike on the upper surface dropping to approximately $C_p = -0.5$. Both curves recover gradually and converge at the trailing edge at approximately $C_p = 0$, indicating clean flow reattachment. The enclosed area between the curves is small but positive — consistent with the modest positive lift coefficient of 0.2139 .

generated purely by camber at zero incidence. The nearly symmetric shape of the two curves confirms the low-lift, low-disturbance nature of the flow at this angle.

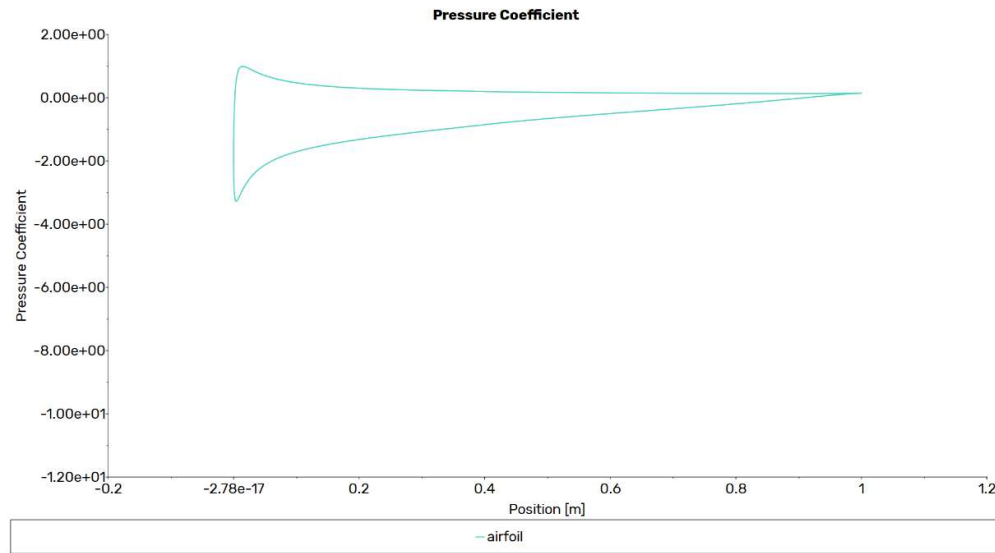


Figure 8: Pressure coefficient plot at 8° angle of attack.

At 8° , the C_p distribution shows a dramatically larger separation between the two curves compared to 0° . The stagnation C_p peak at the leading edge remains at approximately +1.2, which is expected since total pressure is independent of AoA, but the upper surface suction spike has deepened significantly to approximately $C_p = -3.2$, a sixfold increase from 0° . The suction spike is sharp and concentrated at the leading edge, followed by a smooth gradual pressure recovery along the chord toward $C_p \approx 0$ at the trailing edge. The lower surface curve shows a broadly positive C_p throughout the chord length, sitting clearly above the upper surface curve across the full chord. The enclosed area between the two curves is now substantially larger than at 0° , directly consistent with the much higher lift coefficient of 1.0491. The smooth, gradual pressure recovery on the upper surface with no abrupt changes confirms the flow remains fully attached along the entire chord — consistent with the peak aerodynamic efficiency at this angle.

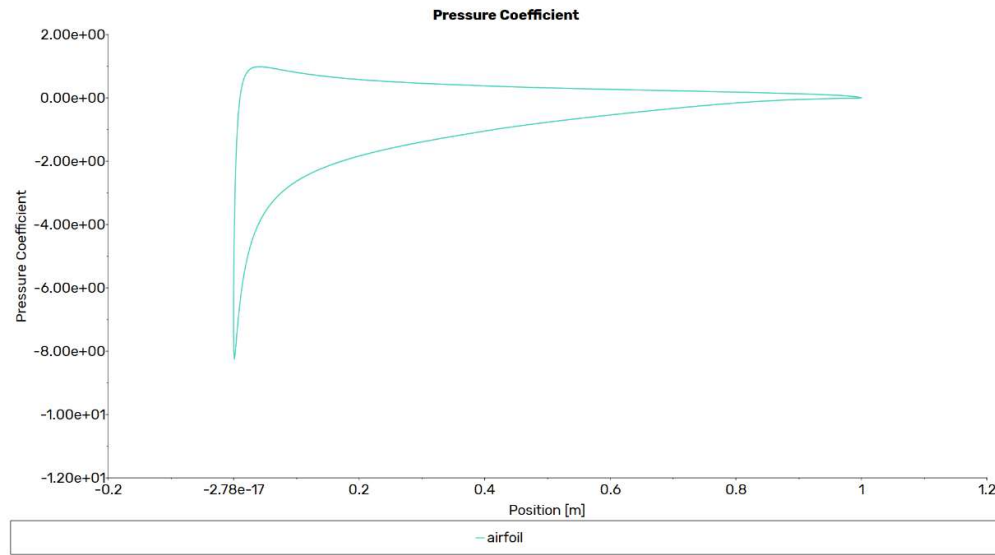


Figure 9: Pressure coefficient plot at 14° angle of attack.

At 14°, the C_p distribution shows further intensification of the suction peak, now reaching approximately $C_p = -8.0$, more than double the 8° value. The spike is sharper and more pronounced, concentrated extremely tightly at the leading edge before recovering steeply. The pressure recovery curve on the upper surface is steeper than at 8°, indicating a stronger adverse pressure gradient that the boundary layer must overcome. The lower surface C_p curve remains broadly positive and has shifted slightly higher than at 8°, increasing the enclosed area further. Both curves converge at the trailing edge at approximately $C_p = 0$, confirming the flow is still attached at the trailing edge despite the steep adverse pressure gradient and is consistent with C_l still increasing at this angle. The significantly larger enclosed area compared to 8° is directly consistent with the higher lift coefficient of 1.5436.

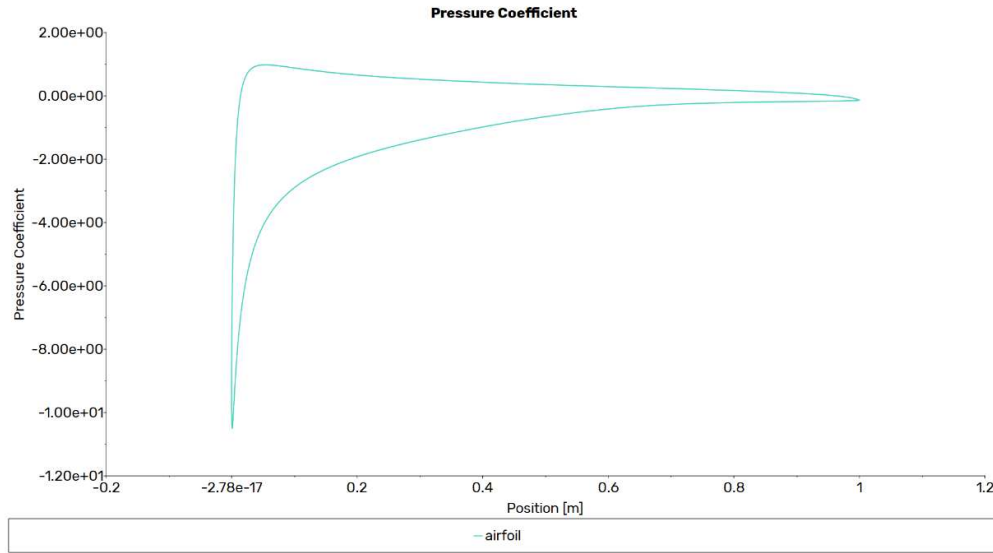


Figure 10: Pressure coefficient plot at 17° angle of attack.

At 17° , the C_p distribution shows the most extreme suction peak of all cases, reaching approximately $C_p = -10.5$, which is the deepest value across all five plots. The spike is extremely sharp and narrow, indicating a highly concentrated suction force at the very tip of the leading edge. The pressure recovery on the upper surface is the steepest of all cases, an almost vertical drop from -10.5 back toward -2.0 within a very short chordwise distance, indicating that the adverse pressure gradient is at its maximum. Despite this, both curves still converge at the trailing edge at approximately $C_p = -0.1$, suggesting the flow is marginally still attached at the trailing edge at exactly the stall point. The enclosed area between the two curves is at its maximum across all cases and is directly consistent with the maximum lift coefficient of 1.6428 at this angle. The extreme sharpness of the suction spike at 17° compared to 14° reflects the boundary layer operating at its absolute limit of attached flow.

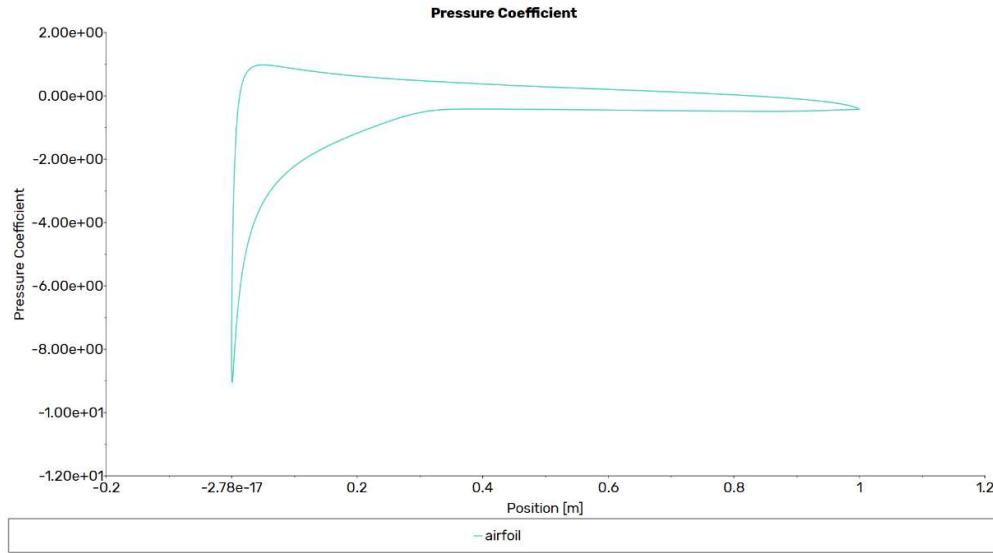


Figure 11: Pressure coefficient plot at 20° angle of attack.

At 20°, the C_p distribution shows a fundamentally different character from all previous cases. The suction spike at the leading edge has reduced to approximately $C_p = -8.8$ — noticeably less extreme than at 17° despite the higher angle of attack — confirming the collapse of the organised leading edge suction that defines post-stall behaviour. Most strikingly, the upper surface pressure recovery curve no longer converges smoothly to the lower surface curve at the trailing edge. Instead, both curves converge at a clearly negative C_p value of approximately -0.4 at the trailing edge, rather than $C_p \approx 0$ as seen in all previous cases. This non-zero trailing edge C_p is the clearest signature of flow separation — the separated flow behind the trailing edge maintains a uniform low pressure that prevents normal pressure recovery. The enclosed area between the two curves has reduced compared to 17°, directly consistent with the drop in lift coefficient to 1.35, while the elevated trailing edge pressure difference explains the dramatically increased drag as the separated wake creates a large momentum deficit behind the airfoil.

Two sets of pressure contours are presented for each angle of attack. The auto-range contours use a colour scale fitted to each individual case, revealing localised pressure features with maximum contrast. The fixed-range contours use a consistent scale of -14,000 Pa to +1,300 Pa across all cases, enabling direct visual comparison of how the pressure field evolves with increasing angle of attack.

Figure 12- presents the pressure coefficient (C_p) distributions along the airfoil chord at selected angles of attack.

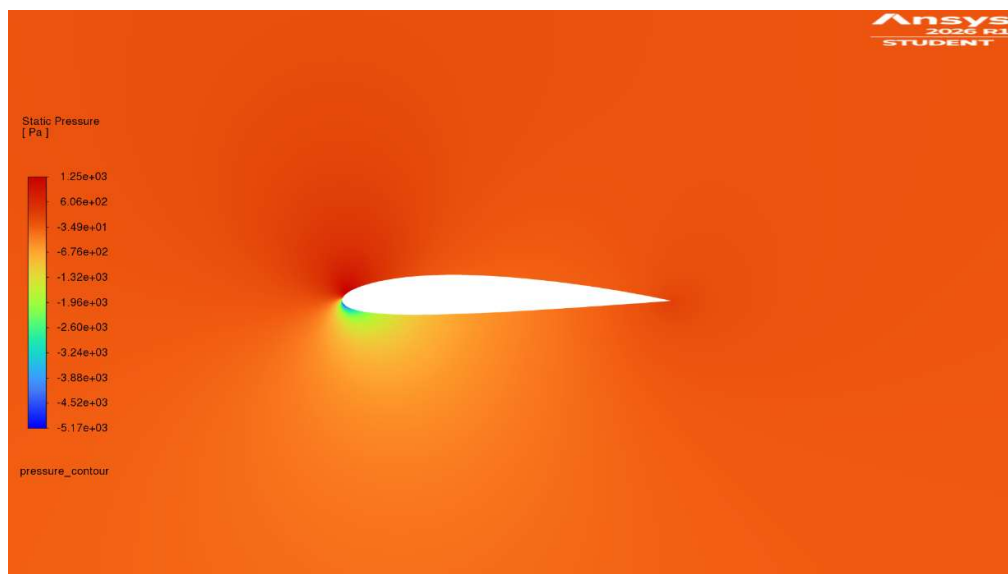


Figure 12: Auto-range pressure contour at -8° angle of attack.



Figure 13: Fixed-range pressure contour at -8° angle of attack.

At -8° , the auto-range pressure contour reveals a strikingly inverted pressure field compared to all positive AoA cases. The stagnation point has migrated to the **upper surface** of the leading edge — visible as the red high-pressure zone now appearing above the chord line rather than below it. Most notably, the suction region has shifted entirely to the **lower surface**, with the green/blue low-pressure zone concentrated at the lower leading edge reaching $-5,170$ Pa. The upper surface shows a broadly positive pressure relative to the lower surface — the complete opposite of the positive AoA cases — directly consistent with the negative lift coefficient of -0.6381 , where the net pressure force acts downward. The fixed-range contour at -8° shows a faint but visible lighter orange glow on the lower surface near the leading edge against the uniform red background — a subtle but clear indicator of the inverted suction region even on the wide fixed scale.

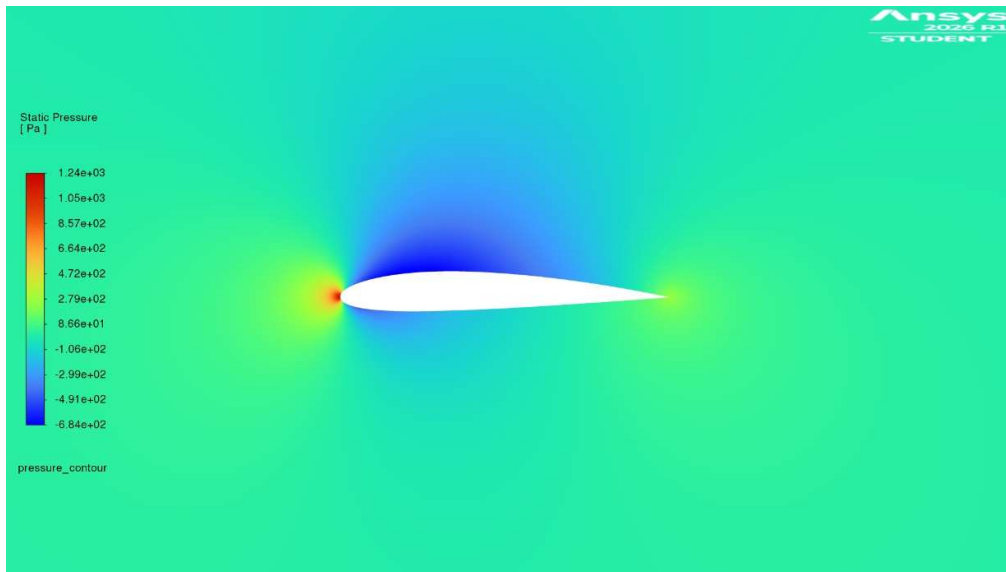


Figure 14: Auto-range pressure contour at 0° angle of attack.



Figure 15: Fixed-range pressure contour at 0° angle of attack.

At 0°, the auto-range contour reveals a rich and detailed pressure field. A strong red stagnation zone at the leading edge reaches +1,240 Pa, transitioning sharply to an organised blue low-pressure region covering the entire upper surface down to -684 Pa. The lower surface shows a moderate positive pressure region distinctly higher than the upper surface, confirming net upward lift generation from camber alone at zero incidence. The fixed-range contour at 0°,

however, appears almost entirely uniform orange with only a faint hint of lighter colouring at the leading edge and upper surface. This is because the pressure values at 0° are ranging only from -684 Pa to +1,240 Pa which is small relative to the fixed scale maximum of -14,000 Pa, causing the colour contrast to be compressed and washed out. This clearly illustrates the limitation of comparing low-AoA cases on the same scale as high-AoA cases.

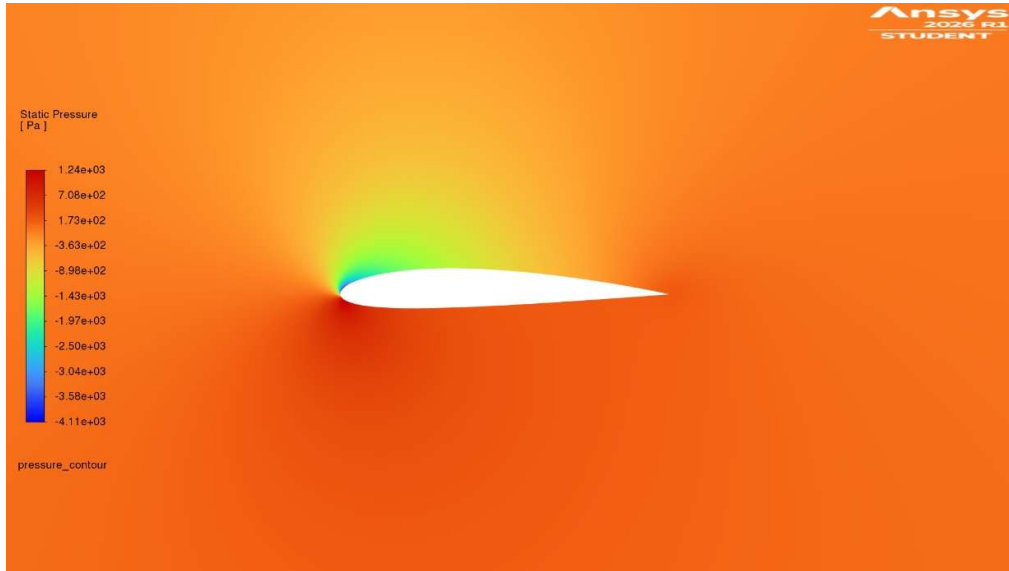


Figure 16: Auto-range pressure contour at 8° angle of attack.

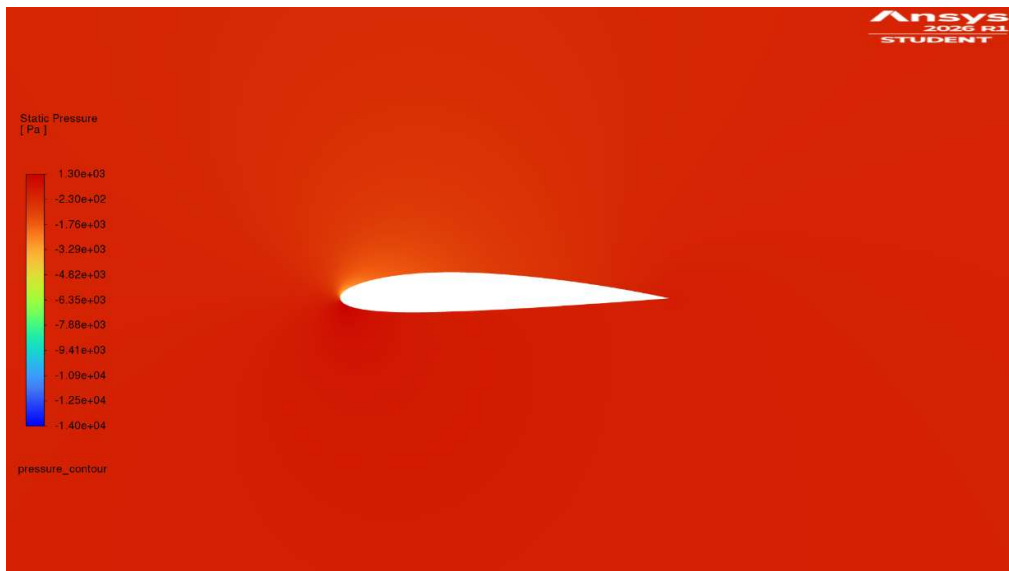


Figure 17: Fixed-range pressure contour at 8° angle of attack.

At 8° (peak efficiency), the auto-range contour shows a markedly more intense pressure field compared to 0° . The suction peak at the leading edge upper surface reaches $-4,110$ Pa — six times more negative than at 0° — with a compact but intense green/blue zone concentrated at the upper leading edge. The stagnation point has shifted visibly toward the lower surface. The fixed-range contour at 8° shows a slight but visible intensification compared to 0° — a faint orange glow around the leading edge upper surface is now discernible against the uniform red background, confirming the growing suction region. While still subtle, this marks a clear visual progression from the completely uniform 0° fixed-range contour, demonstrating the growing pressure differential driving the increasing lift coefficient.

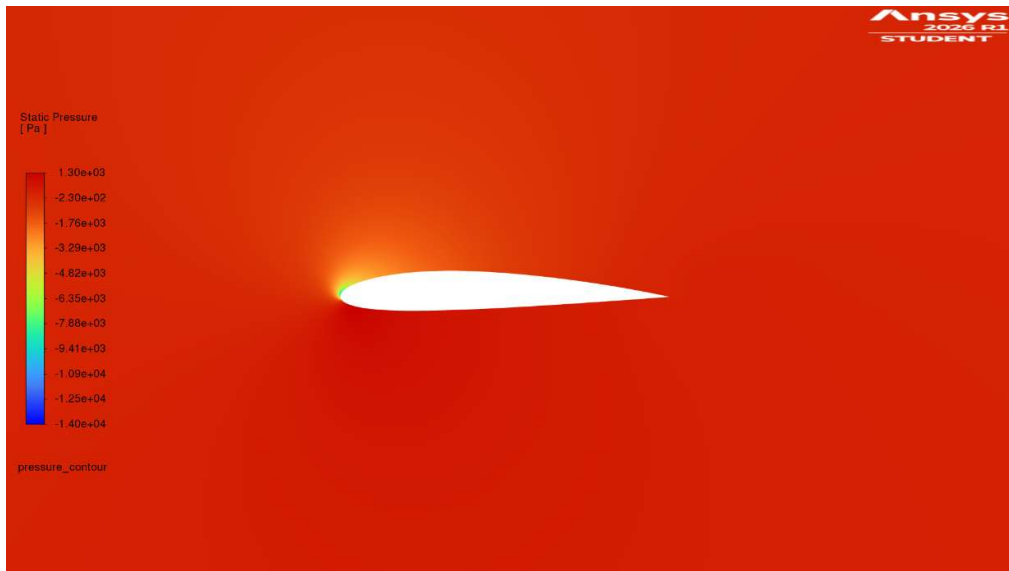


Figure 18: Fixed-range pressure contour at 14° angle of attack.

At 14° , only the fixed-range pressure contour is discussed, as the auto-range contour produces a nearly identical result — the actual pressure values at this angle approach the fixed-range limits closely enough that both colour scales are essentially the same. The fixed-range contour shows a distinct bright green/yellow spot at the leading edge upper surface against the red background, confirming the suction peak has reached a magnitude that registers meaningfully on the wider fixed scale. The stagnation point has migrated noticeably toward the lower surface,

and a faint darkening of the red below the airfoil confirms a growing lower surface positive pressure region. The pressure values across the rest of the airfoil body remain within the orange/red range of the fixed scale, indicating the pressure disturbance is still concentrated primarily at the leading edge rather than spread across the entire surface.

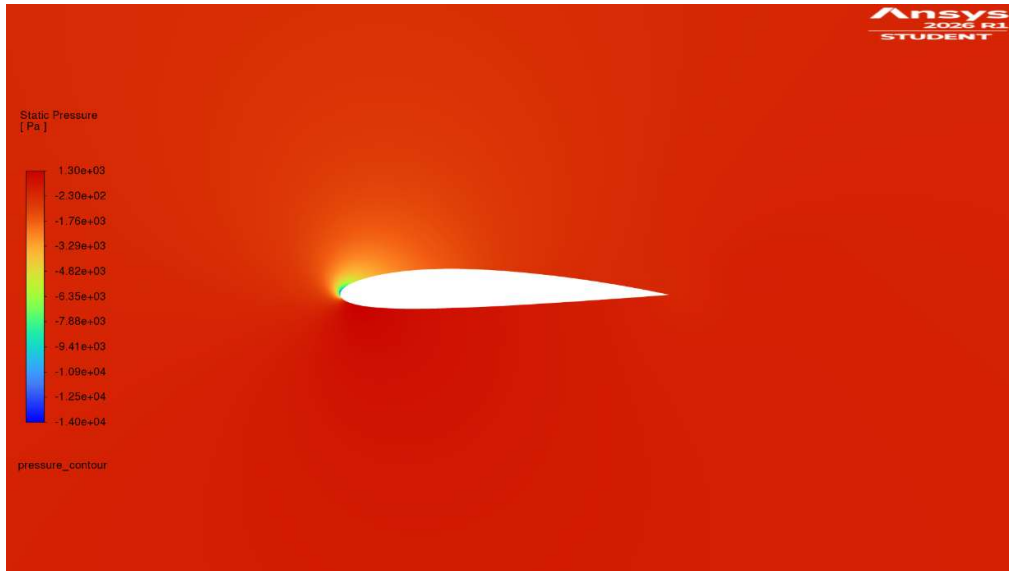


Figure 19: Fixed-range pressure contour at 17° angle of attack.

At 17° (stall angle), the fixed-range contour shows the most prominent leading edge feature of all cases — a visibly brighter and larger green/yellow suction spot compared to 14°, now the most intense feature visible across the entire fixed-scale series. The suction peak of -13,200 Pa approaches the fixed-range minimum of -14,000 Pa, explaining why the auto-range and fixed-range contours are virtually identical at this angle. The stagnation region on the lower leading edge is clearly visible as an orange/yellow zone, and the pressure gradient between the intense leading edge suction and the uniform orange airfoil body is at its steepest — confirming the boundary layer is under maximum adverse pressure stress at the stall condition



Figure 20: Fixed-range pressure contour at 20° angle of attack.

At 20° (post-stall), the fixed-range contour shows a subtle but physically meaningful change compared to 17°. The leading edge suction spot, while still prominent, is visibly slightly less intense than at 17° — consistent with the reduction in suction peak from -13,200 Pa to -11,400 Pa as the organised leading edge suction collapses post-stall. This slight reduction in the green/yellow intensity at the leading edge, despite the higher geometric angle of attack, is the clearest visual indicator of post-stall behaviour in the pressure contours — the flow can no longer sustain the organised suction peak it achieved at 17°, directly explaining the drop in lift coefficient from 1.6428 to 1.35.

As with pressure, both auto-range and fixed-range velocity contours are presented. The auto-range contours reveal localised velocity features at each AoA, while the fixed-range contours (0 to 160 m/s) enable direct cross-AoA comparison of flow acceleration and wake development.

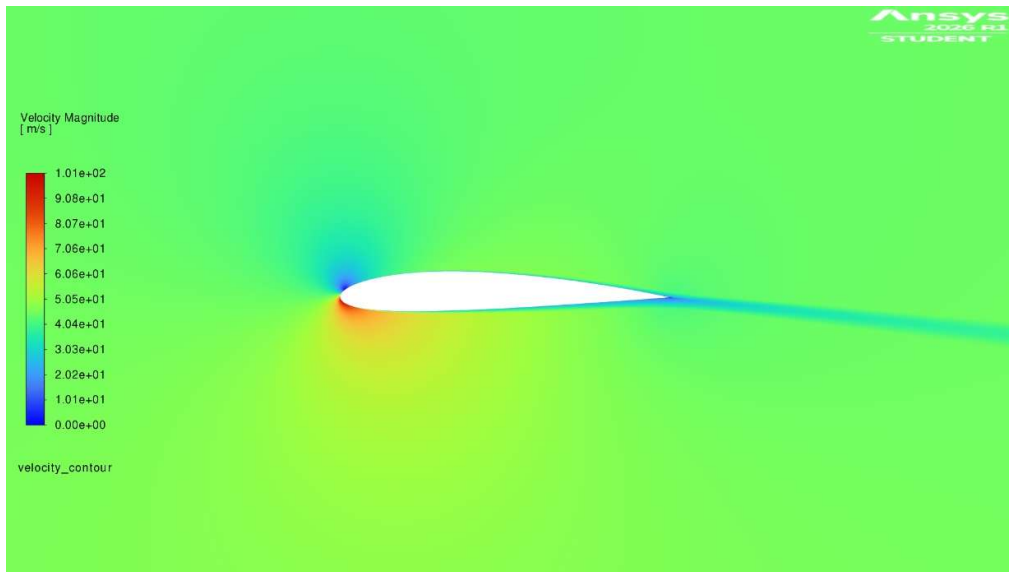


Figure 24: Auto-range velocity contour at -8° angle of attack.

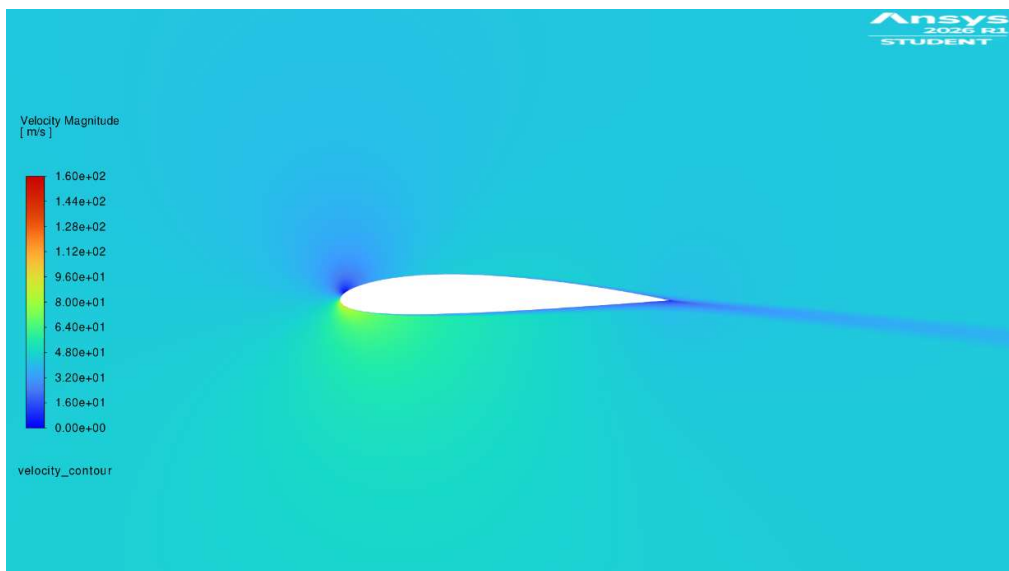


Figure 25: Fixed-range velocity contour at -8° angle of attack.



Figure 26: Wide-shot velocity contour at -8° angle of attack.

At -8° , the auto-range velocity contour shows a clear inversion of the velocity asymmetry compared to positive AoA cases. The maximum velocity of 101 m/s is now concentrated on the lower surface at the leading edge — visible as the red/orange zone below the chord line rather than above it — as the flow accelerates around the lower surface to generate the negative lift. The upper surface shows slower, more uniform flow. The fixed-range velocity contour makes this inversion clearly visible on the common scale — the green/yellow acceleration zone appears on the lower surface, in direct contrast to all positive AoA cases where it appears on the upper surface. The wide-shot contour shows the wake angling **downward** at approximately -8° consistent with the negative incidence and is noticeably thicker than the 0° wake — consistent with the higher drag coefficient at -8° compared to 0° .

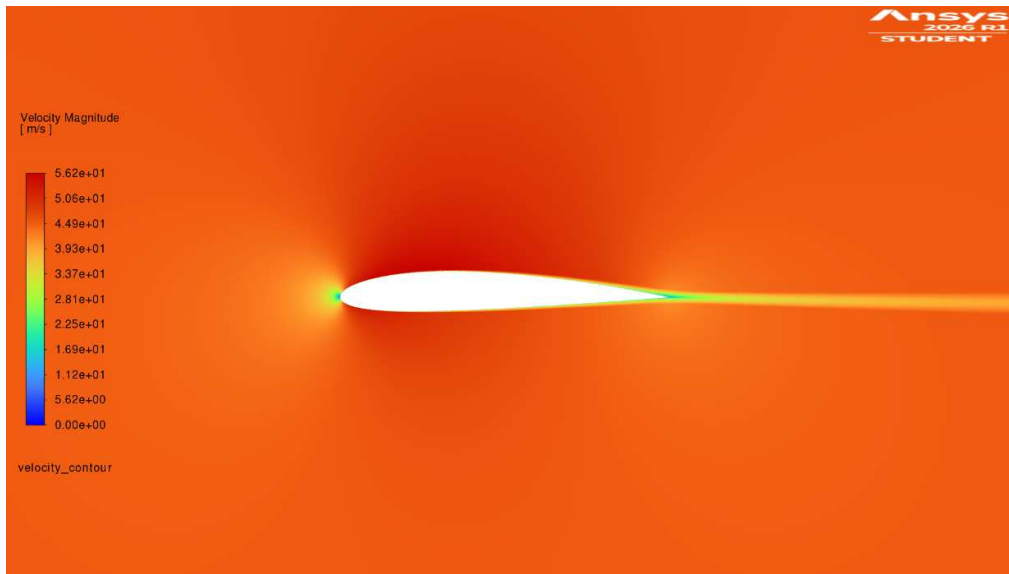


Figure 27: Auto-range velocity contour at 0° angle of attack.

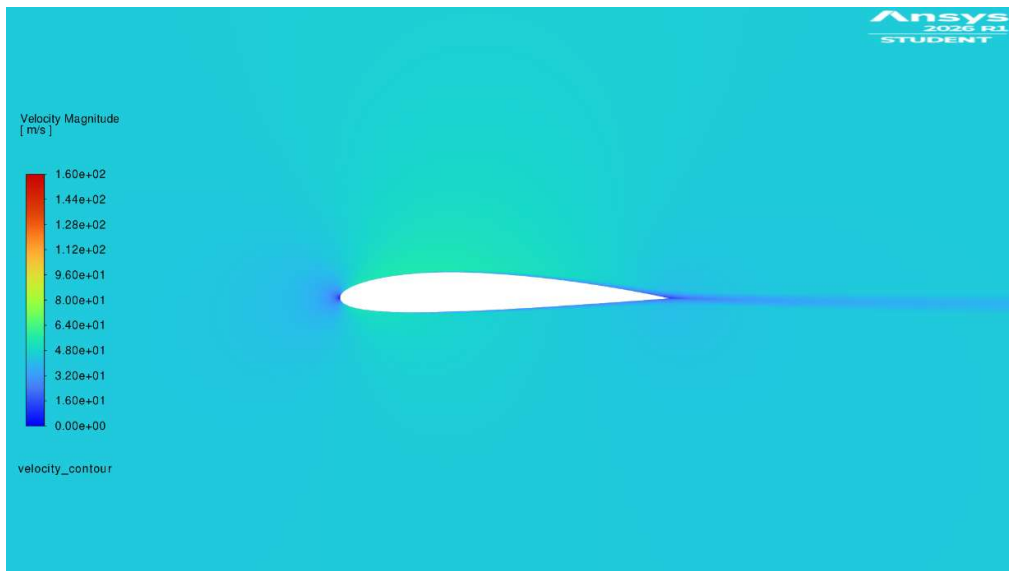


Figure 28: Fixed-range velocity contour at 0° angle of attack.



Figure 29: Wide-shot velocity contour at 0° angle of attack.

At 0° , the auto-range velocity contour shows a maximum velocity of only 56.2 m/s — the lowest peak of all positive AoA cases — with mildly asymmetric flow distribution consistent with the small positive lift from camber. The fixed-range contour places this in context clearly: against the 0–160 m/s scale, the entire flow field appears a uniform cyan/teal colour with only a faint green tinge at the leading edge upper surface, confirming that at 0° the flow velocities are all close to the freestream value of 45.28 m/s with minimal acceleration. The wake trailing behind the trailing edge is visible as a thin, barely-perceptible blue line confirming the low momentum deficit and minimum drag condition.

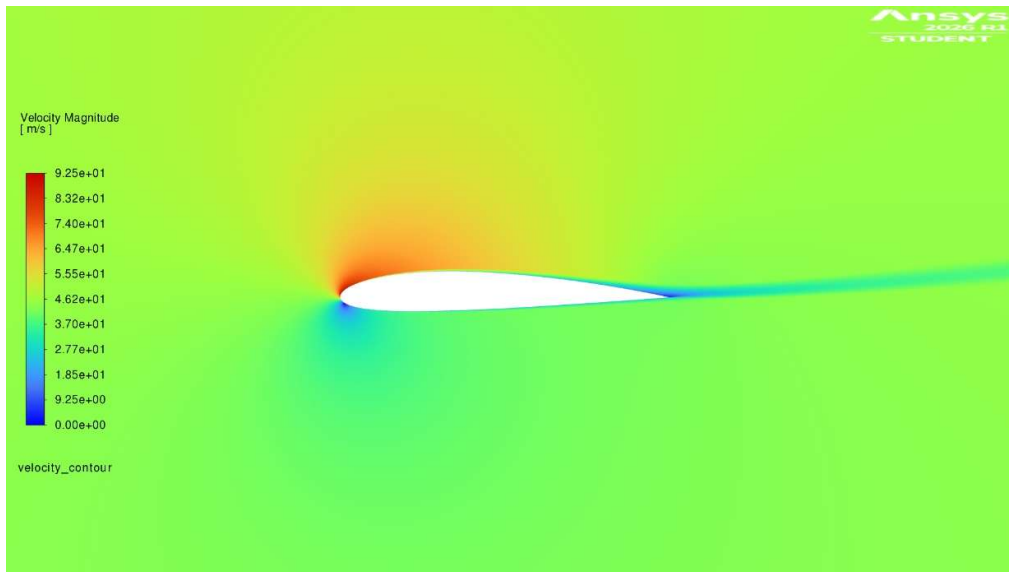


Figure 30: Auto-range velocity contour at 8° angle of attack.

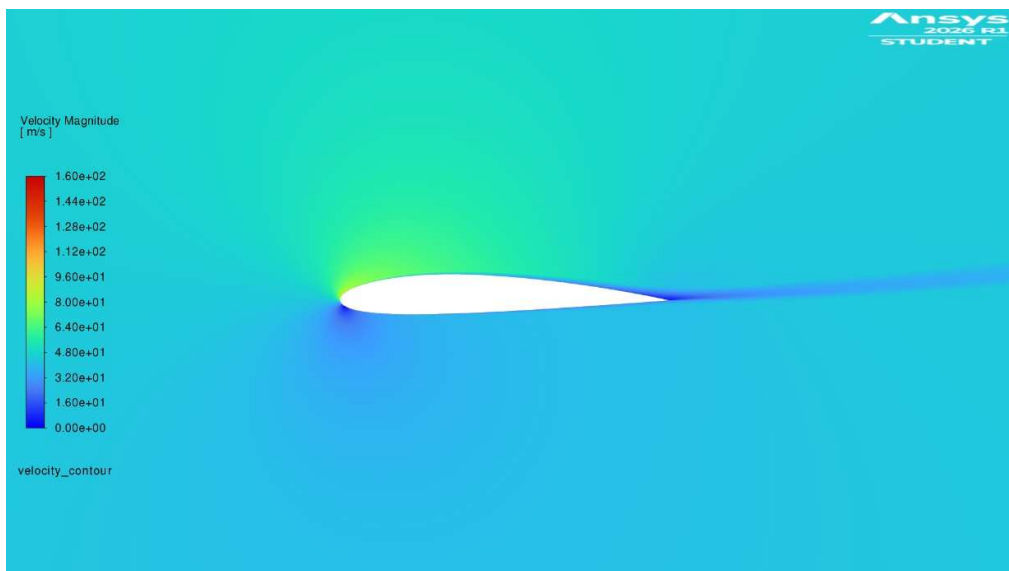


Figure 31: Fixed-range velocity contour at 8° angle of attack.



Figure 32: Wide-shot velocity contour at 8° angle of attack.

At 8°, the auto-range contour reveals significant flow acceleration with peak velocities reaching 92.5 m/s at the leading edge, clearly visible as a red/orange zone. The velocity field is now strongly asymmetric — fast flow over the upper surface contrasting with slower flow beneath. The fixed-range contour shows this clearly in comparison to 0°: a bright green/yellow zone now occupies the leading edge upper surface, standing out distinctly against the cyan background. The wake behind the trailing edge has widened and angles upward at approximately 8° consistent with the inclined freestream, appearing as a more prominent blue streak than at 0°.

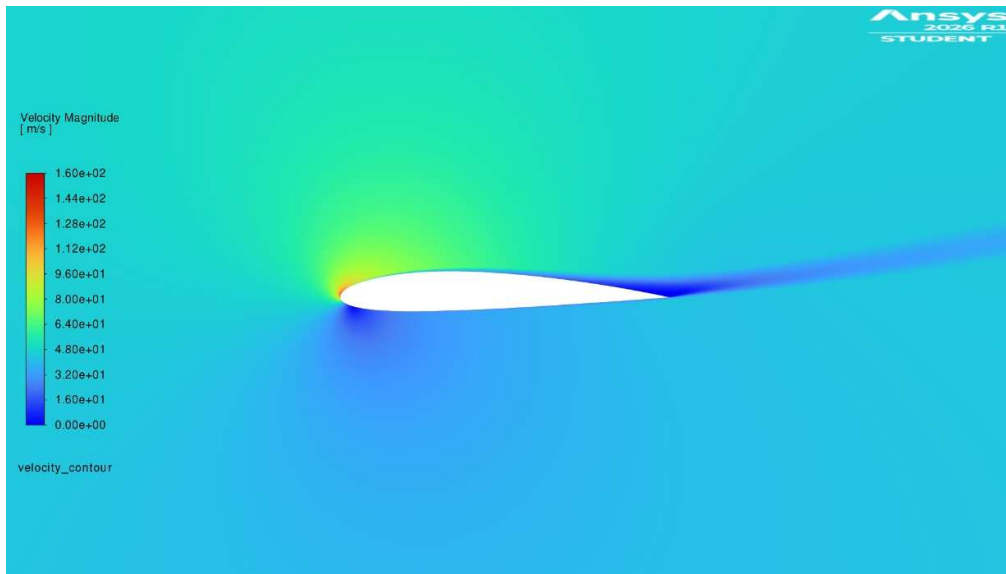


Figure 33: Fixed-range velocity contour at 14° angle of attack.

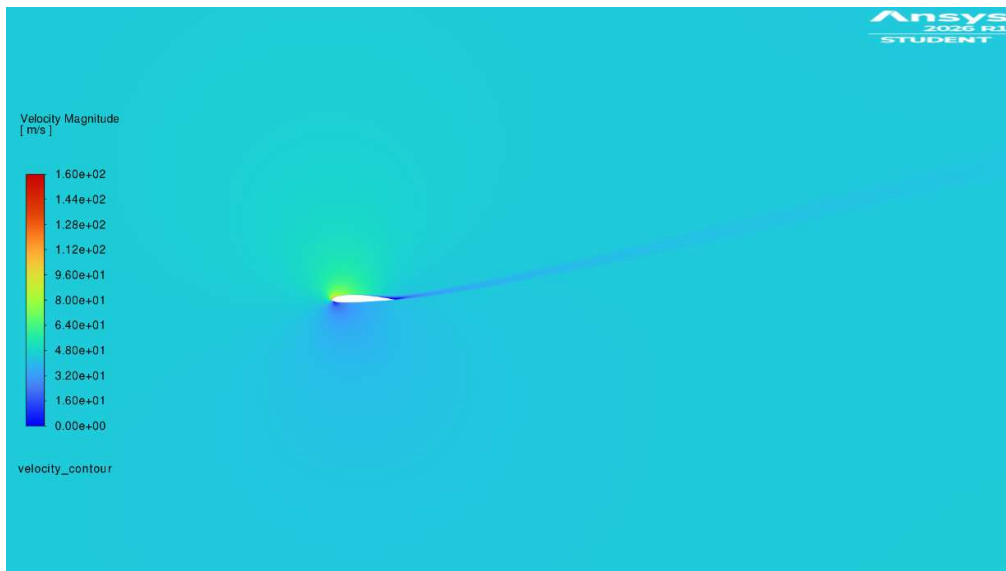


Figure 34: Wide-shot velocity contour at 14° angle of attack.

At 14°, only the fixed-range velocity contour is discussed, as the auto-range produces a nearly identical result — the peak velocity of 135 m/s at 14° is close enough to the fixed-range maximum of 160 m/s that both scales produce essentially the same colour distribution. The fixed-range contour shows a clear yellow/green acceleration zone at the leading edge upper surface, substantially larger and more intense than at 8°. A prominent blue low-velocity region

has developed immediately behind the trailing edge on the upper surface, indicating onset of boundary layer thickening under the adverse pressure gradient. The wake trailing downstream is noticeably wider than at 8° and angles upward at 14° consistent with the inclined freestream.

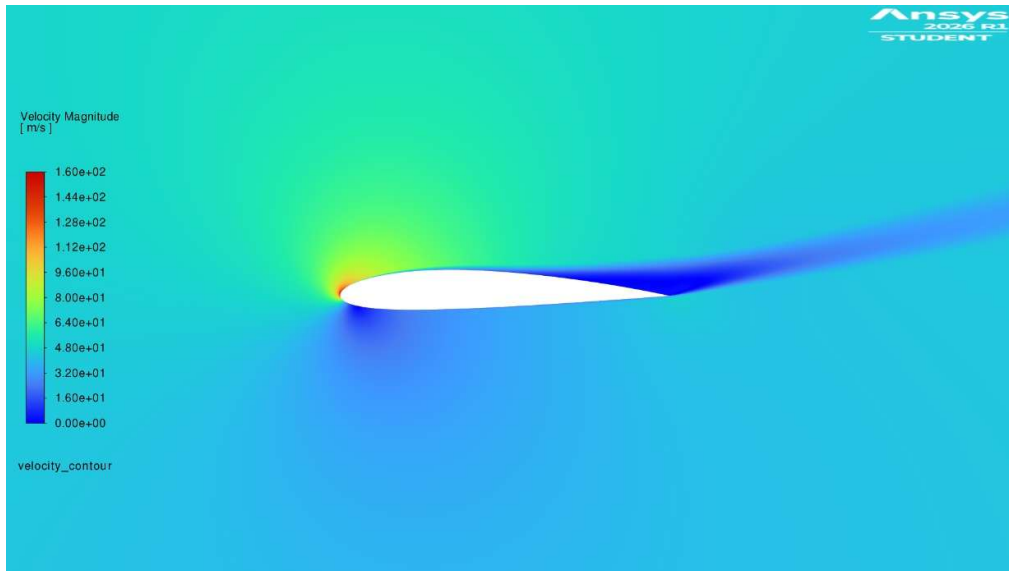


Figure 35: Fixed-range velocity contour at 17° angle of attack.

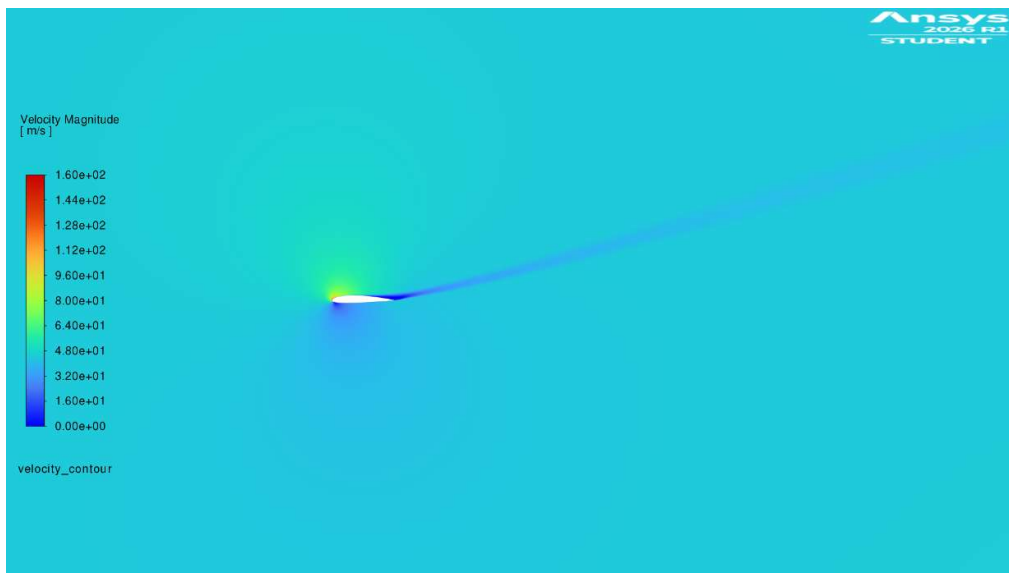


Figure 36: Wide-shot velocity contour at 17° angle of attack.

At 17° , the fixed-range contour shows the peak velocity has increased to 151 m/s — approaching the fixed-range maximum of 160 m/s, again explaining why auto and fixed

contours are nearly identical. The leading edge yellow acceleration zone is at its most intense across all cases, while simultaneously the blue low-velocity region behind the trailing edge has grown dramatically compared to 14° , now covering a much larger portion of the upper surface rear and extending further downstream. This simultaneous presence of maximum leading edge acceleration and a large trailing edge separation region captured on the same fixed scale provides the clearest visual representation of the airfoil at its aerodynamic limit — maximum suction at the front driving maximum lift, while the boundary layer at the rear is on the verge of full separation.

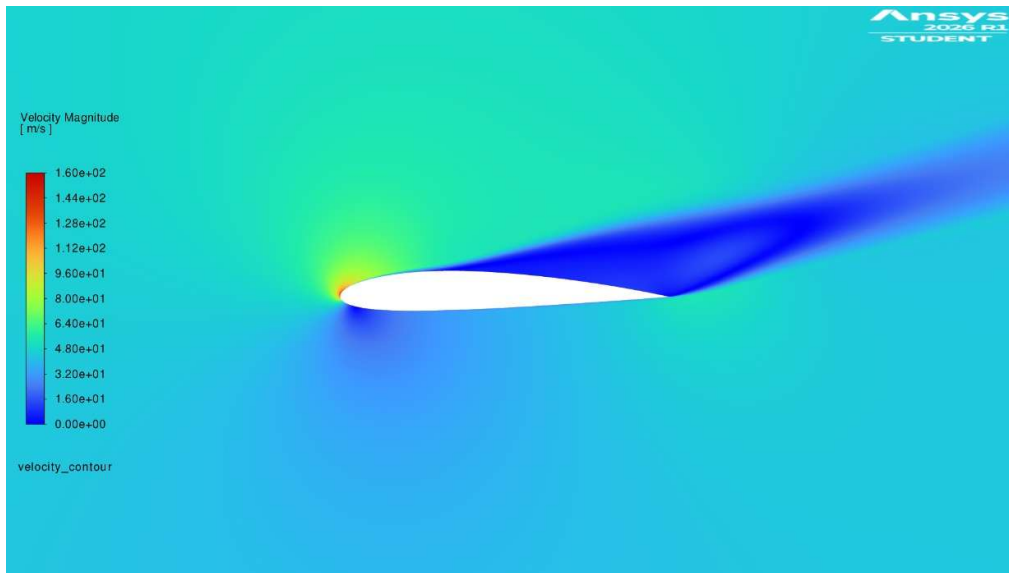


Figure 37: Fixed-range velocity contour at 20° angle of attack.

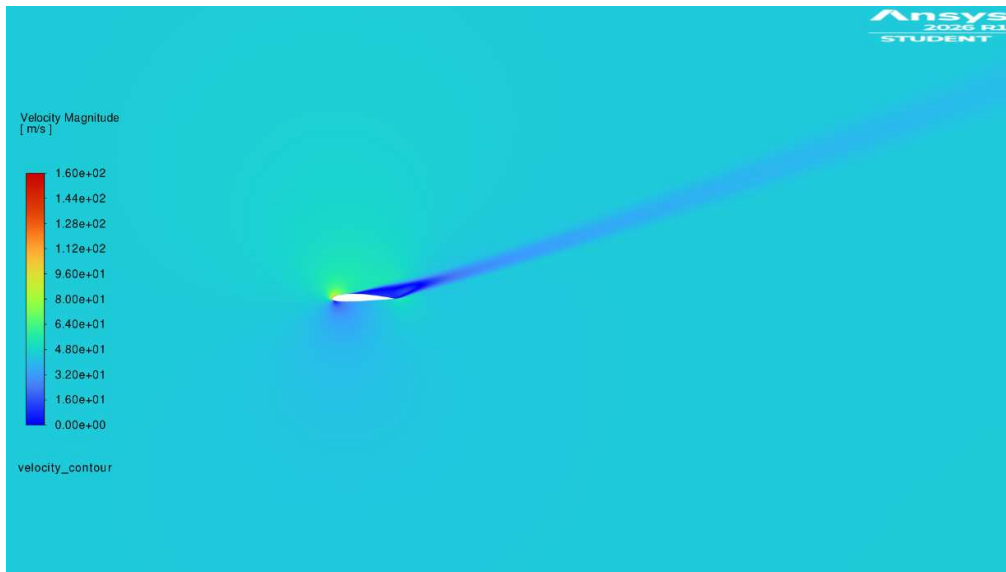


Figure 38: Wide-shot velocity contour at 20° angle of attack.

At 20° , the fixed-range contour captures the post-stall flow field most clearly. The leading edge acceleration zone has slightly reduced in intensity compared to 17° — consistent with the reduced peak velocity of 140 m/s — while the blue separated region now dominates the entire upper surface and extends far downstream into a massive wake. The wide-shot contour makes the full scale of this post-stall wake unmistakably clear — a broad, diffuse blue region fills the entire downstream domain at a steep angle consistent with 20° incidence. This massive momentum deficit is the direct physical cause of the drag coefficient reaching 0.098 at 20° — more than ten times the minimum value recorded at 0° .

Table 10: Summary of Key Results.

Parameter	Value	AoA
Maximum Cl	1.6428	17°
Minimum Cd	0.009193	-2°
Maximum Cl/Cd	73.28	8°
Stall angle	17°	—
Zero lift angle	≈ -2°	—
Peak suction pressure	-13,200 Pa	17°
Peak flow velocity	151 m/s	17°

6. Conclusion

This study presented a two-dimensional steady-state CFD analysis of the NACA 2412 airfoil at $Re = 3.1 \times 10^6$ using the SST k- ω turbulence model in ANSYS Fluent 2026 R1, covering angles of attack from -8° to 20°.

A mesh independence study was conducted across three mesh levels using the Grid Convergence Index method based on Richardson extrapolation with a refinement ratio of $r = \sqrt{2}$. Both lift and drag coefficients exhibited monotonic convergence, with GCI fine values of 0.172% and 0.936% respectively — confirming the 400,020 cell fine mesh produces results within 1% of the grid-independent solution. The fine mesh was selected for all production simulations.

Validation against NACA TR-824 experimental data at 14° angle of attack yielded a lift coefficient deviation of 1.69% and a drag coefficient deviation of 12.2%. The close agreement in lift coefficient confirms the simulation accurately captures the pressure field and lift generation mechanism. The drag coefficient overprediction is attributed to the fully turbulent assumption of SST k- ω , which overestimates skin friction drag by treating the laminar portion

of the boundary layer as turbulent — a known and well-documented limitation of fully turbulent RANS solvers at these Reynolds numbers.

The aerodynamic analysis revealed a maximum lift coefficient of $Cl_{max} = 1.6428$ at a stall angle of 17° , beyond which lift decreases as flow separates progressively from the upper surface. Maximum aerodynamic efficiency of $Cl/Cd = 73.28$ was achieved at 8° angle of attack, representing the optimal operating condition at this Reynolds number. The minimum drag coefficient of 0.009193 occurred at -2° , where the airfoil operates at its most streamlined condition. Post-stall behaviour at 18° – 20° was characterised by oscillatory residuals consistent with unsteady massively separated flow, with coefficient values obtained by averaging over the final 50 iterations.

Flow field visualisation confirmed the progressive development of the leading edge suction peak with increasing angle of attack, reaching maximum intensity at the stall angle before collapsing post-stall. The wide-shot velocity contours clearly demonstrated the dramatic growth of the wake from a barely perceptible thin line at 0° to a massive, separated region at 20° , directly explaining the tenfold increase in drag coefficient across the simulated range.

The primary limitations of this study are the fully turbulent RANS assumption, the relatively compact domain size of 7.5c upstream and 14c downstream compared to the recommended 20c, and the use of steady-state simulation at post-stall angles where the flow is inherently unsteady. Future work could address these limitations by employing the γ - $Re\theta$ transitional SST model to better capture laminar-turbulent transition, extending the domain size, and applying unsteady RANS or Large Eddy Simulation at post-stall conditions for more physically accurate post-stall predictions.

7. References

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Roache, P. J. (1994). Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116(3), 405–413. <https://doi.org/10.1115/1.2910291>

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Thomas, C. (2023). *Mesh independence in CFD: NACA2412 example (Ansys student)* [Video]. YouTube. <https://www.youtube.com/watch?v=Bzr35PLVupA&t=3132s>

8. Appendix

Appendix A - Complete Aerodynamic Data Table

Table A1: the complete lift coefficient, drag coefficient and lift-to-drag ratio for all simulated angles of attack at $Re = 3.1 \times 10^6$.

Angle of Attack (°)	Drag Coefficient	Lift Coefficient	C_l/C_d
-8	0.012128201	-0.6381138	-52.61404804
-6	0.010537241	-0.4284668	-40.66213917
-4	0.009594548	-0.2154380	-22.45420571
-2	0.009192682	-0.0003250	-0.03535230
0	0.009282084	0.2139360	23.04827376
2	0.009818312	0.4282772	43.62024972
4	0.010811717	0.6408320	59.27199075
6	0.012282055	0.8486603	69.09758424
8	0.014316319	1.0491118	73.28083427
10	0.017057789	1.2367771	72.50512361
12	0.020829597	1.4056329	67.48248178
13	0.023299212	1.4796730	63.50742677
14	0.026339863	1.5435918	58.60287884
15	0.030263255	1.5952709	52.71313016
16	0.035522715	1.6303877	45.89704644
17	0.043014954	1.6428282	38.19202503
18	0.054056436	1.6233462	30.03058137
19	0.070810000	1.5580000	22.00254201
20	0.098000000	1.3500000	13.77551020

Appendix B - Pressure Contours

Figures B1 through B38 present static pressure contours for all simulated angles of attack. For each angle of attack, two contours are presented — an auto-range contour scaled to the local

minimum and maximum pressure values to reveal localised flow features, and a fixed-range contour using a consistent scale of -14,000 Pa to +1,300 Pa to enable direct cross-AoA comparison.

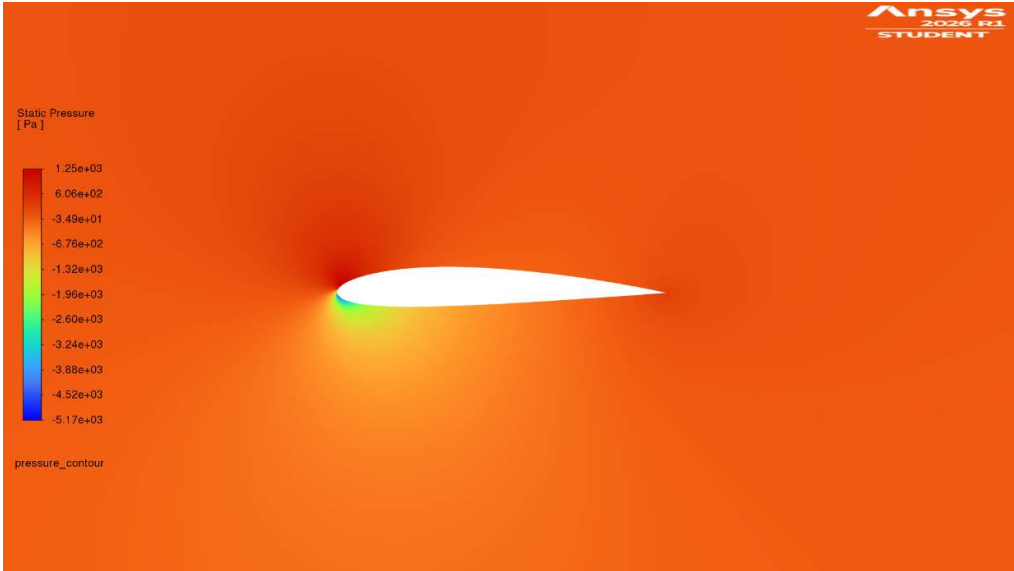


Figure B1: Auto-range pressure contour at -8° angle of attack.



Figure B2: Fixed-range pressure contour at -8° angle of attack.

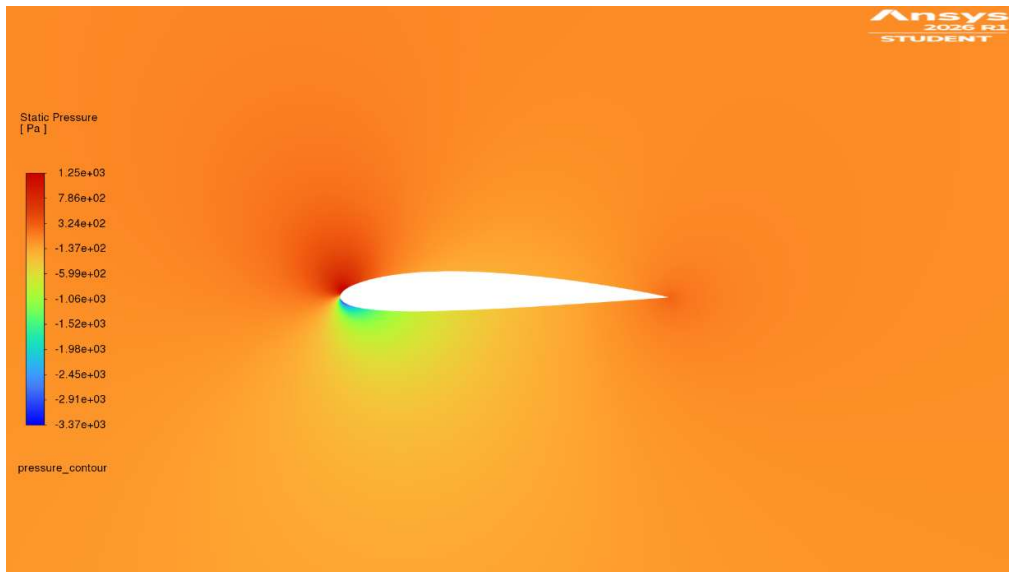


Figure B3: Auto-range pressure contour at -6° angle of attack.



Figure B4: Fixed-range pressure contour at -6° angle of attack.

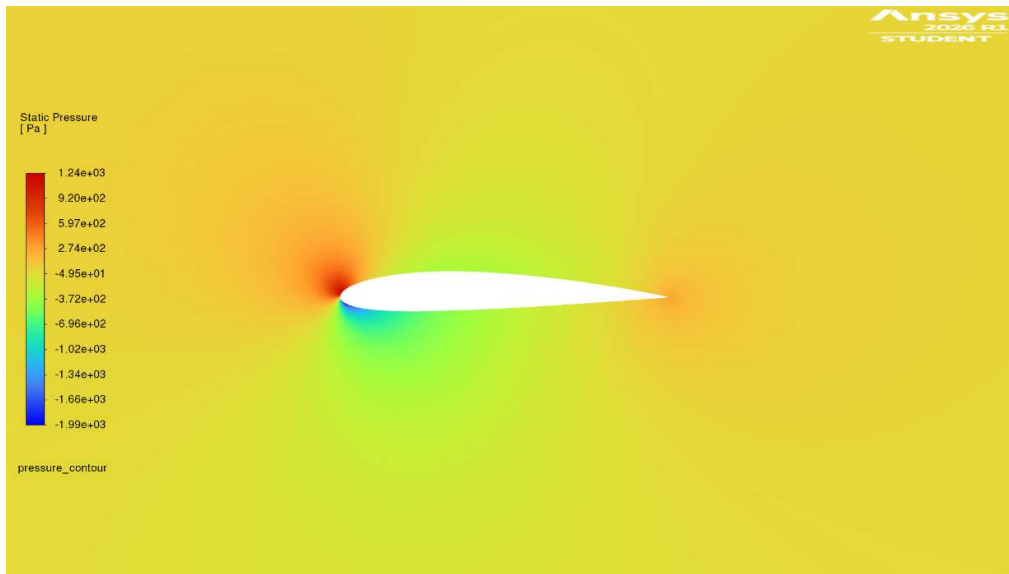


Figure B5: Auto-range pressure contour at -4° angle of attack.



Figure B6: Fixed-range pressure contour at -4° angle of attack.

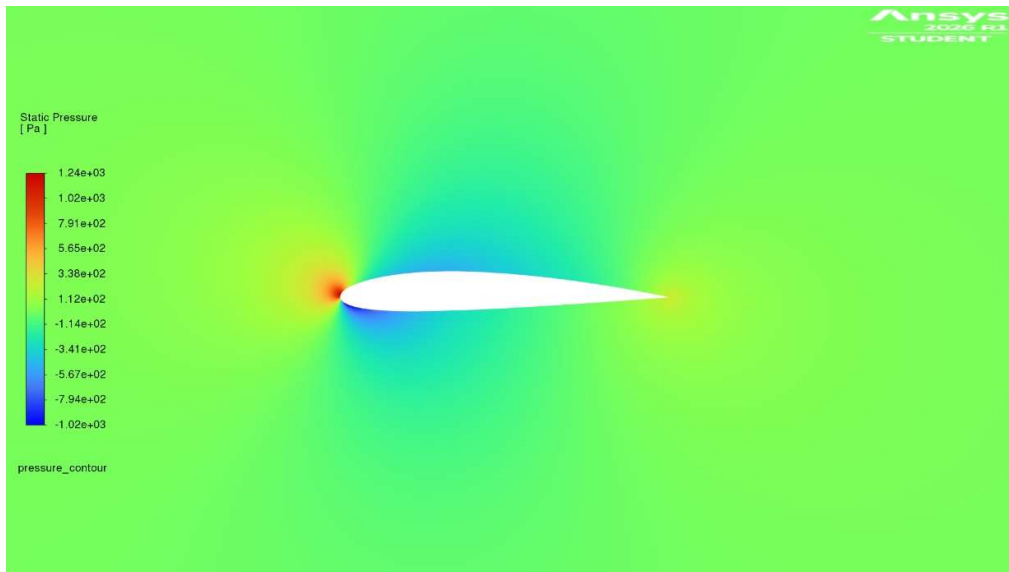


Figure B7: Auto-range pressure contour at -2° angle of attack.



Figure B8: Fixed-range pressure contour at -2° angle of attack.

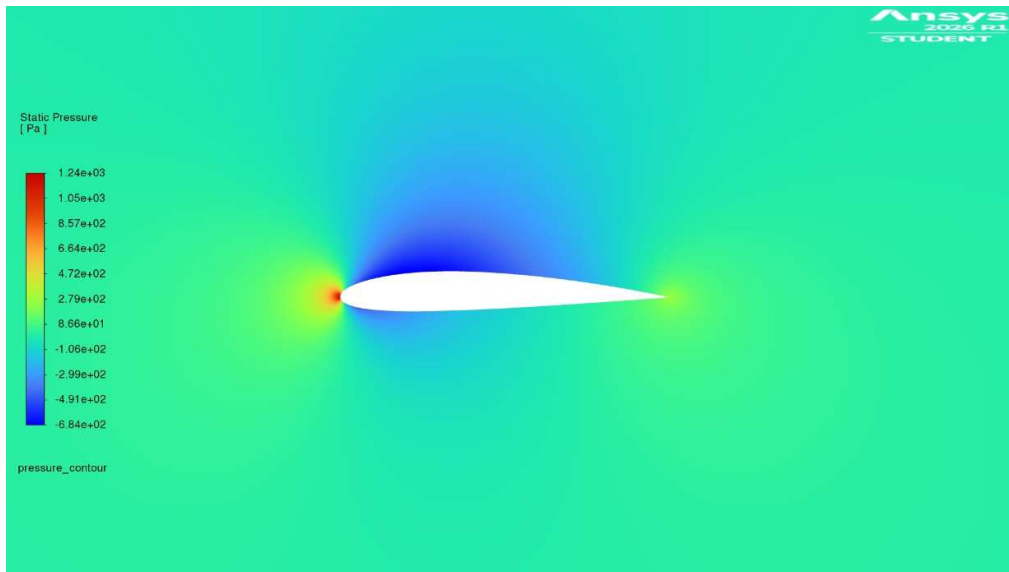


Figure B9: Auto-range pressure contour at 0° angle of attack.



Figure B10: Fixed-range pressure contour at 0° angle of attack.

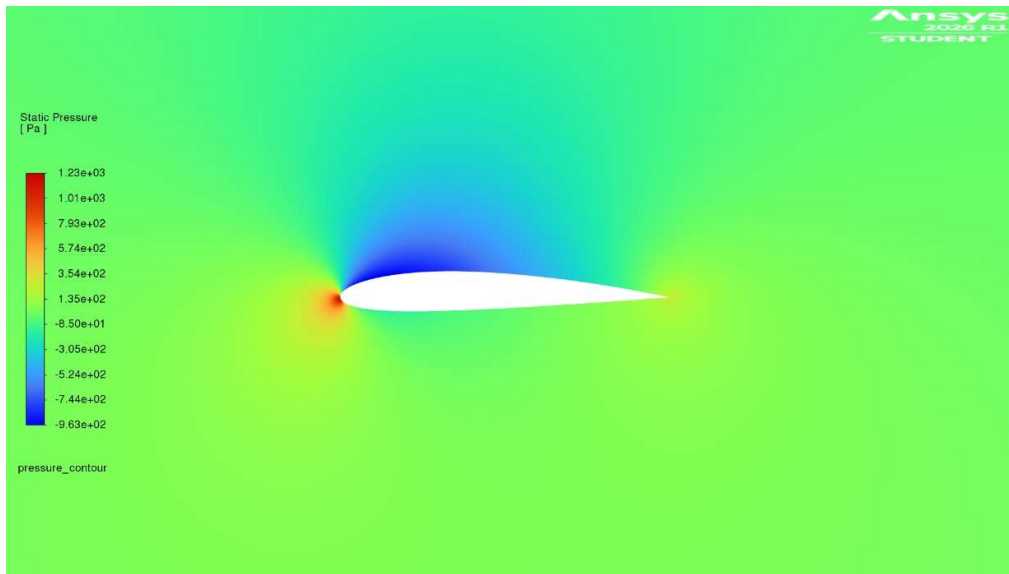


Figure B11: Auto-range pressure contour at 2° angle of attack.



Figure B12: Fixed-range pressure contour at 2° angle of attack.

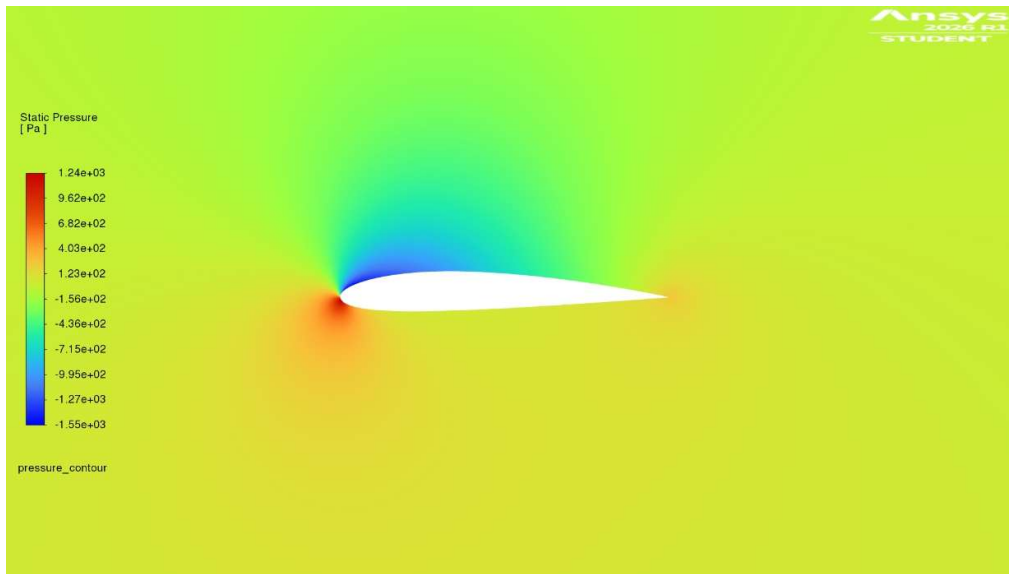


Figure B13: Auto-range pressure contour at 4° angle of attack.



Figure B14: Fixed-range pressure contour at 4° angle of attack.

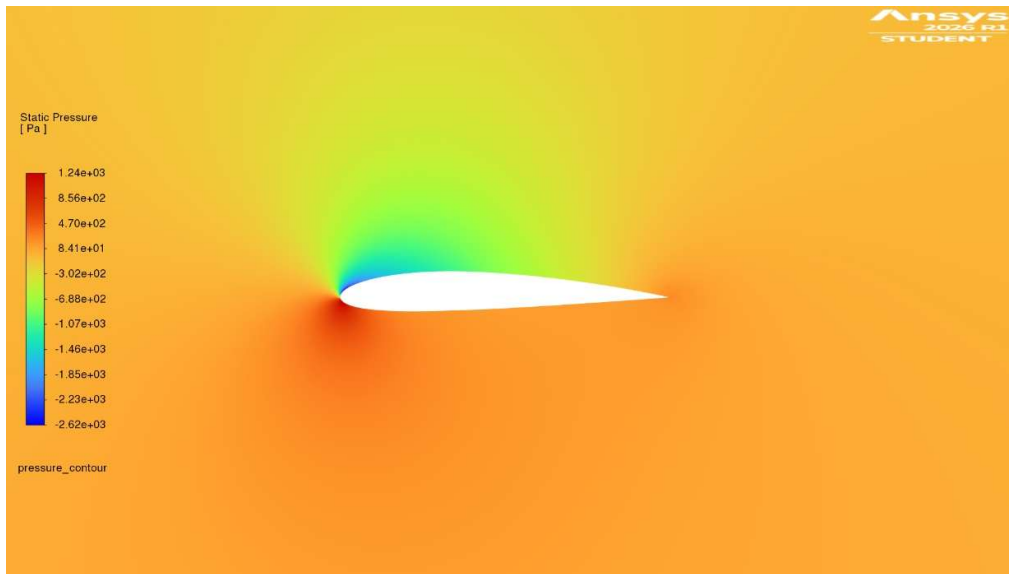


Figure B15: Auto-range pressure contour at 6° angle of attack.



Figure B16: Fixed-range pressure contour at 6° angle of attack.

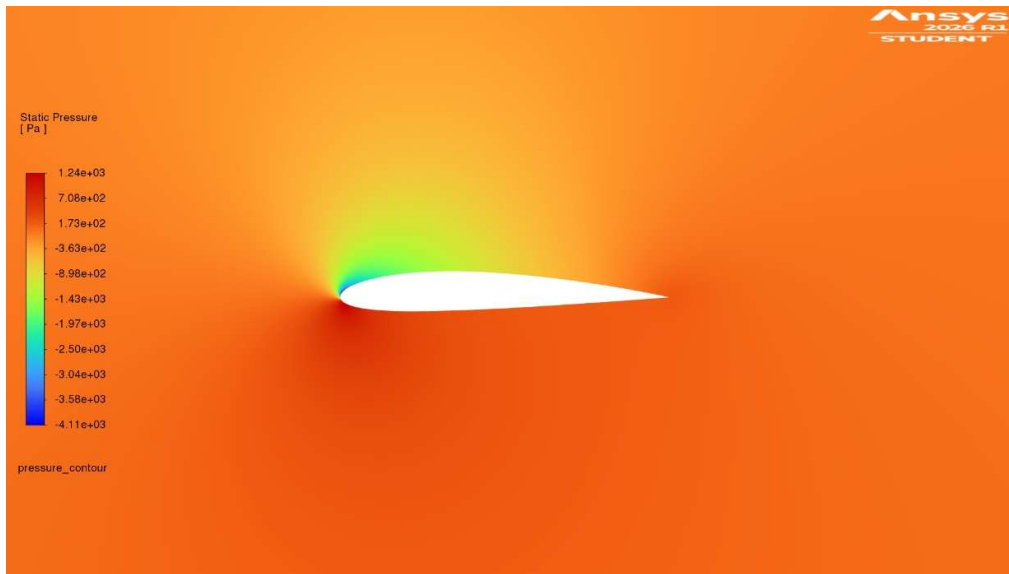


Figure B17: Auto-range pressure contour at 8° angle of attack.



Figure B18: Fixed-range pressure contour at 8° angle of attack.

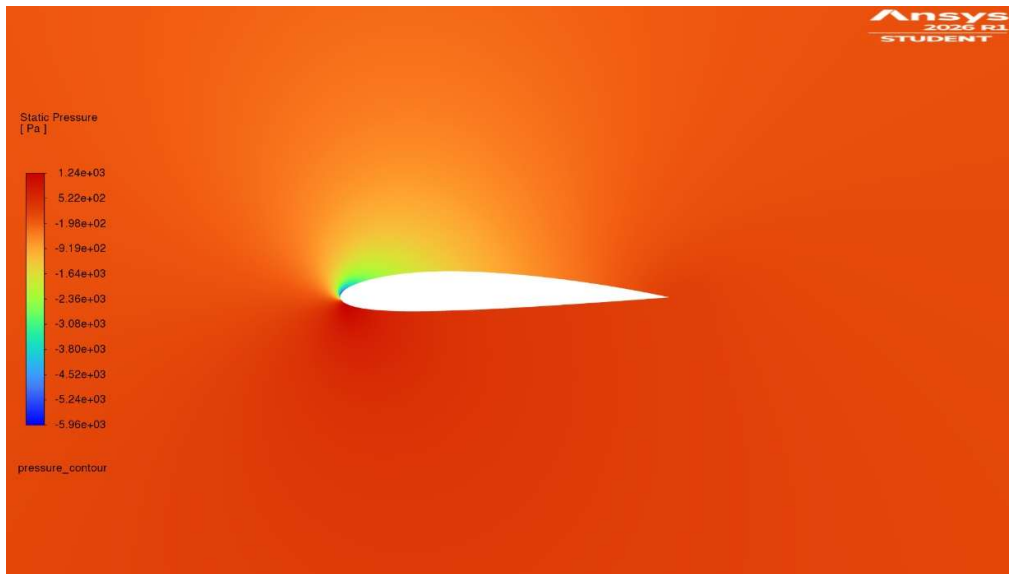


Figure B19: Auto-range pressure contour at 10° angle of attack.

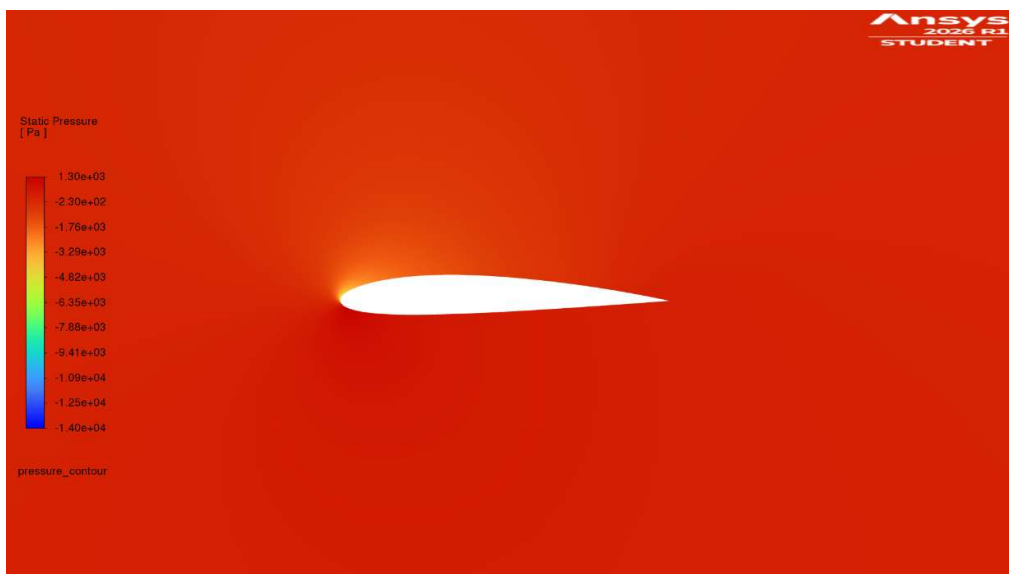


Figure B20: Fixed-range pressure contour at 10° angle of attack.

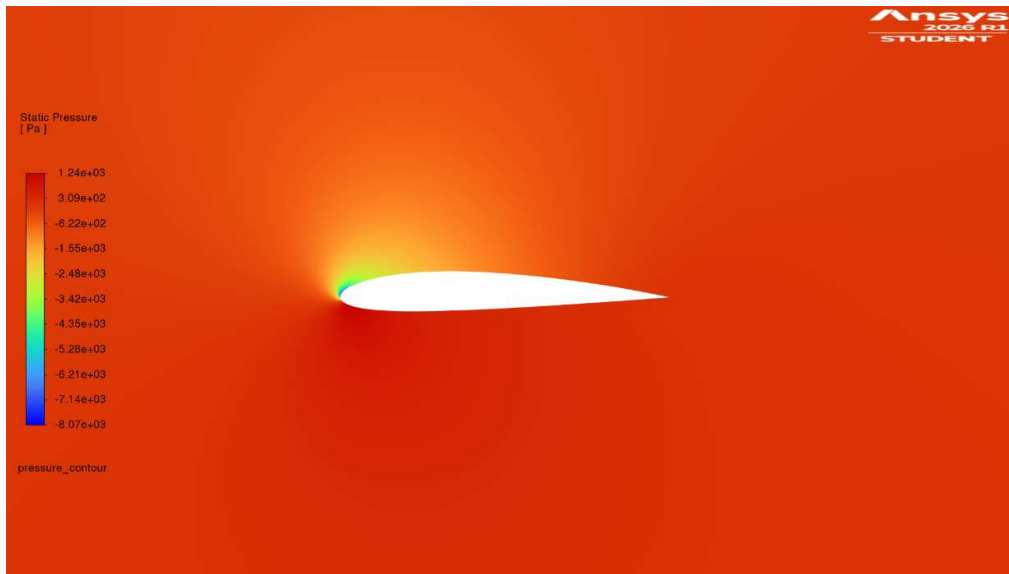


Figure B21: Auto-range pressure contour at 12° angle of attack.

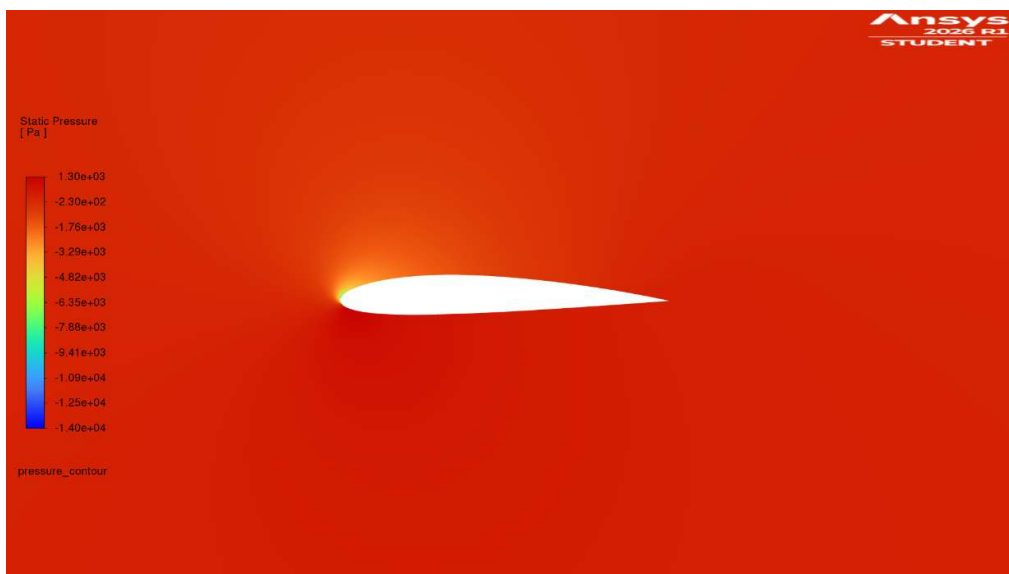


Figure B22: Fixed-range pressure contour at 12° angle of attack.

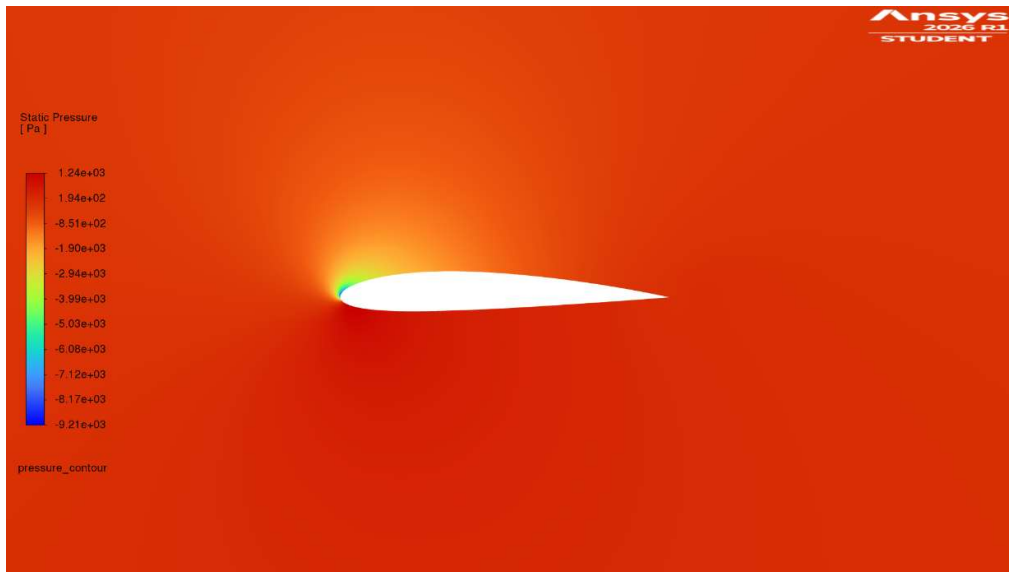


Figure B23: Auto-range pressure contour at 13° angle of attack.

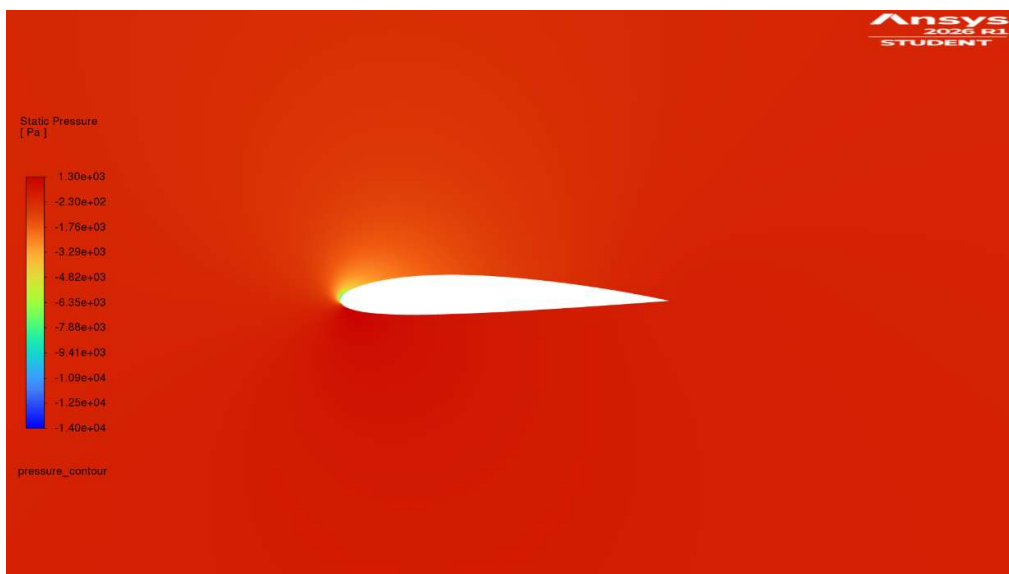


Figure B24: Fixed-range pressure contour at 13° angle of attack.

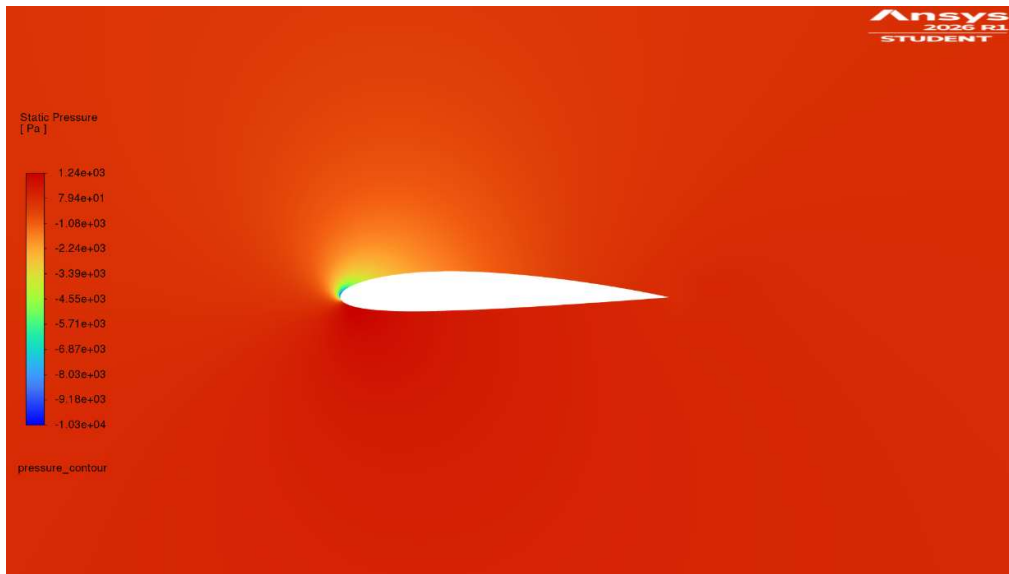


Figure B25: Auto-range pressure contour at 14° angle of attack.

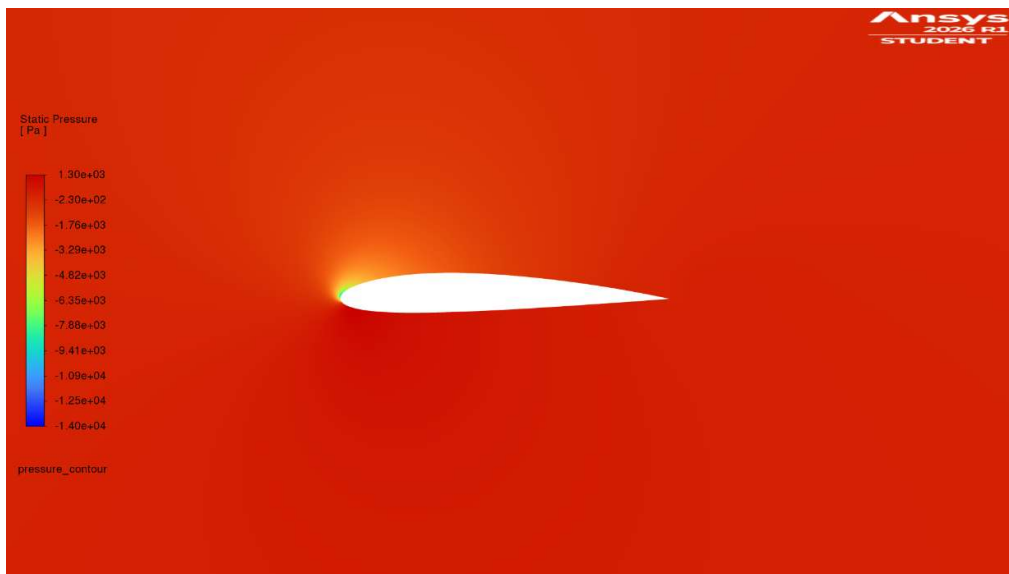


Figure B26: Fixed-range pressure contour at 14° angle of attack.

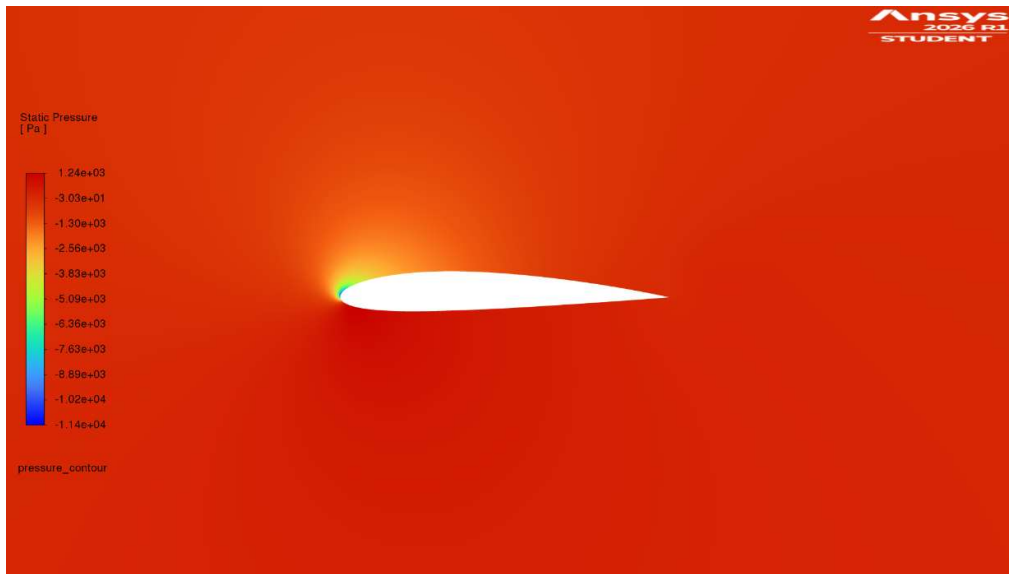


Figure B27: Auto-range pressure contour at 15° angle of attack.

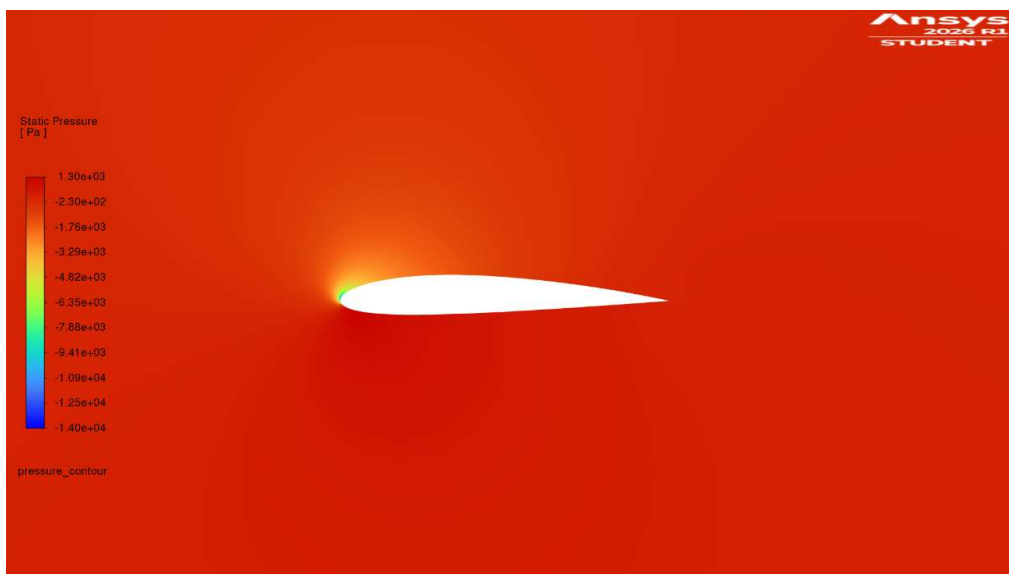


Figure B28: Fixed-range pressure contour at 15° angle of attack.

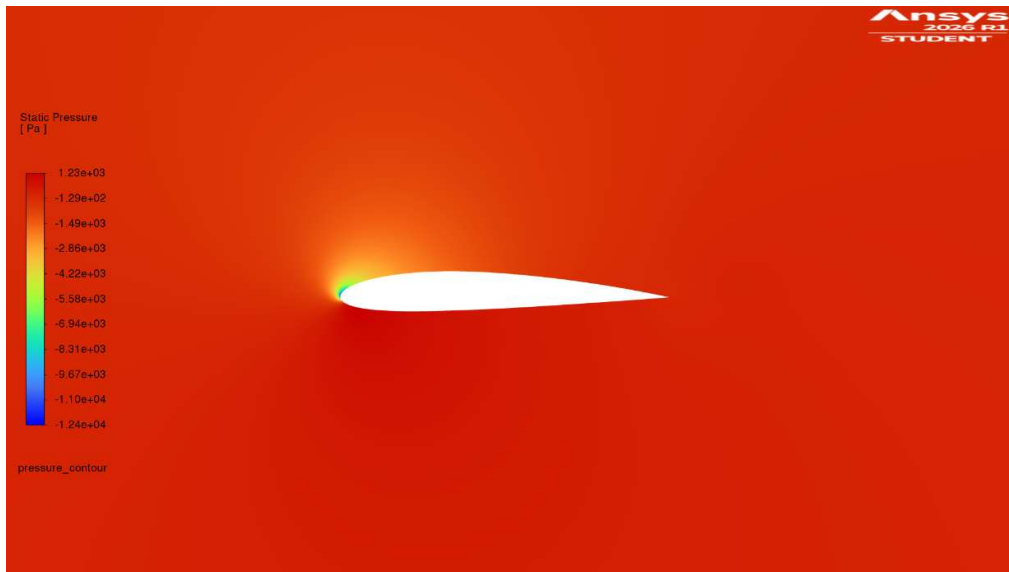


Figure B29: Auto-range pressure contour at 16° angle of attack.

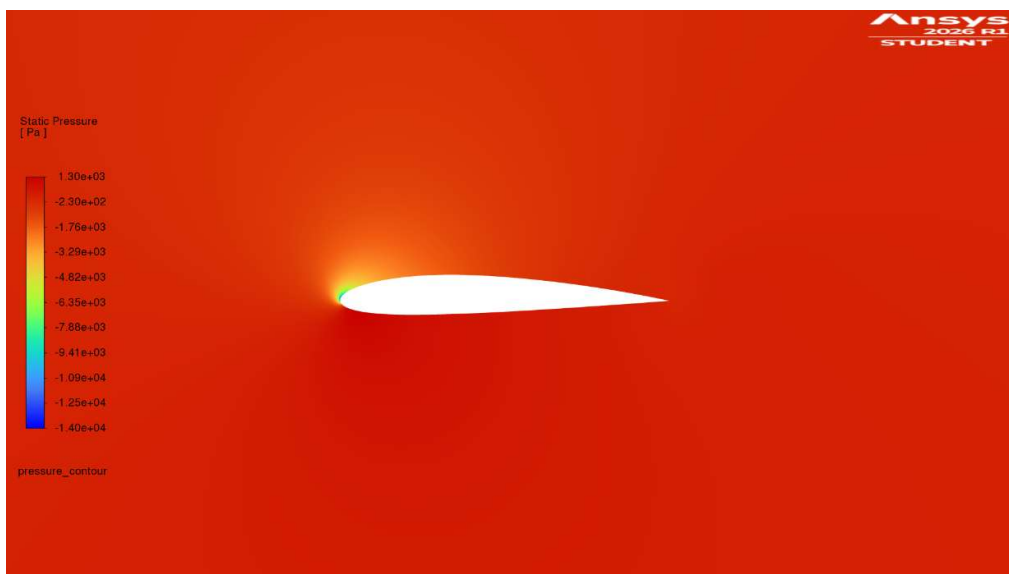


Figure B30: Fixed-range pressure contour at 16° angle of attack.

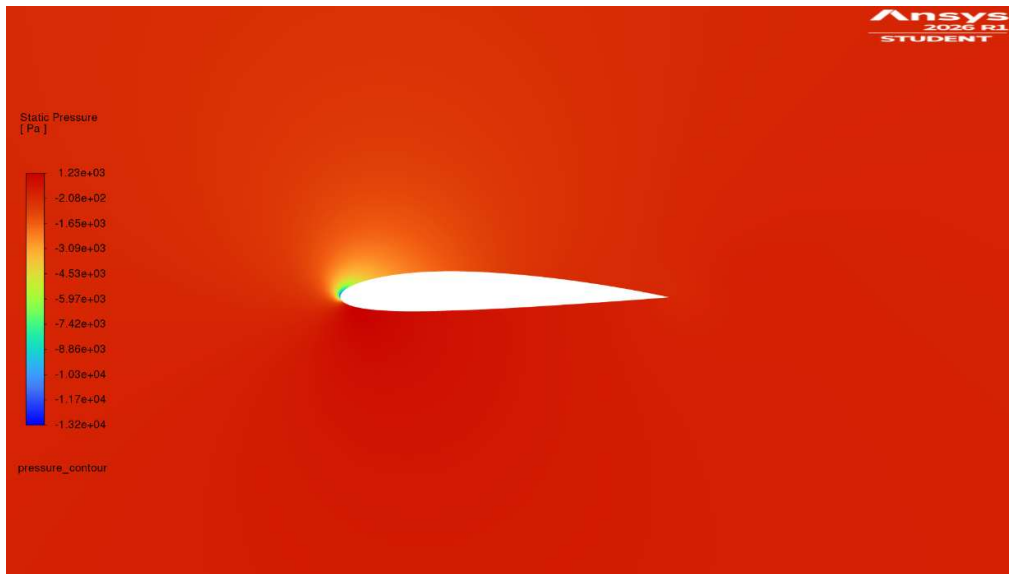


Figure B31: Auto-range pressure contour at 17° angle of attack.

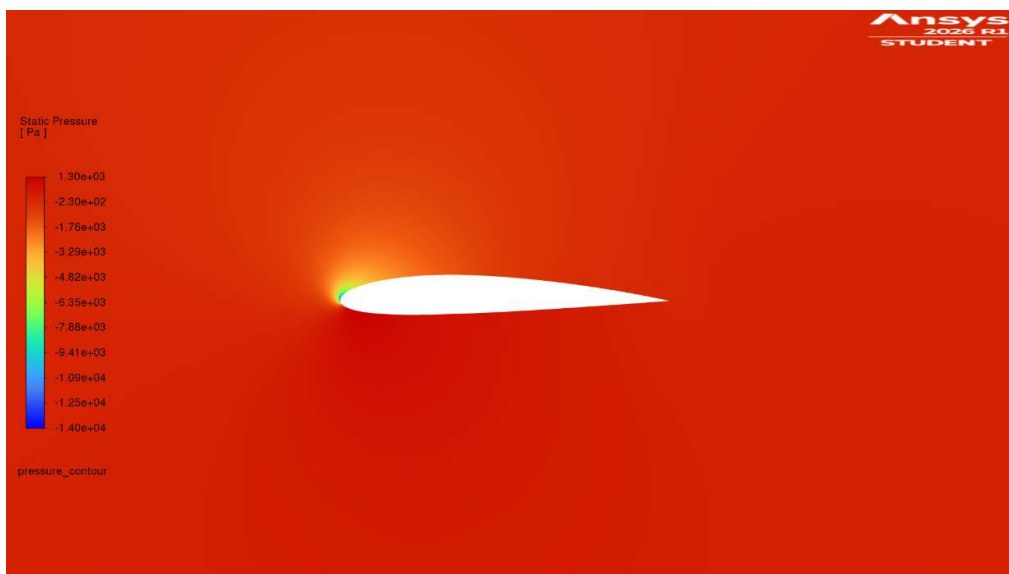


Figure B32: Fixed-range pressure contour at 17° angle of attack.

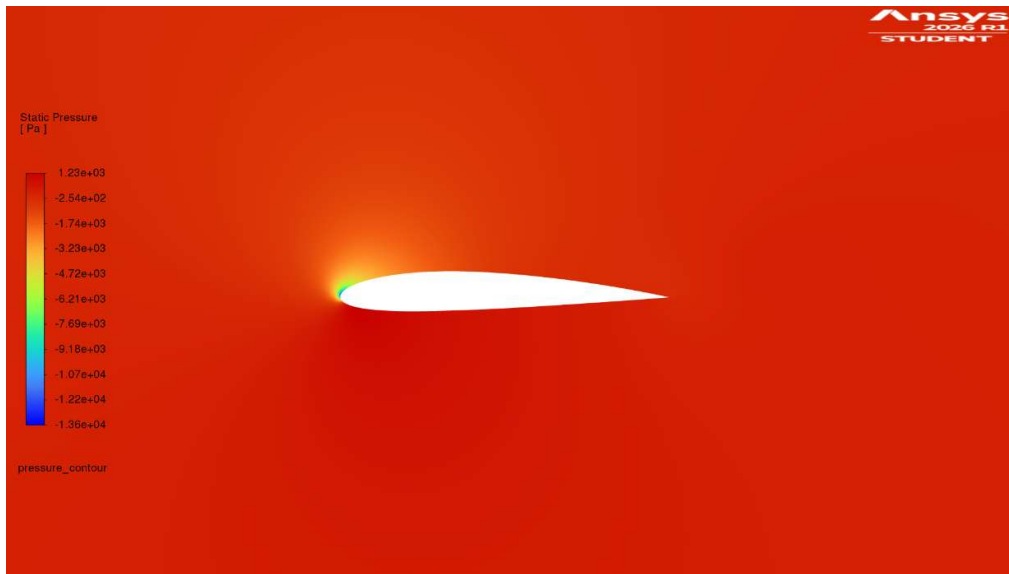


Figure B33: Auto-range pressure contour at 18° angle of attack.

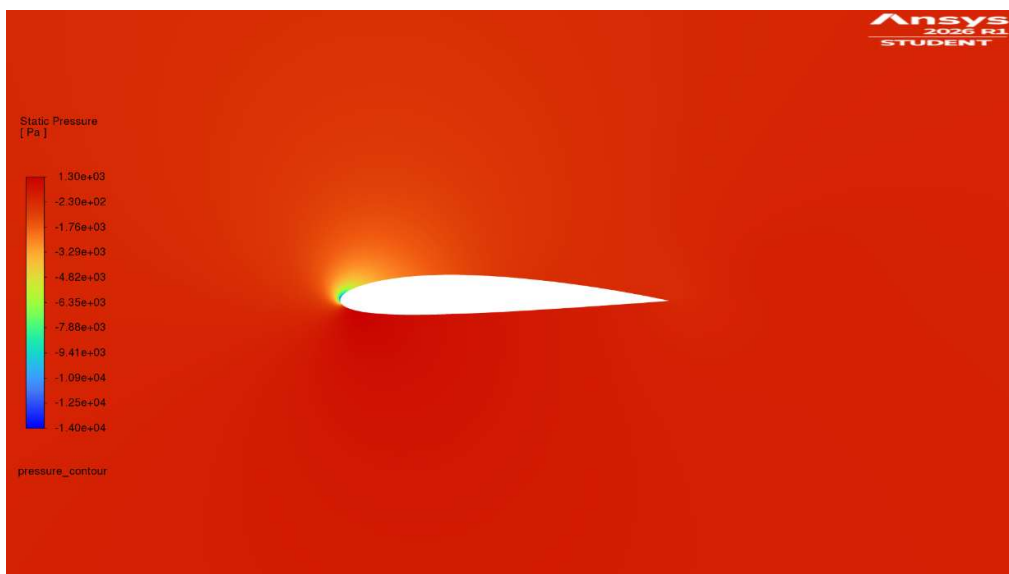


Figure B34: Fixed-range pressure contour at 18° angle of attack.

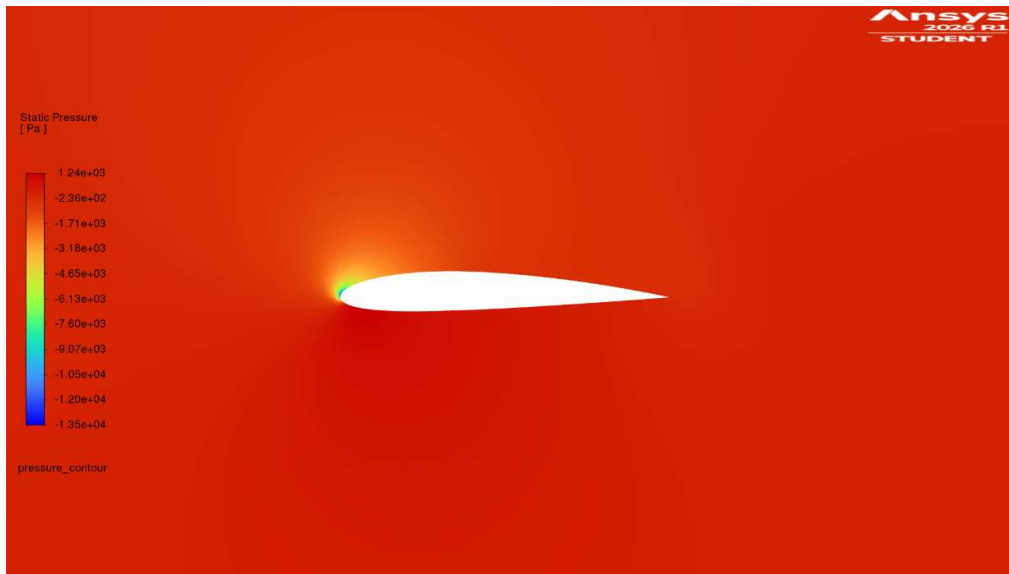


Figure B35: Auto-range pressure contour at 19° angle of attack.

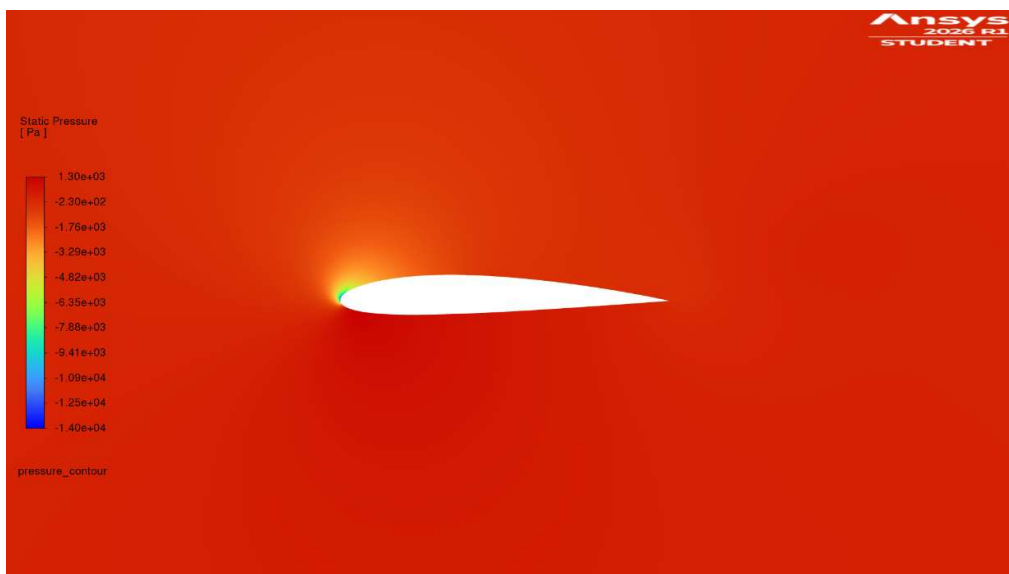


Figure B36: Fixed-range pressure contour at 19° angle of attack.

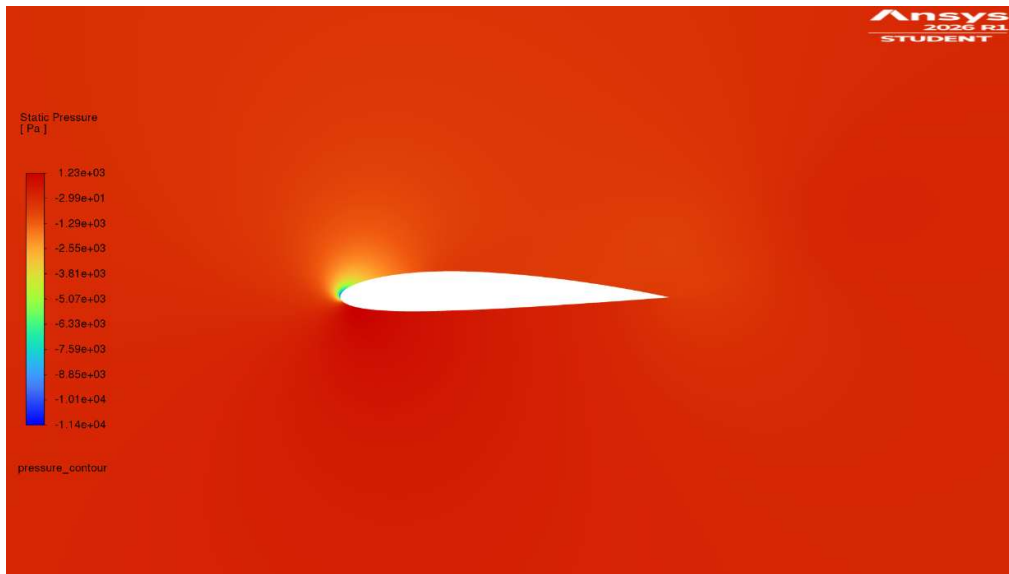


Figure B37: Auto-range pressure contour at 20° angle of attack.



Figure B38: Fixed-range pressure contour at 20° angle of attack.

Appendix C - Velocity Contours

Figures C1 through C57 present velocity magnitude contours for all simulated angles of attack. For each angle of attack, three contours are presented — an auto-range contour, a fixed-range contour using a consistent scale of 0 to 160 ms^{-1} , and a wide-shot contour showing the full downstream wake development.

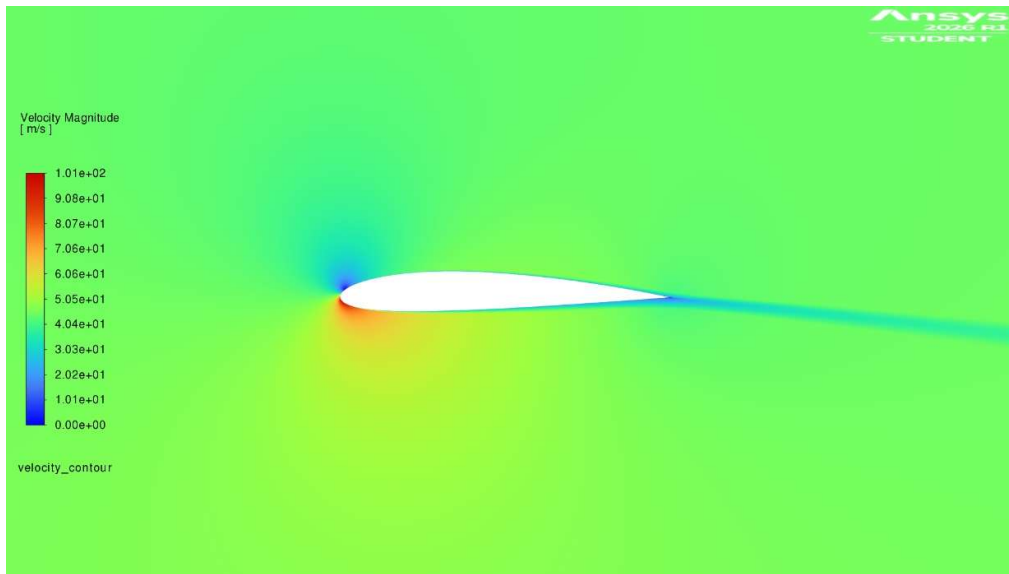


Figure C1: Auto-range velocity contour at -8° angle of attack.

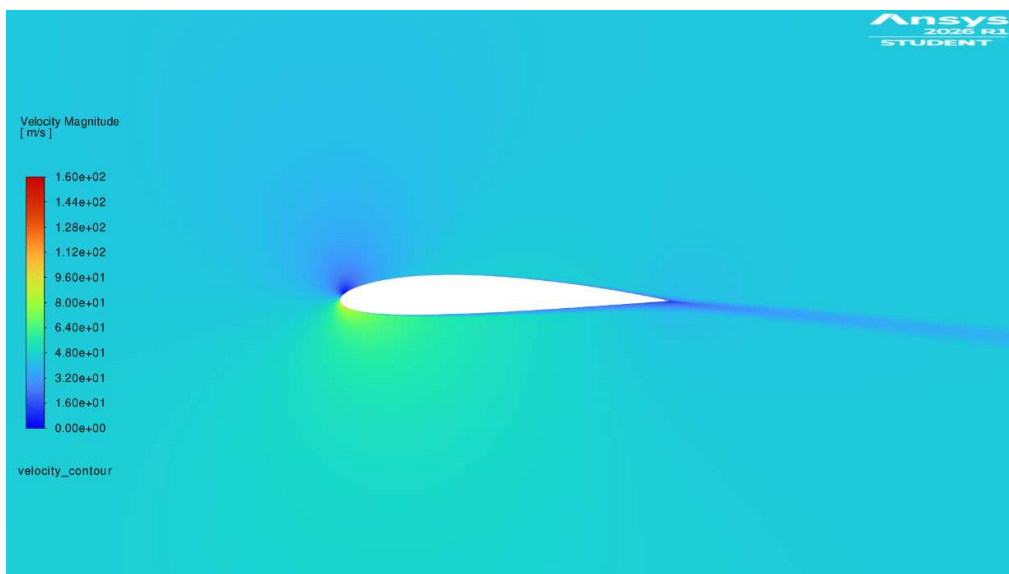


Figure C2: Fixed-range velocity contour at -8° angle of attack.



Figure C3: Wide-shot velocity contour at -8° angle of attack.

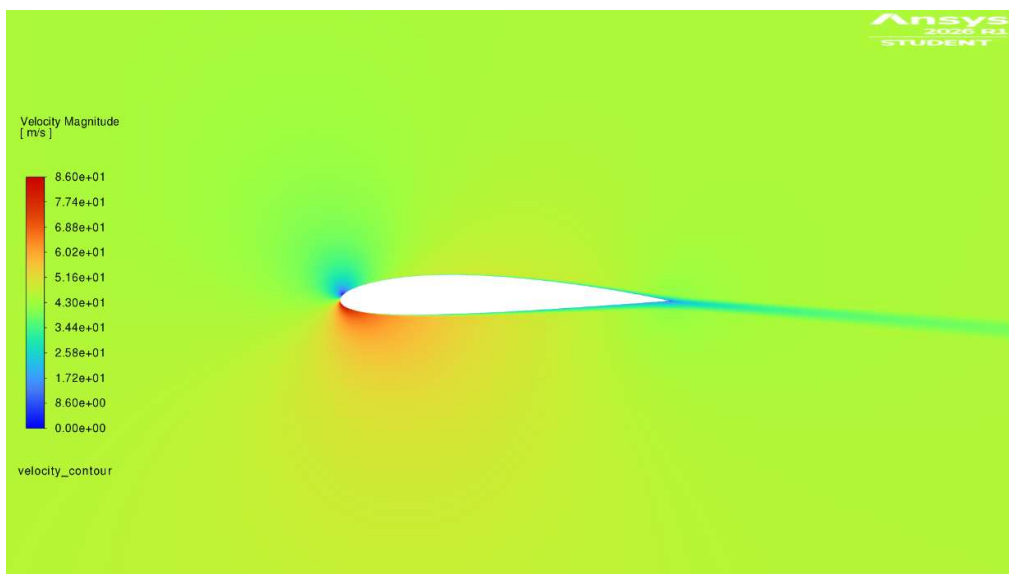


Figure C4: Auto-range velocity contour at -6° angle of attack.

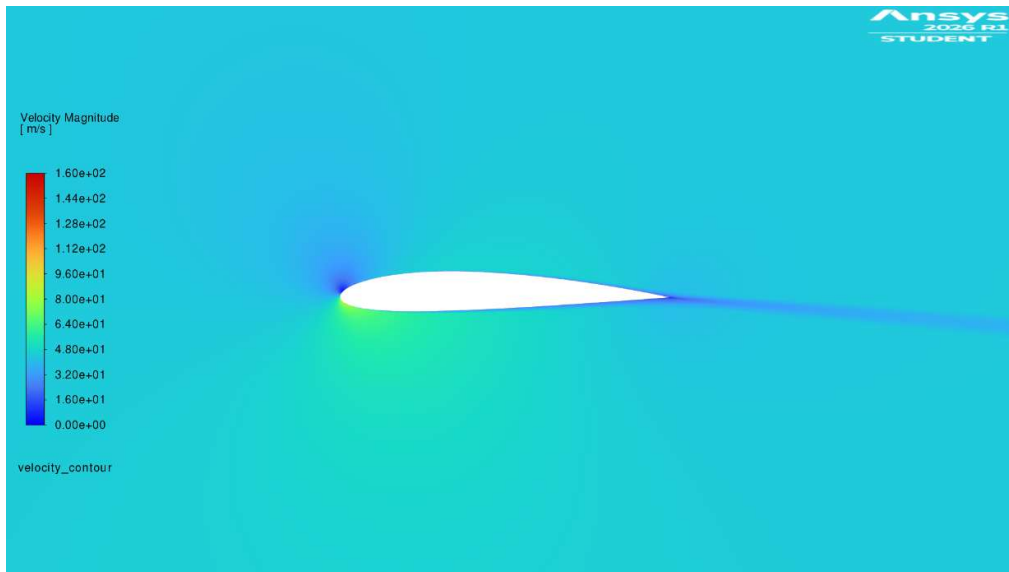


Figure C5: Fixed-range velocity contour at -6° angle of attack.

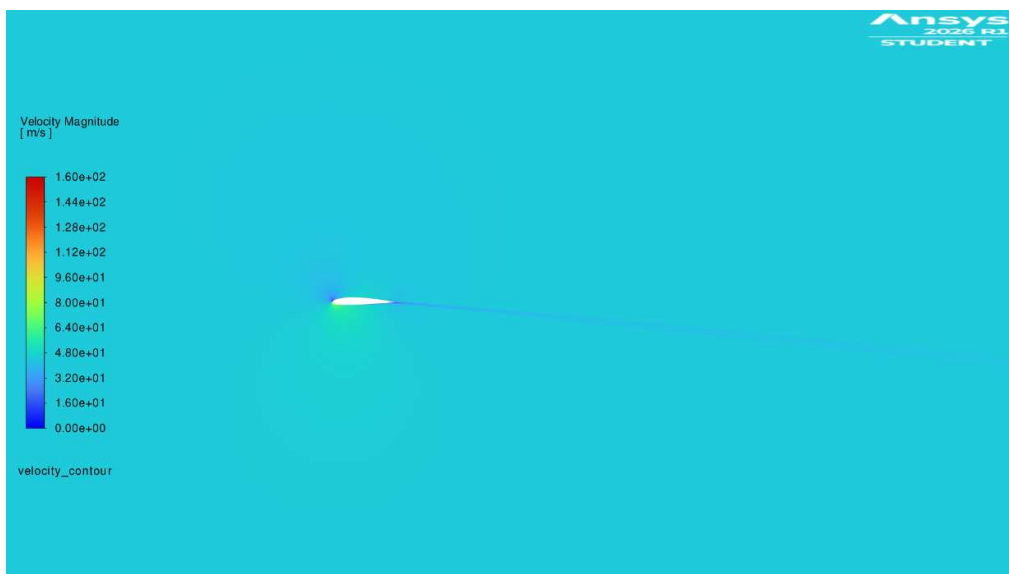


Figure C6: Wide-shot velocity contour at -6° angle of attack.

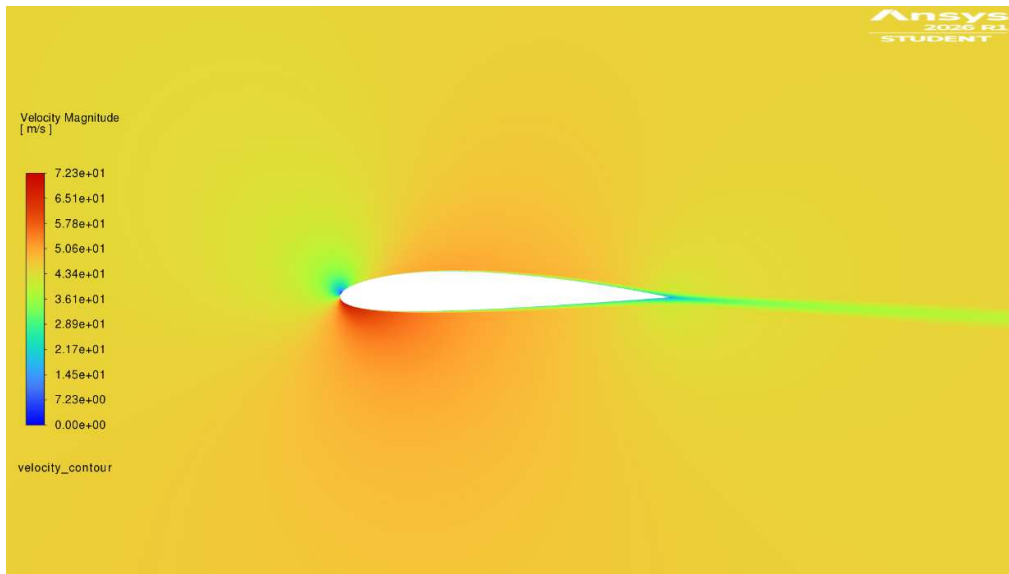


Figure C7: Auto-range velocity contour at -4° angle of attack.

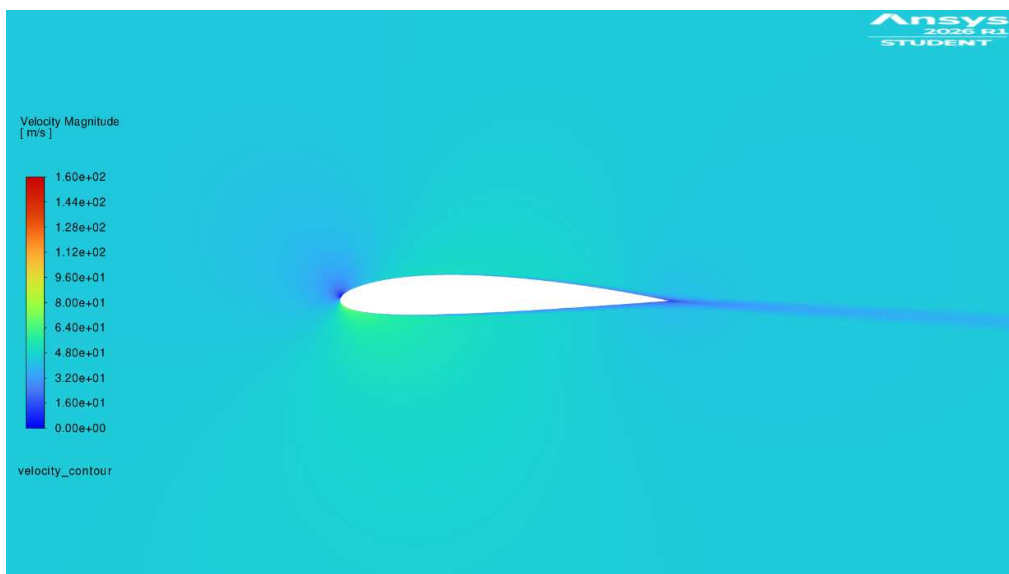


Figure C8: Fixed-range velocity contour at -4° angle of attack.



Figure C9: Wide-shot velocity contour at -4° angle of attack.

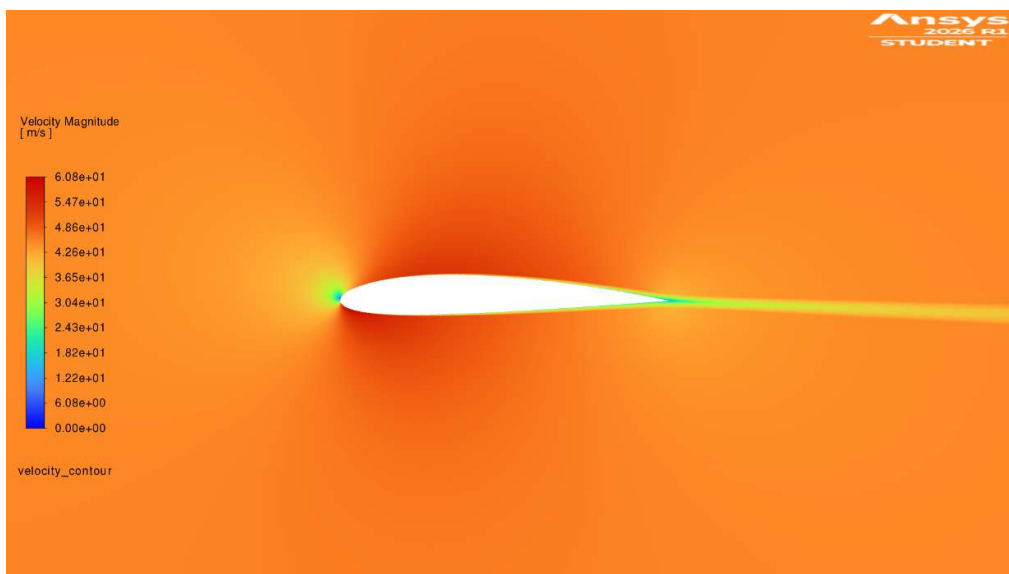


Figure C10: Auto-range velocity contour at -2° angle of attack.



Figure C11: Fixed-range velocity contour at -2° angle of attack.

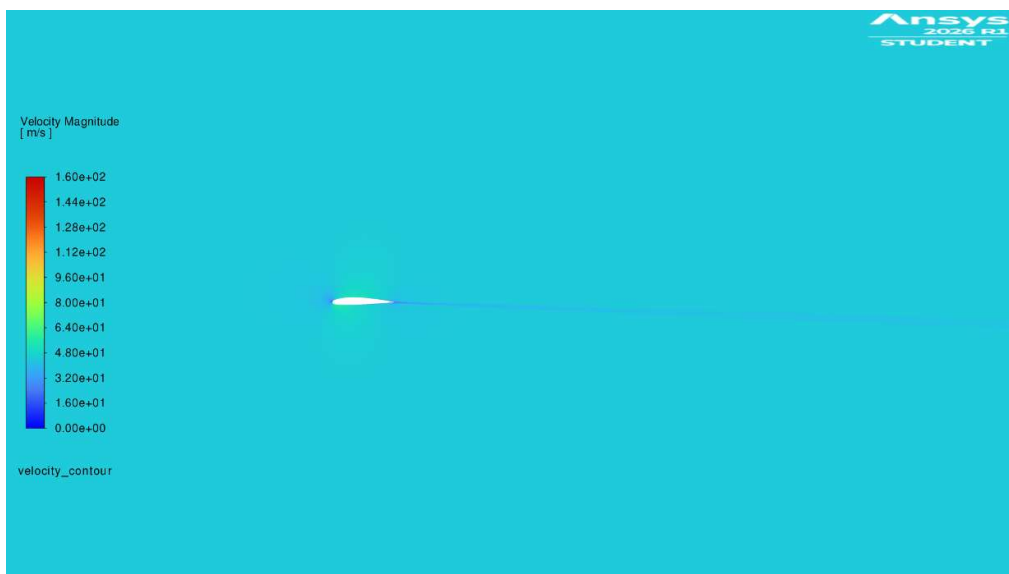


Figure C12: Wide-shot velocity contour at -2° angle of attack.

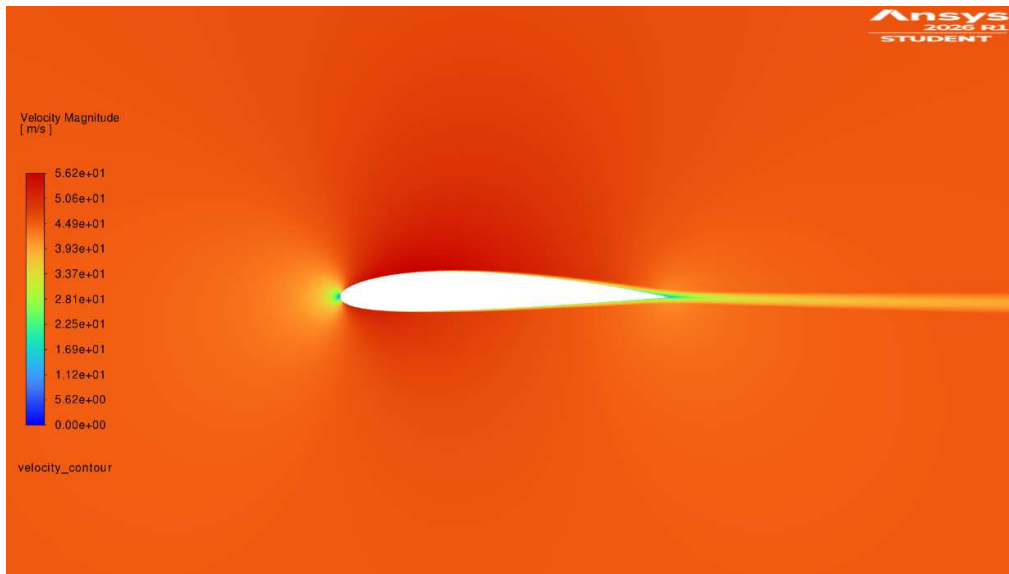


Figure C13: Auto-range velocity contour at 0° angle of attack.

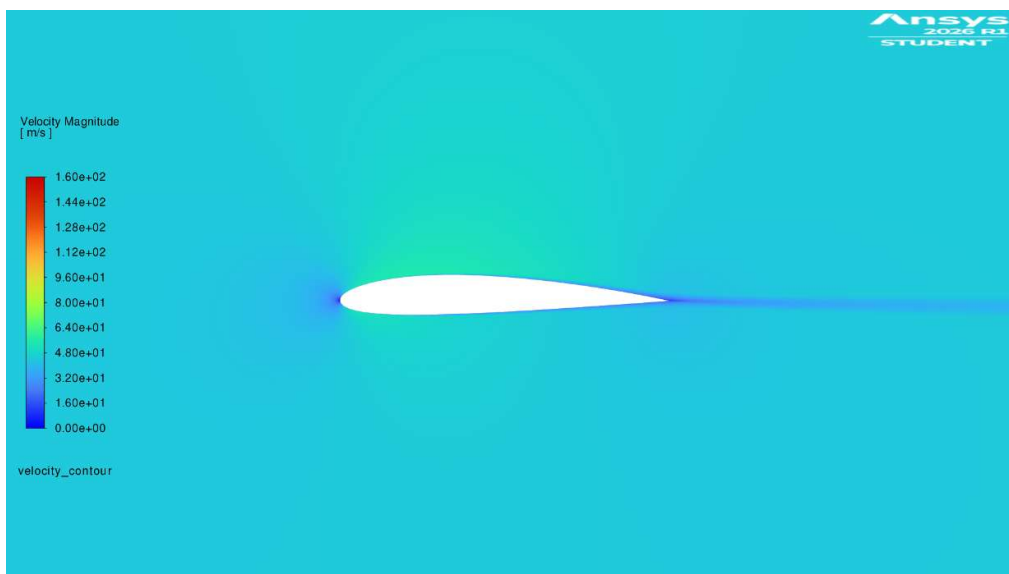


Figure C14: Fixed-range velocity contour at 0° angle of attack.



Figure C15: Wide-shot velocity contour at 0° angle of attack.

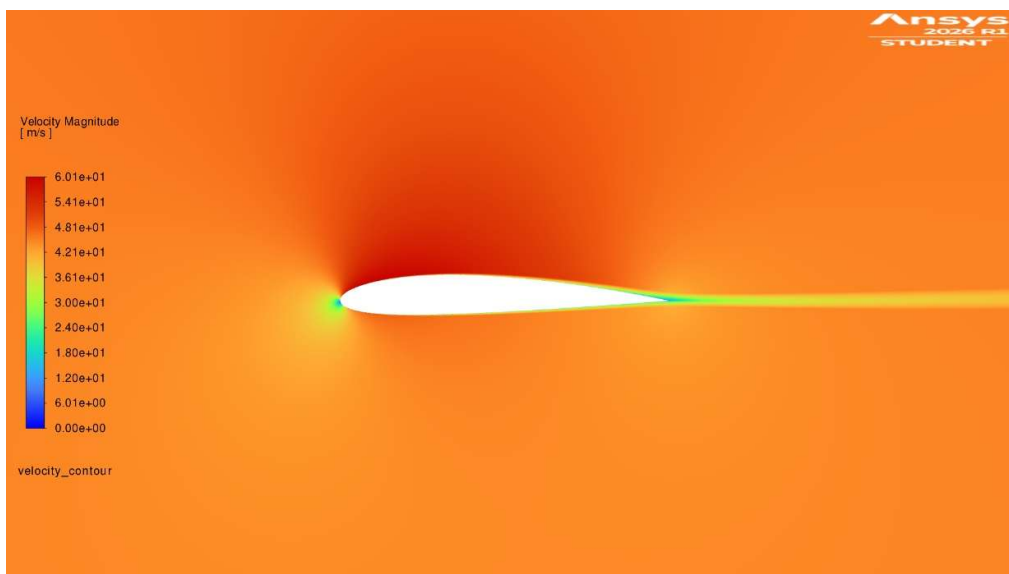


Figure C16: Auto-range velocity contour at 2° angle of attack.

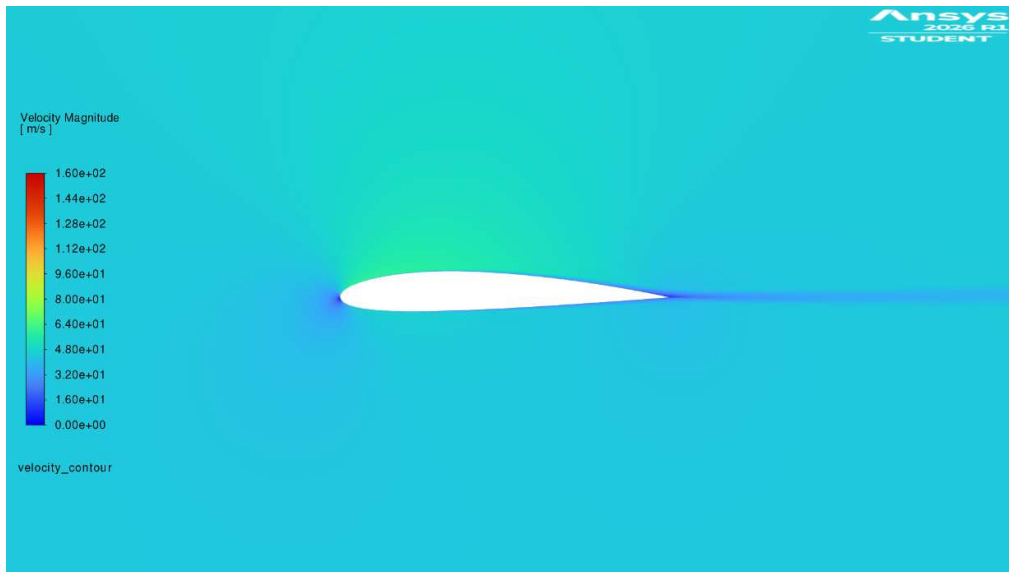


Figure C17: Fixed-range velocity contour at 2° angle of attack.

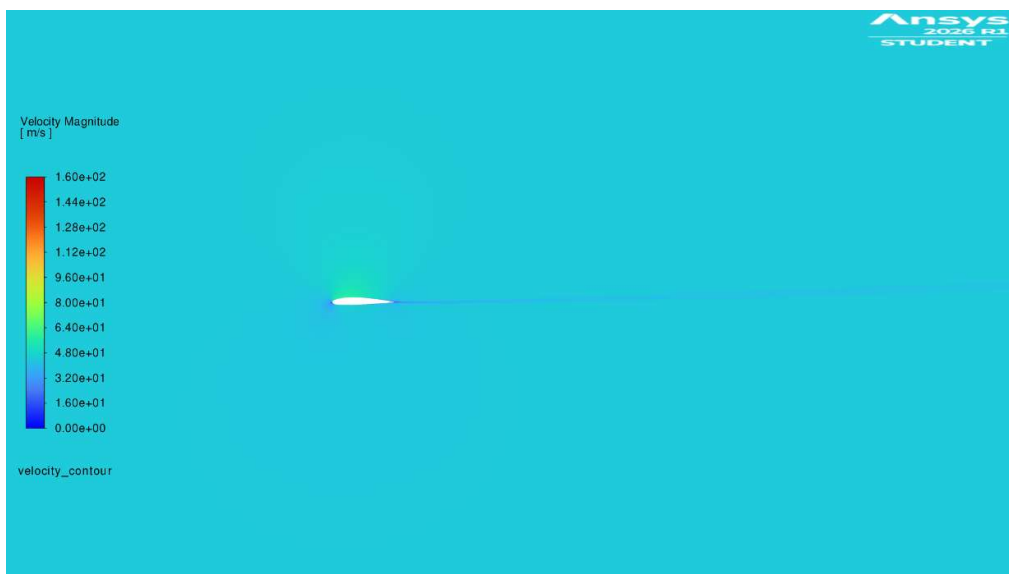


Figure C18: Wide-shot velocity contour at 2° angle of attack.

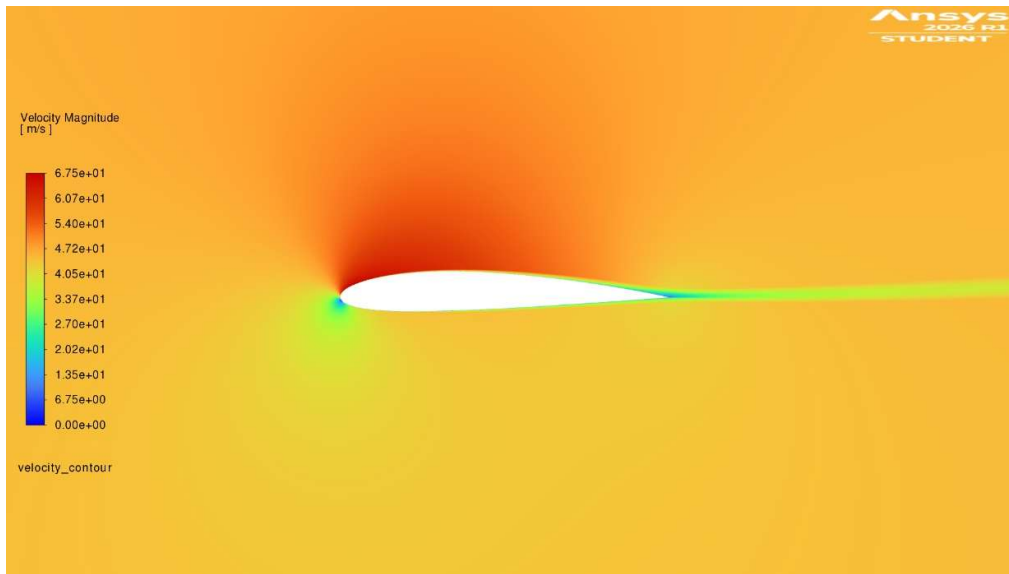


Figure C19: Auto-range velocity contour at 4° angle of attack.

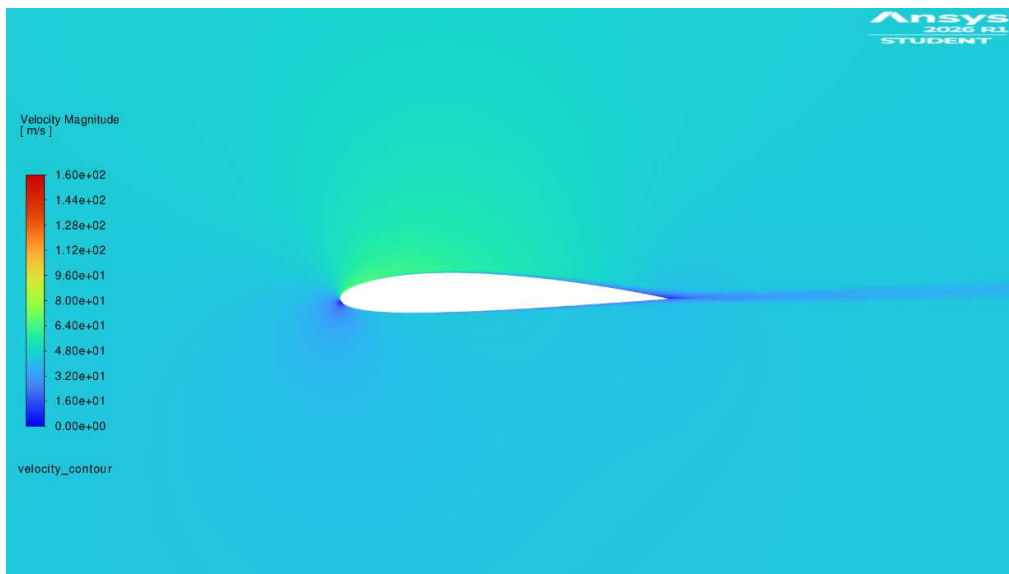


Figure C20: Fixed-range velocity contour at 4° angle of attack.



Figure C21: Wide-shot velocity contour at 4° angle of attack.

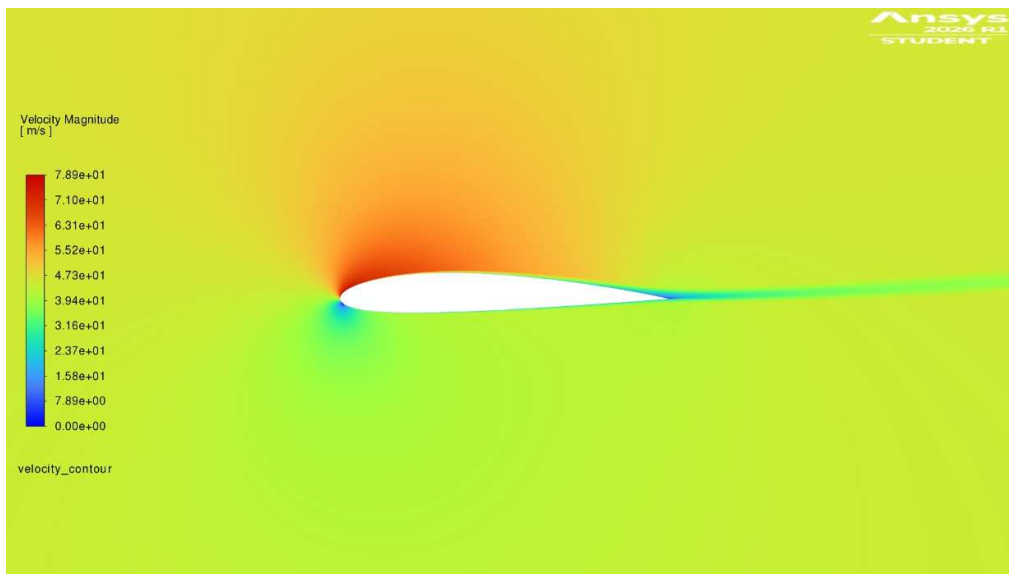


Figure C22: Auto-range velocity contour at 6° angle of attack.

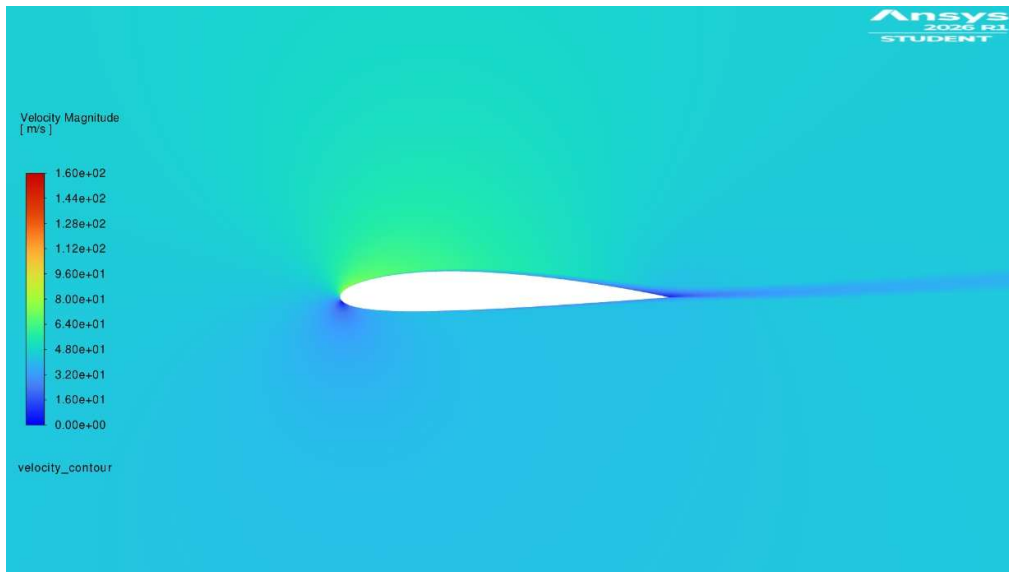


Figure C23: Fixed-range velocity contour at 6° angle of attack.

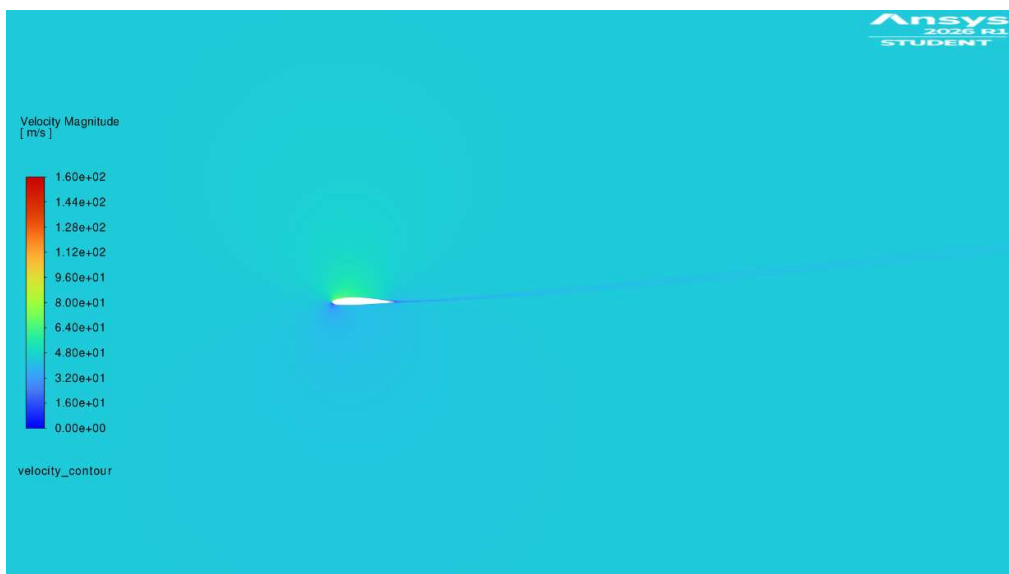


Figure C24: Wide-shot velocity contour at 6° angle of attack.

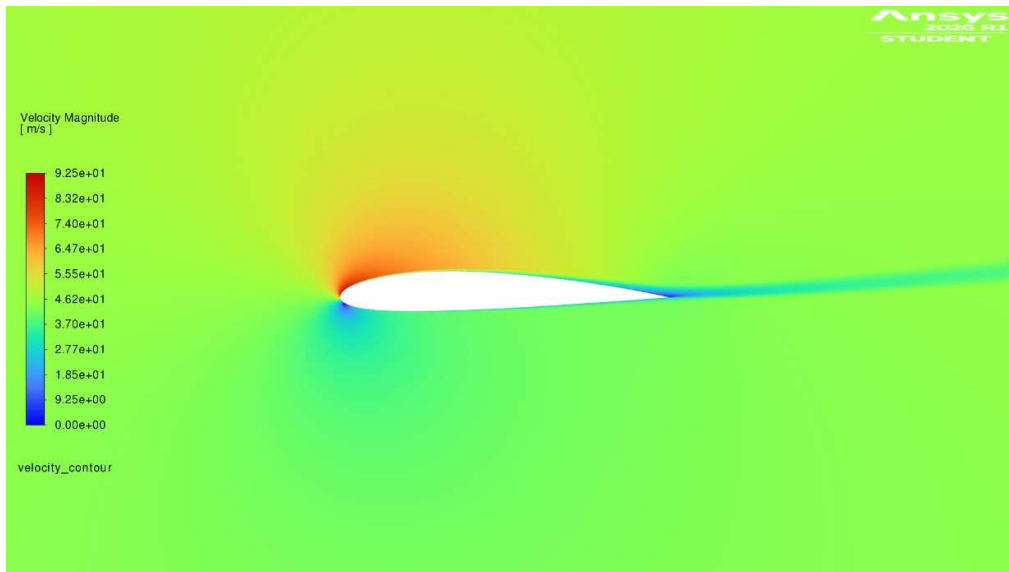


Figure C25: Auto-range velocity contour at 8° angle of attack.

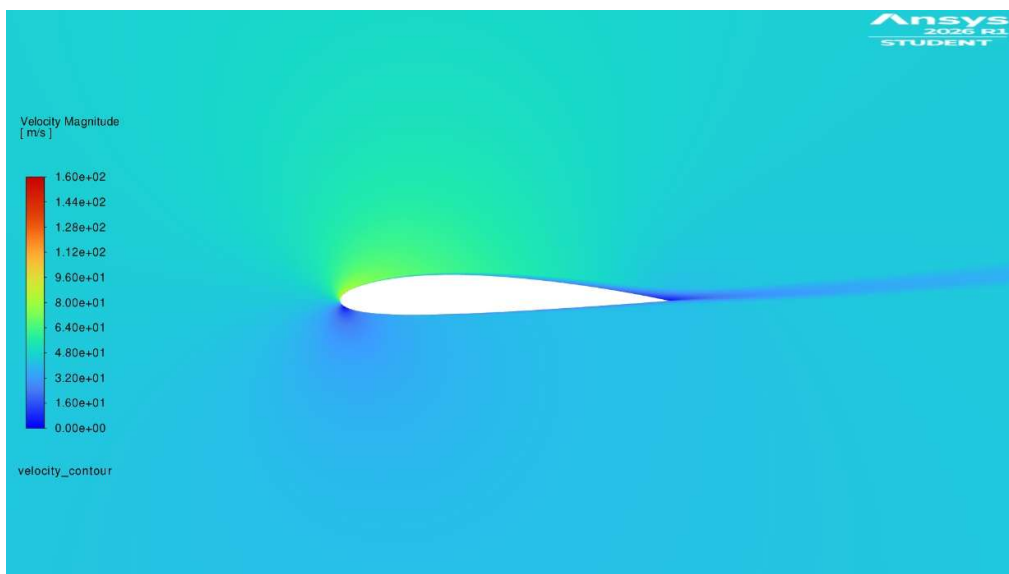


Figure C26: Fixed-range velocity contour at 8° angle of attack.

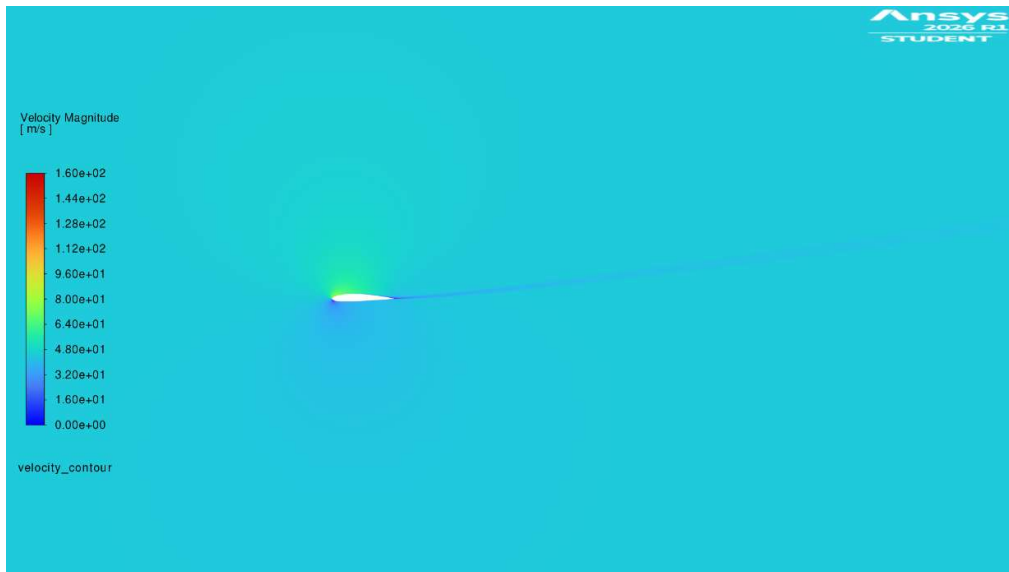


Figure C27: Wide-shot velocity contour at 8° angle of attack.

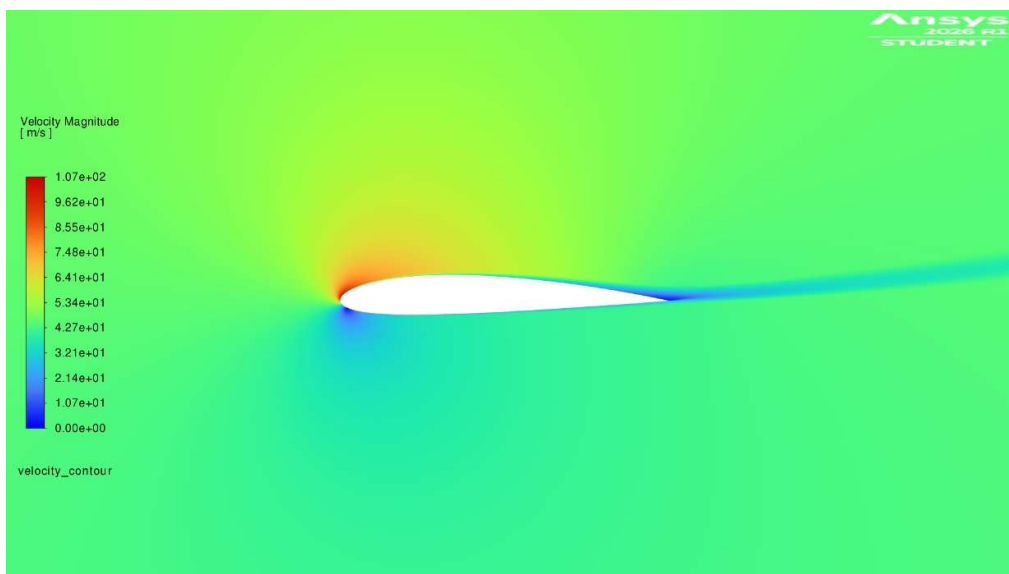


Figure C28: Auto-range velocity contour at 10° angle of attack.

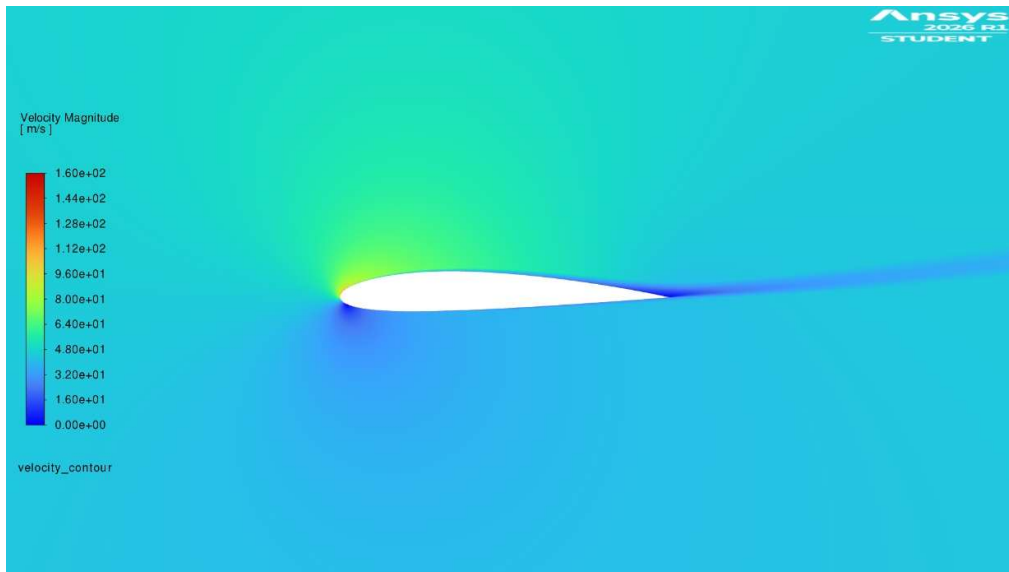


Figure C29: Fixed-range velocity contour at 10° angle of attack.



Figure C30: Wide-shot velocity contour at 10° angle of attack.

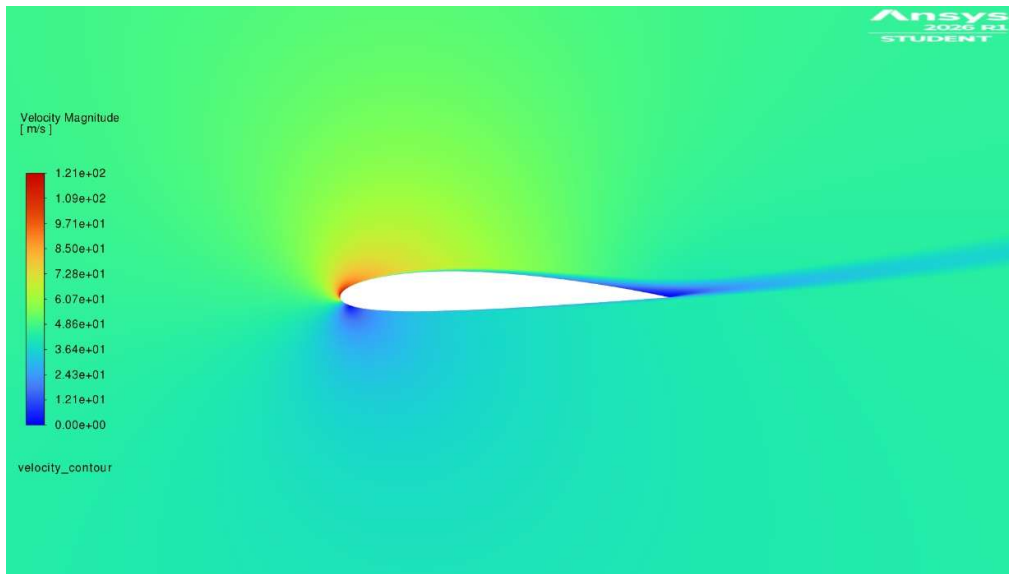


Figure C31: Auto-range velocity contour at 12° angle of attack.

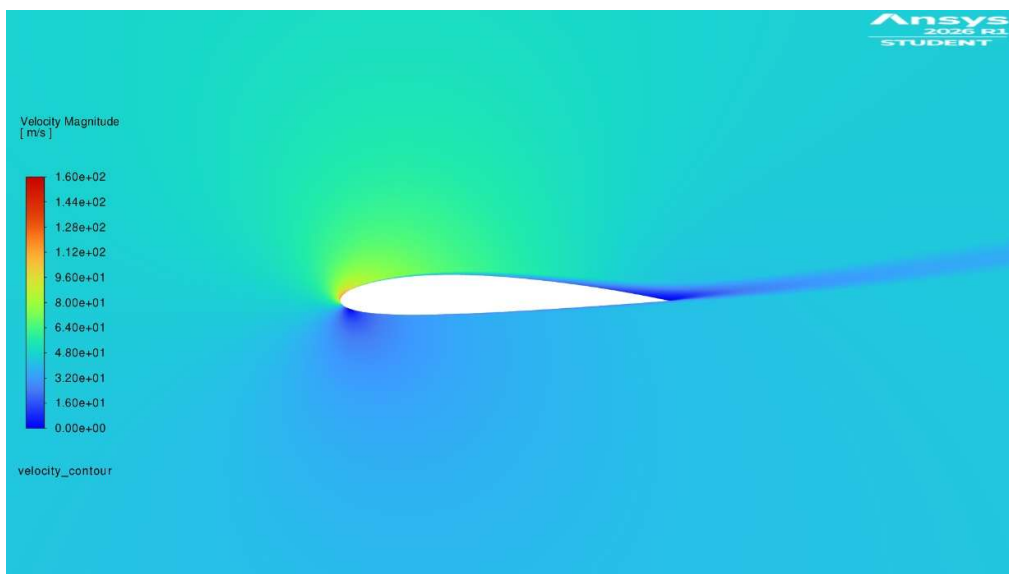


Figure C32: Fixed-range velocity contour at 12° angle of attack.



Figure C33: Wide-shot velocity contour at 12° angle of attack.

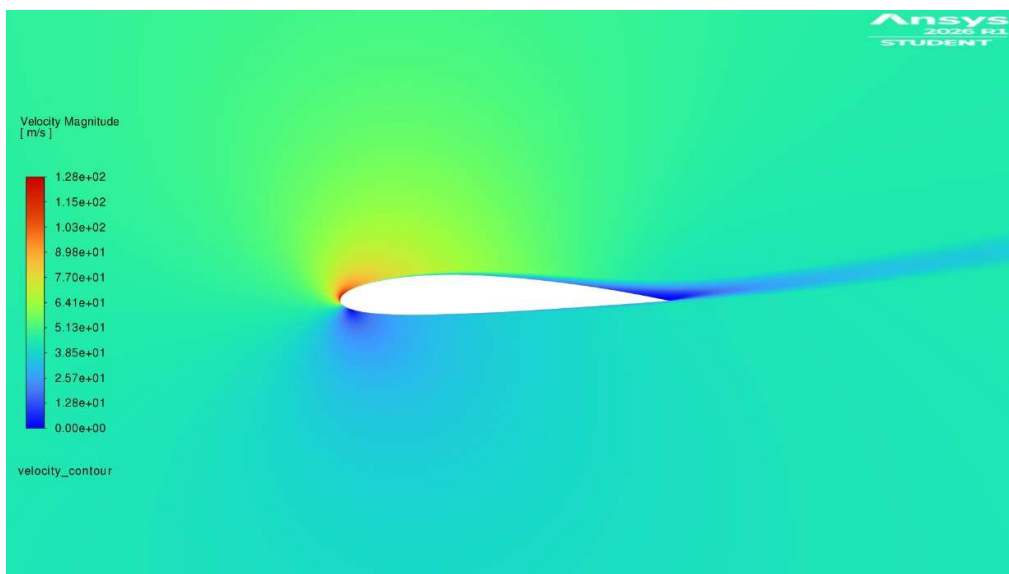


Figure C34: Auto-range velocity contour at 13° angle of attack.

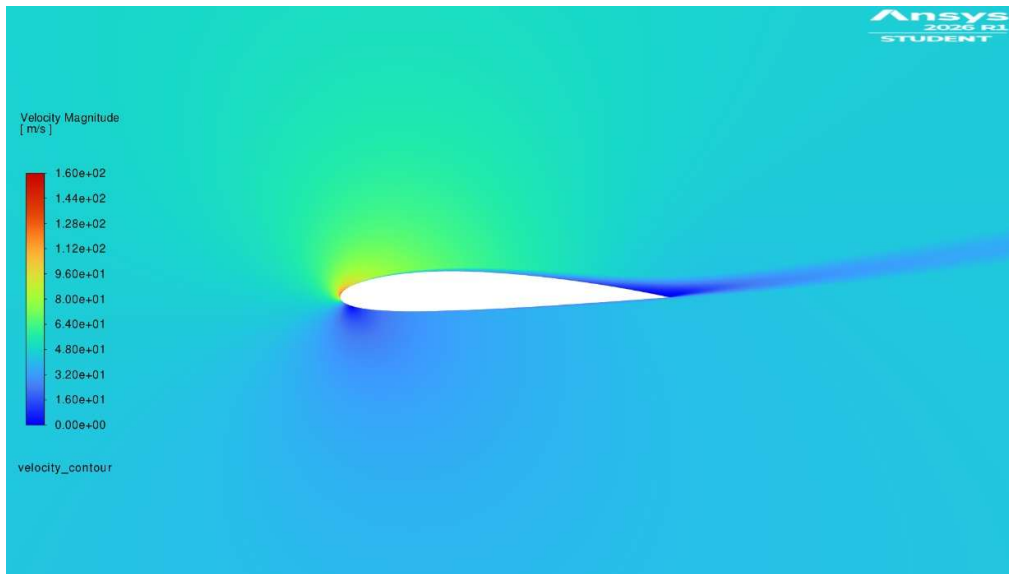


Figure C35: Fixed-range velocity contour at 13° angle of attack.

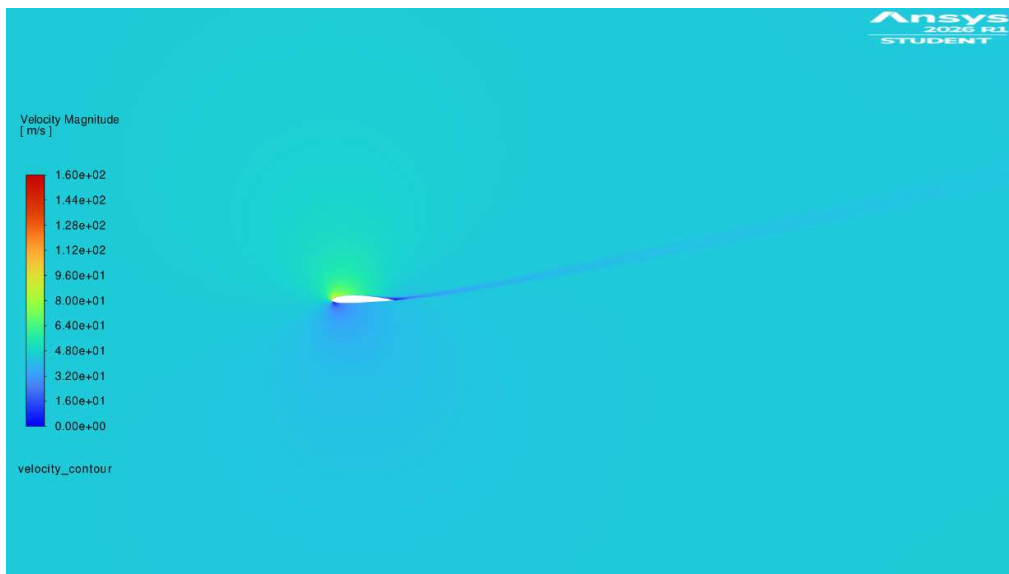


Figure C36: Wide-shot velocity contour at 13° angle of attack.

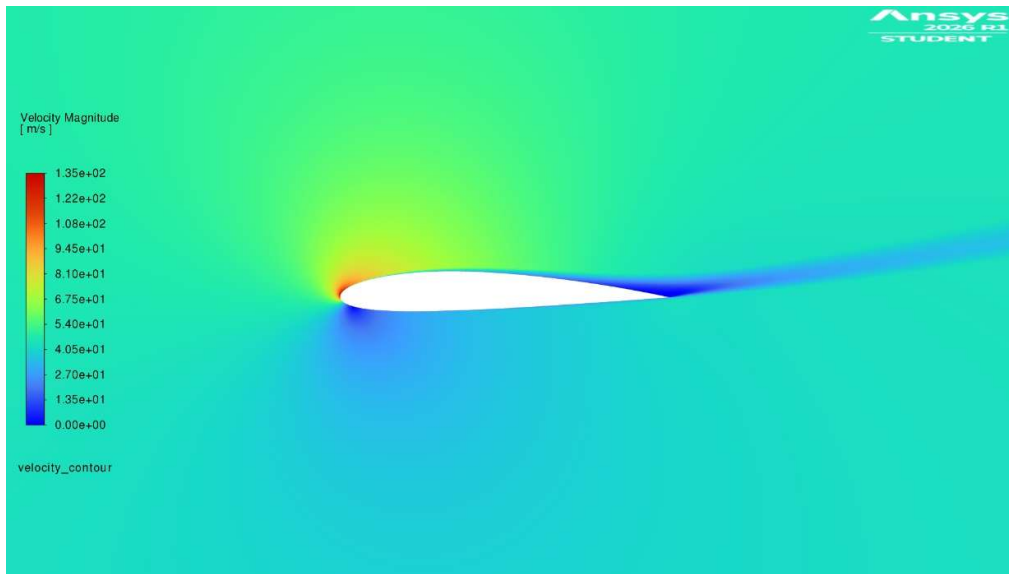


Figure C37: Auto-range velocity contour at 14° angle of attack.

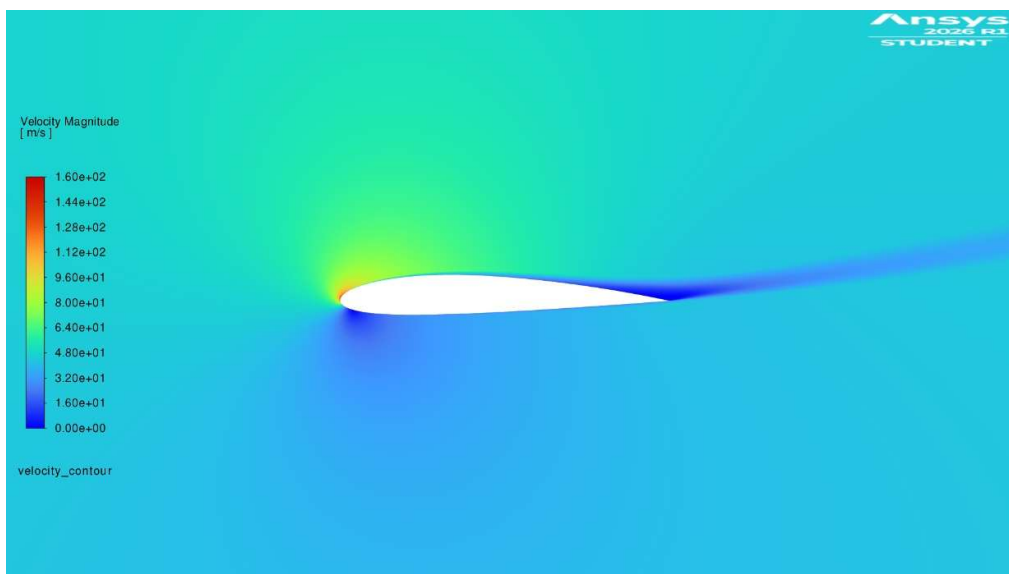


Figure C38: Fixed-range velocity contour at 14° angle of attack.

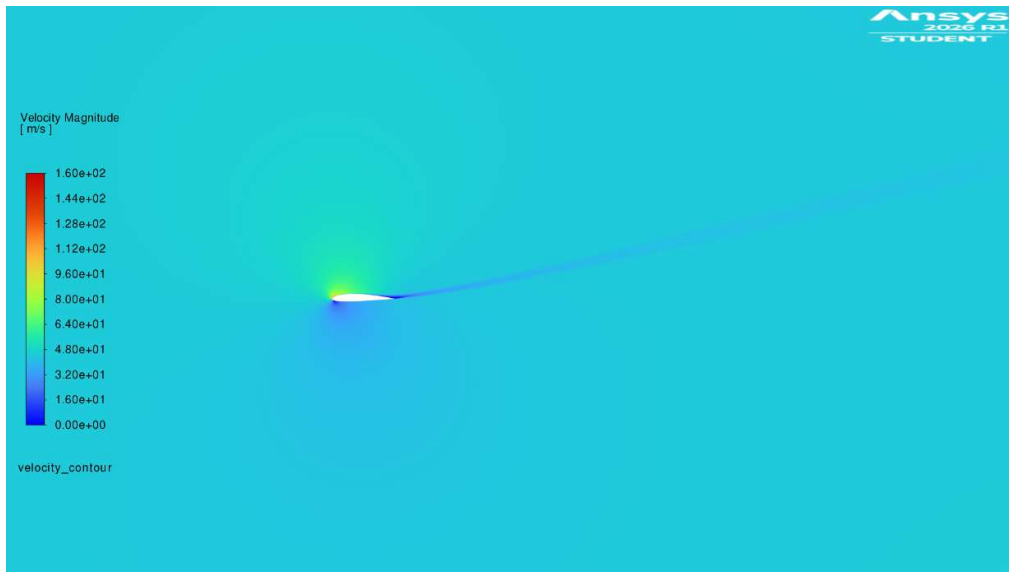


Figure C39: Wide-shot velocity contour at 14° angle of attack.

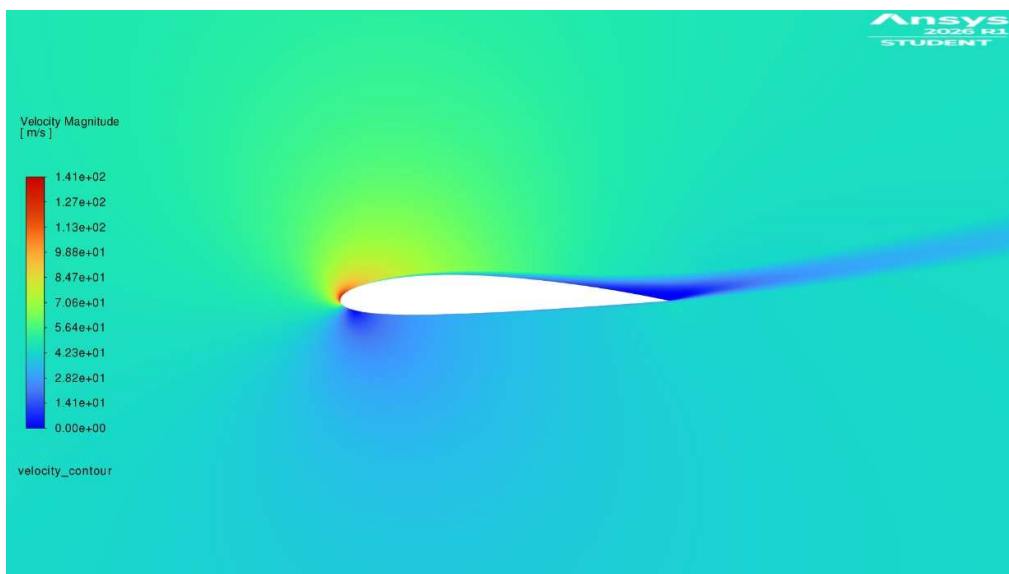


Figure C40: Auto-range velocity contour at 15° angle of attack.

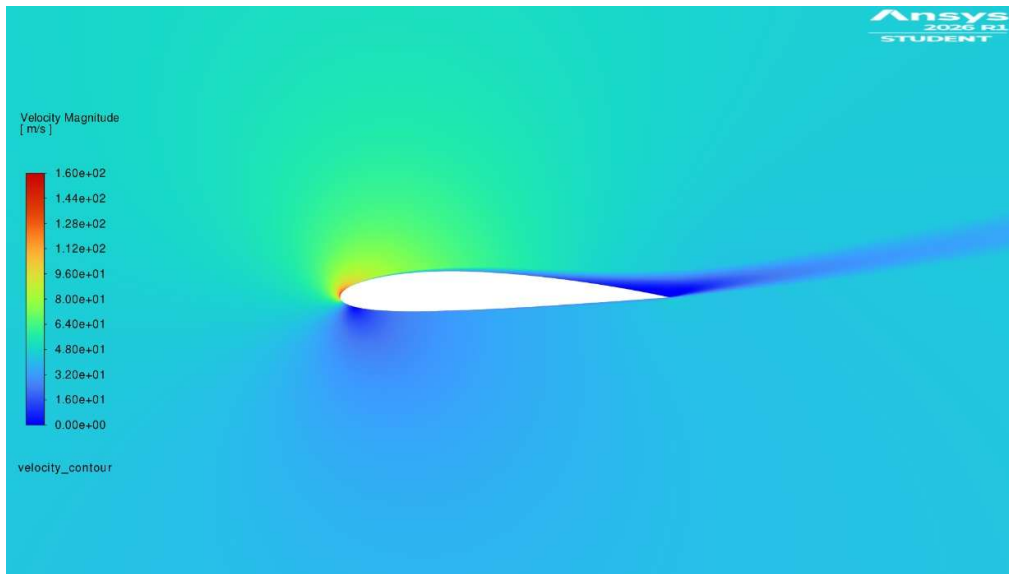


Figure C41: Fixed-range velocity contour at 15° angle of attack.

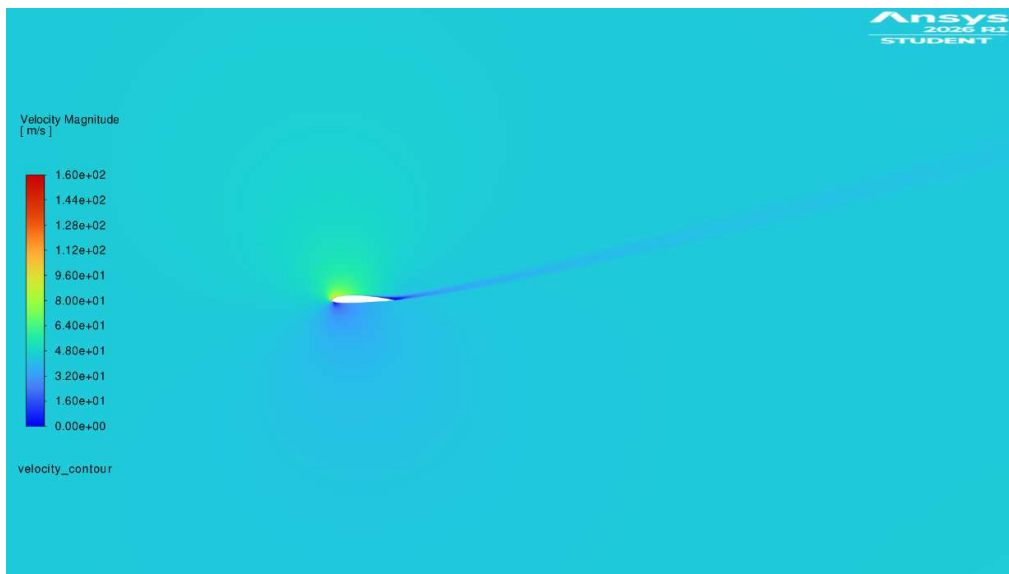


Figure C42: Wide-shot velocity contour at 15° angle of attack.

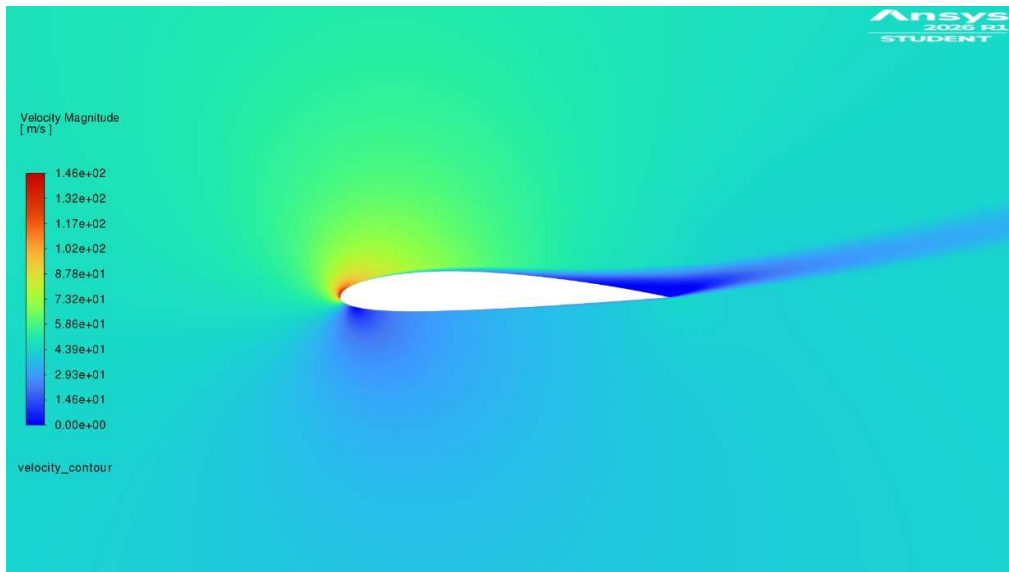


Figure C43: Auto-range velocity contour at 16° angle of attack.

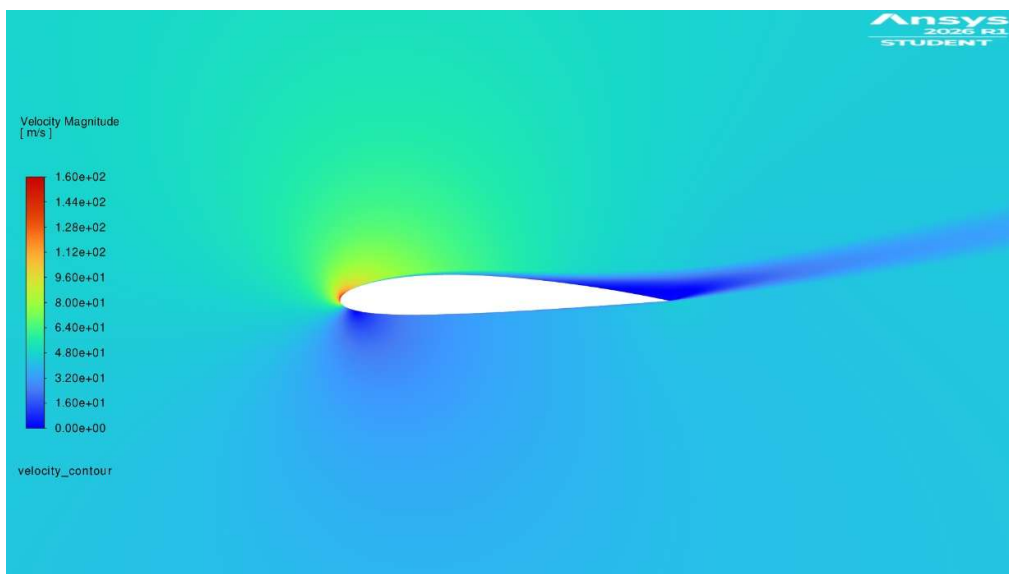


Figure C44: Fixed-range velocity contour at 16° angle of attack.

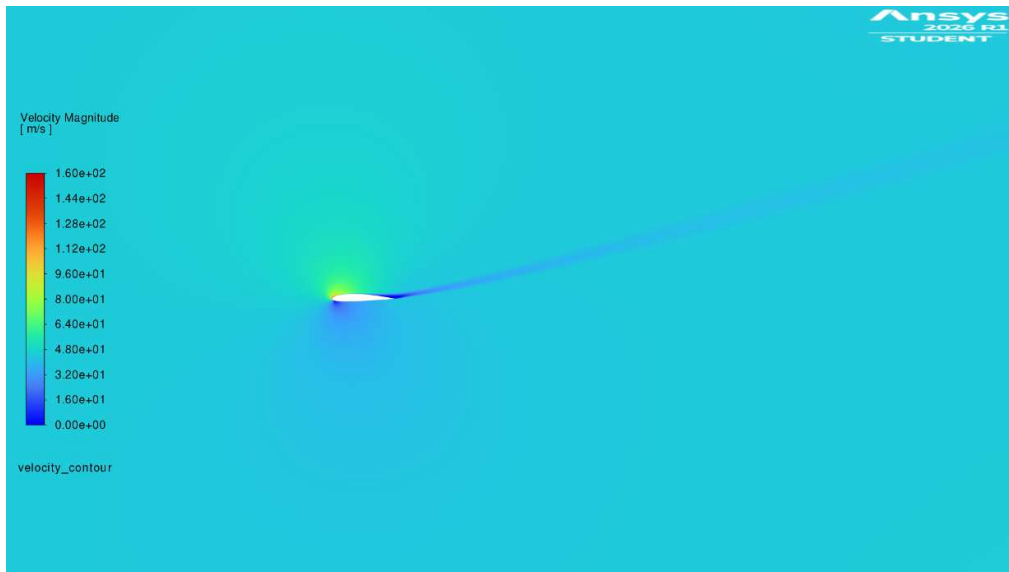


Figure C45: Wide-shot velocity contour at 16° angle of attack.

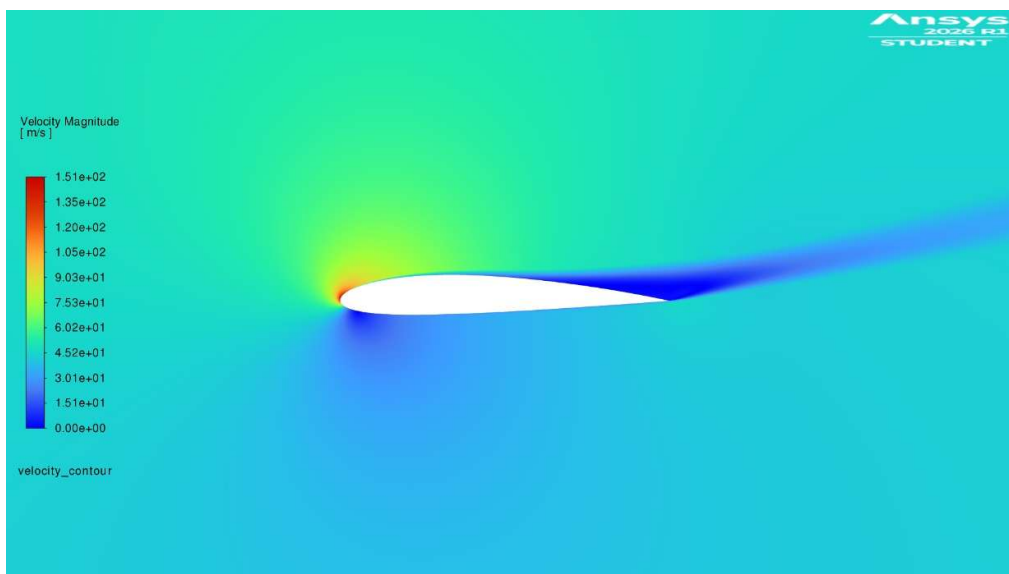


Figure C46: Auto-range velocity contour at 17° angle of attack.

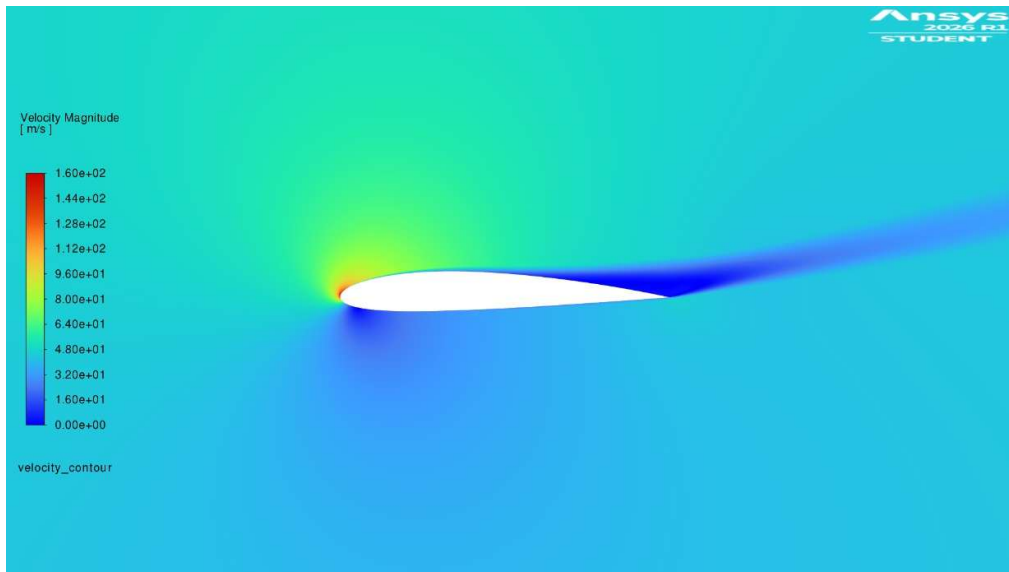


Figure C47: Fixed-range velocity contour at 17° angle of attack.

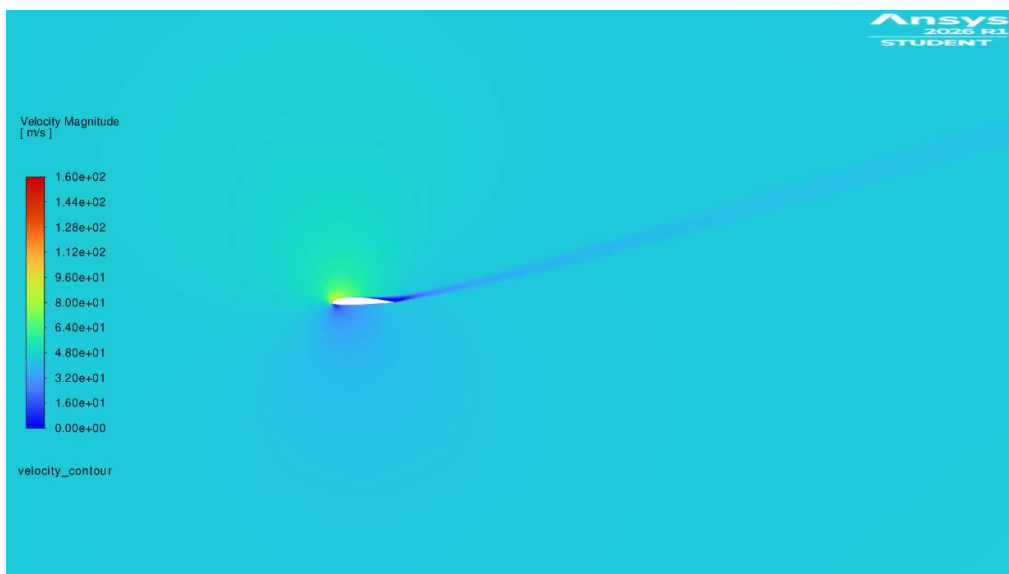


Figure C48: Wide-shot velocity contour at 17° angle of attack.

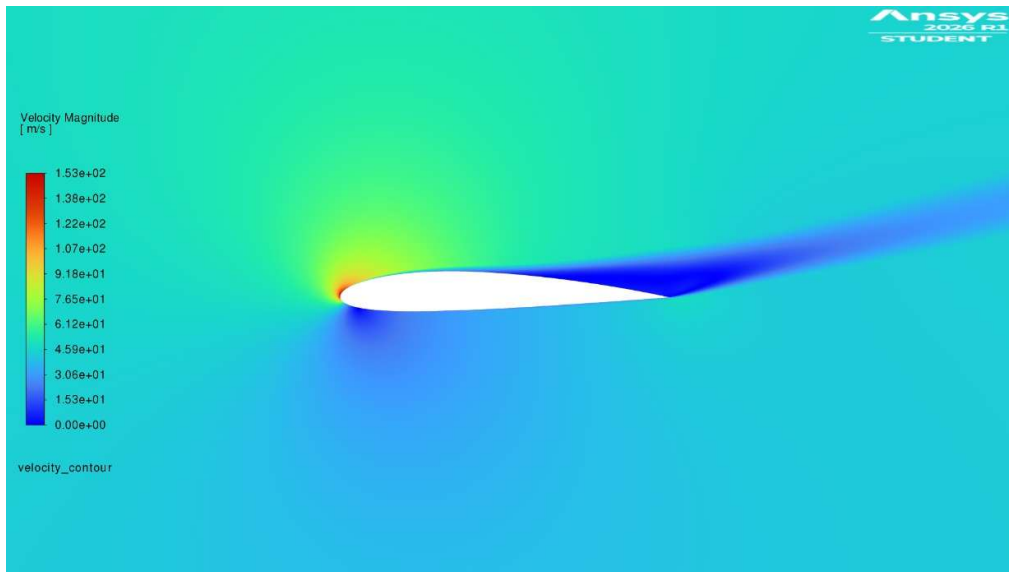


Figure C49: Auto-range velocity contour at 18° angle of attack.

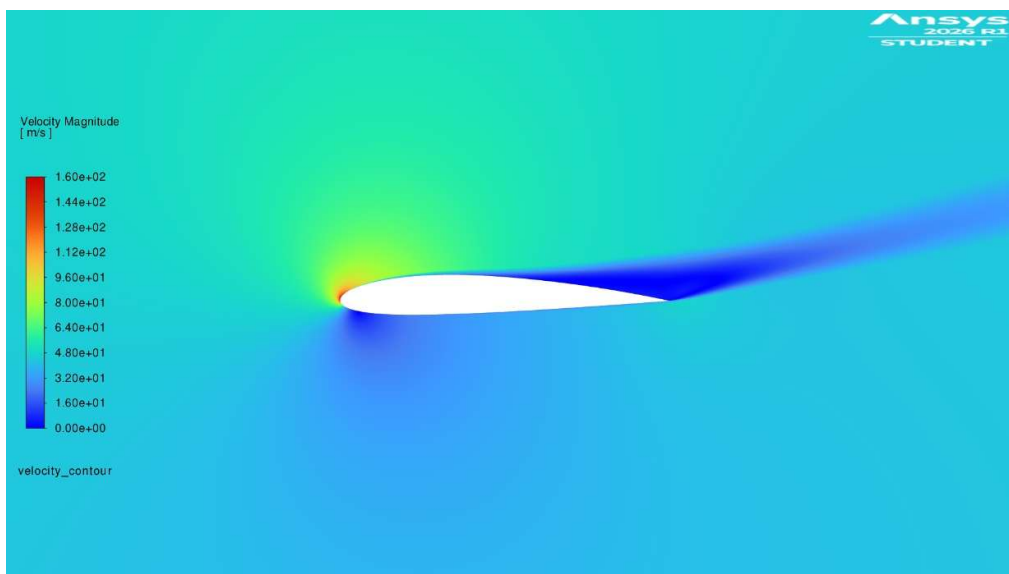


Figure C50: Fixed-range velocity contour at 18° angle of attack.

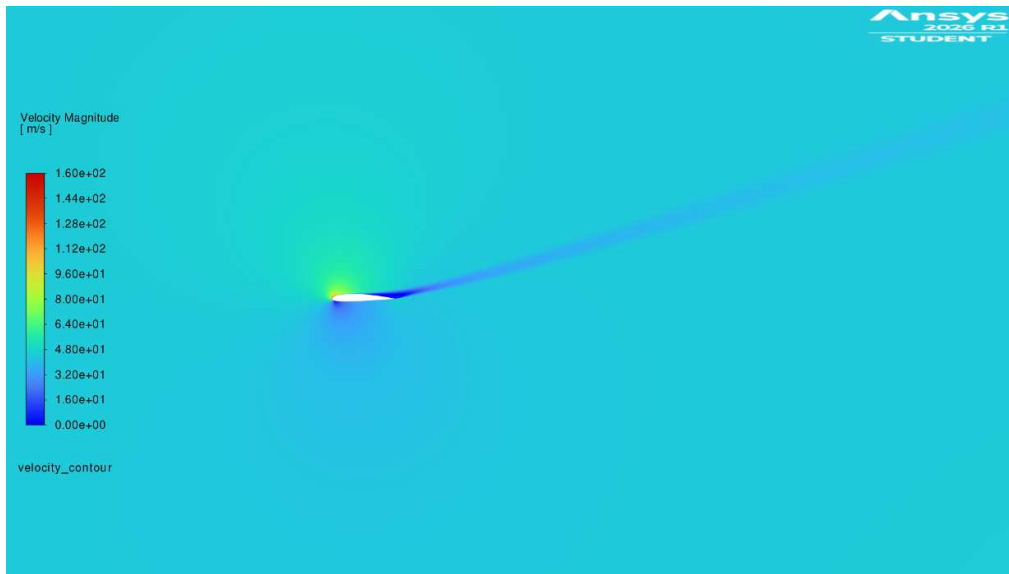


Figure C51: Wide-shot velocity contour at 18° angle of attack.

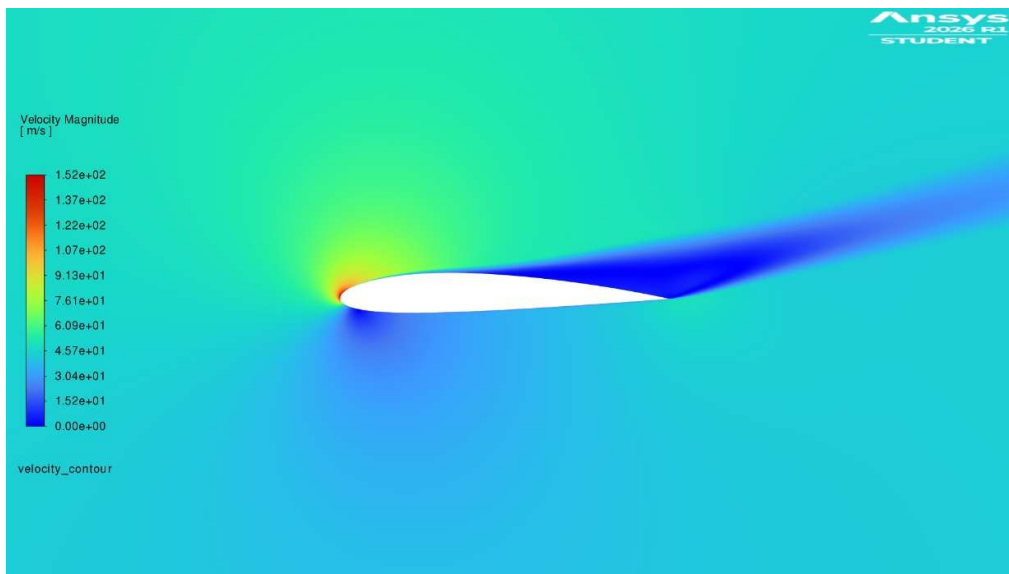


Figure C52: Auto-range velocity contour at 19° angle of attack.

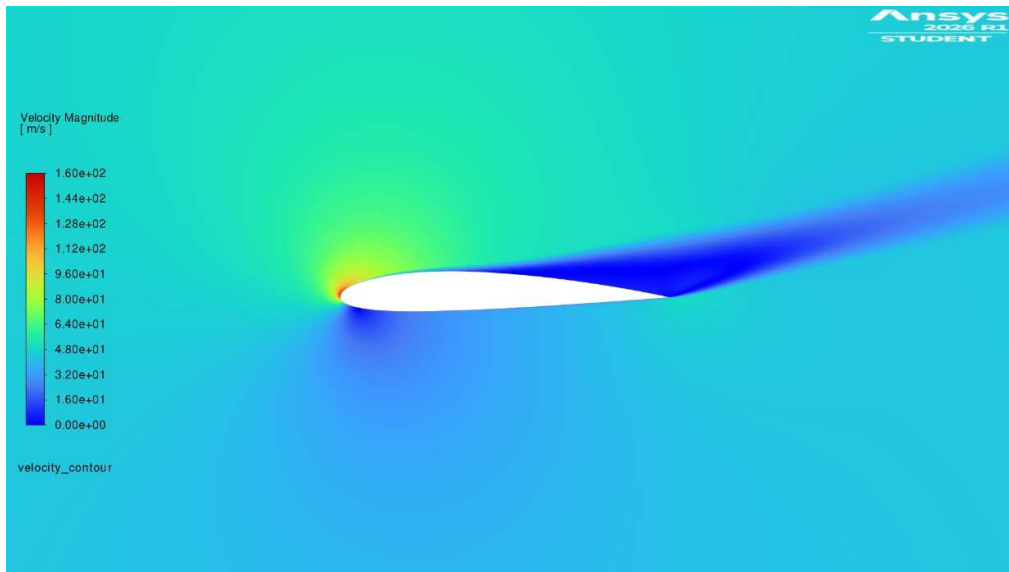


Figure C53: Fixed-range velocity contour at 19° angle of attack.

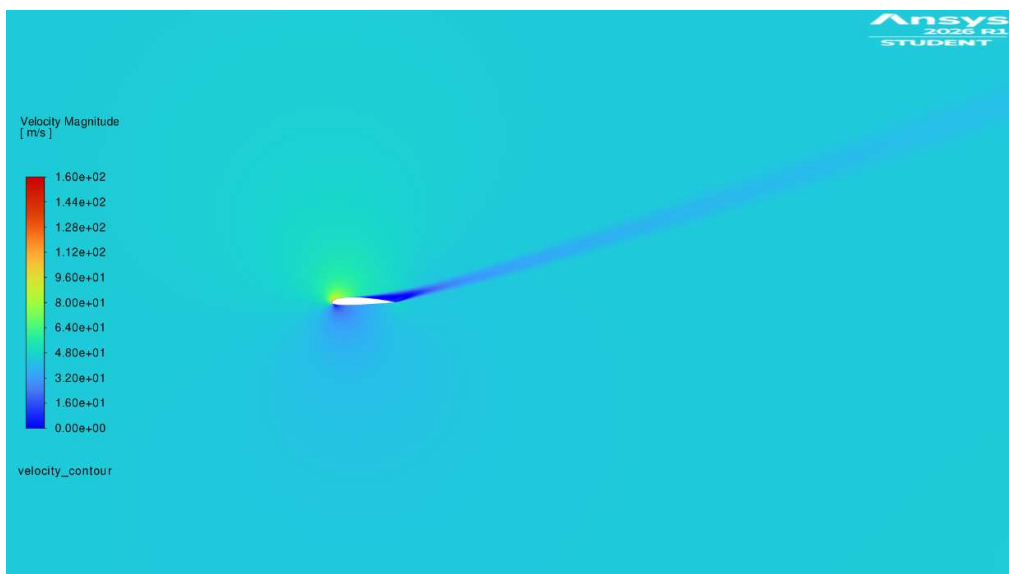


Figure C54: Wide-shot velocity contour at 19° angle of attack.

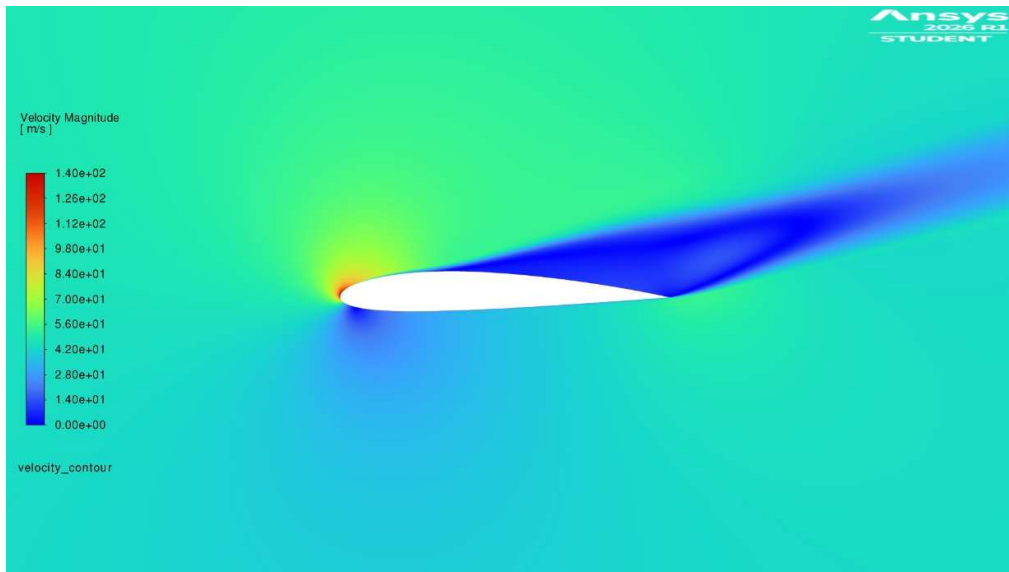


Figure C55: Auto-range velocity contour at 20° angle of attack.

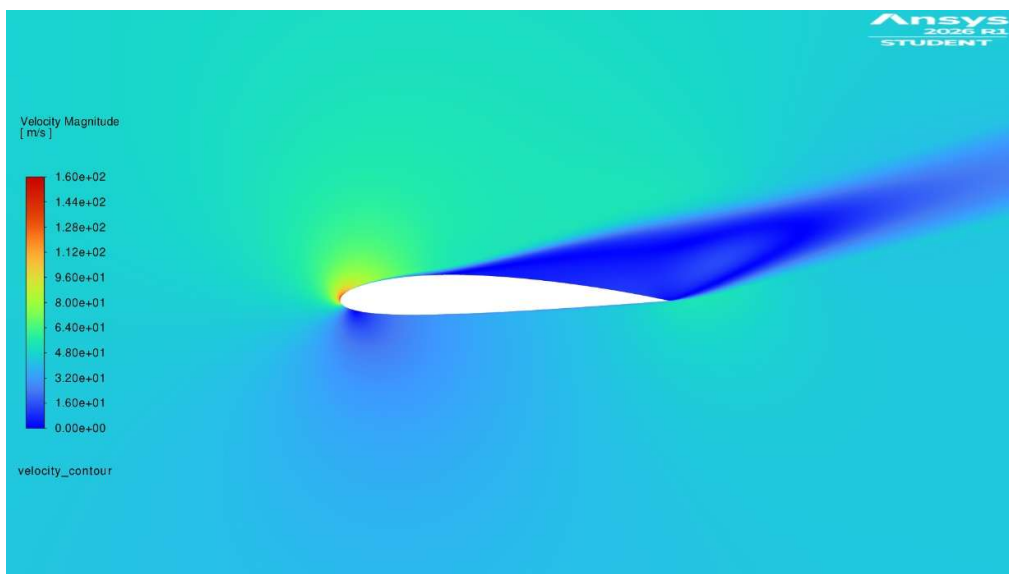


Figure C56: Fixed-range velocity contour at 20° angle of attack.

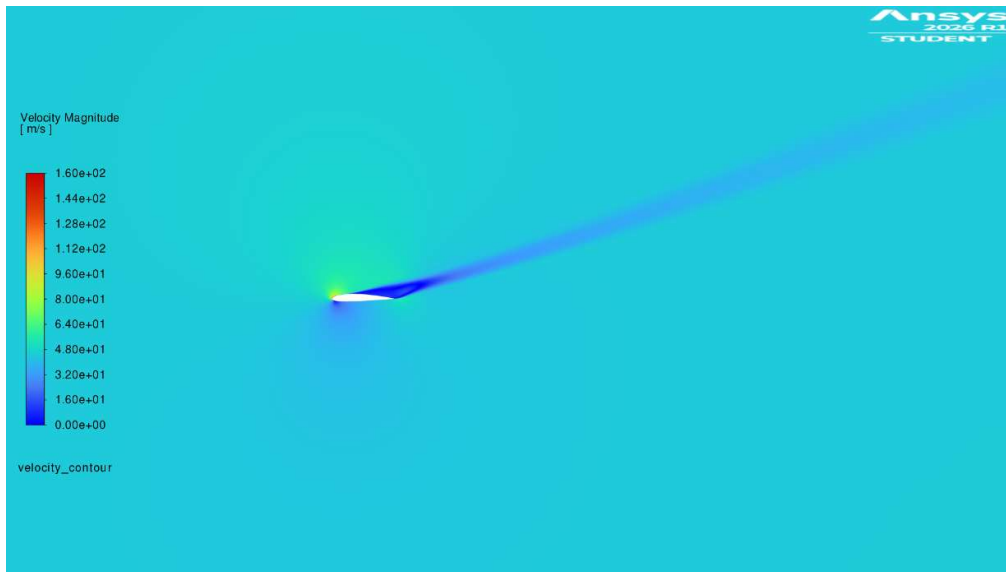


Figure C57: Wide-shot velocity contour at 20° angle of attack.

Appendix D - Scaled Residuals and Convergence Monitors

Figures D1 through D19 present the scaled residual plots for all simulated angles of attack, confirming solution convergence across the full AoA sweep.

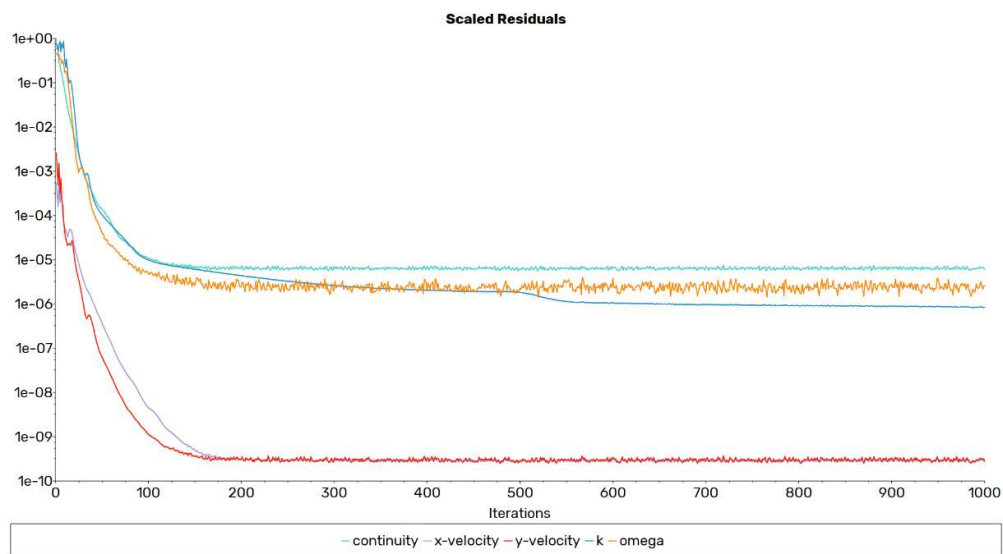


Figure D1: Scaled residuals plot for -8° angle of attack.

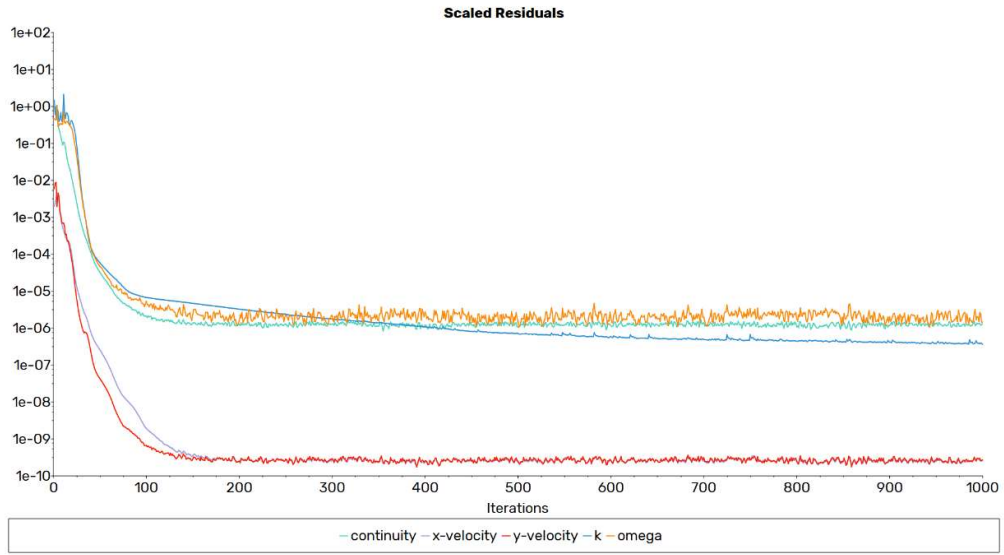


Figure D2: Scaled residuals plot for -6° angle of attack.

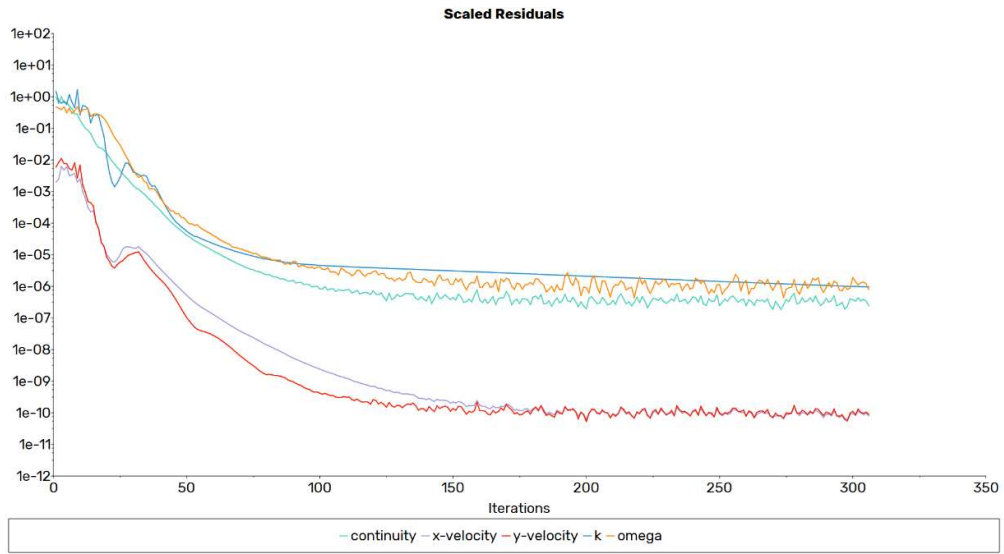


Figure D3: Scaled residuals plot for -4° angle of attack.

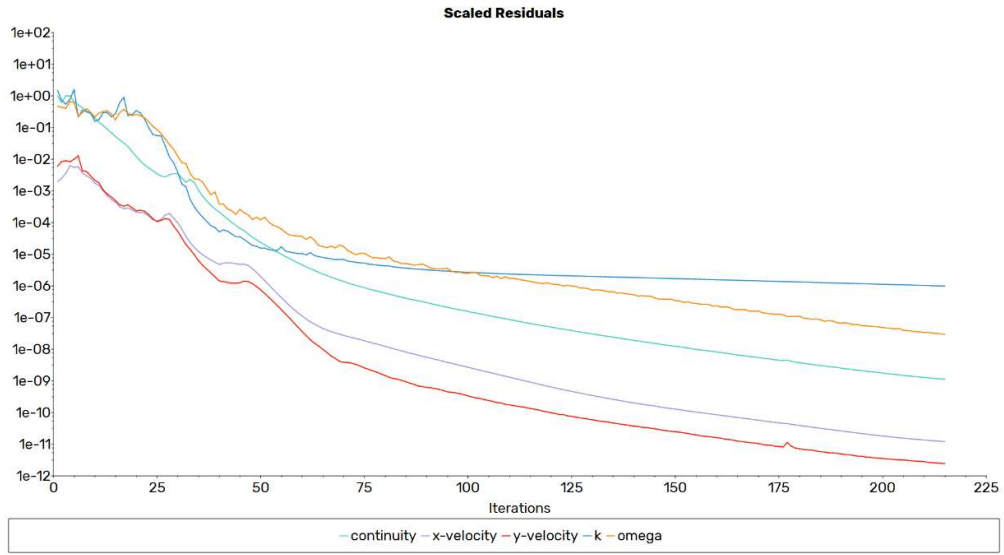


Figure D4: Scaled residuals plot for -2° angle of attack.

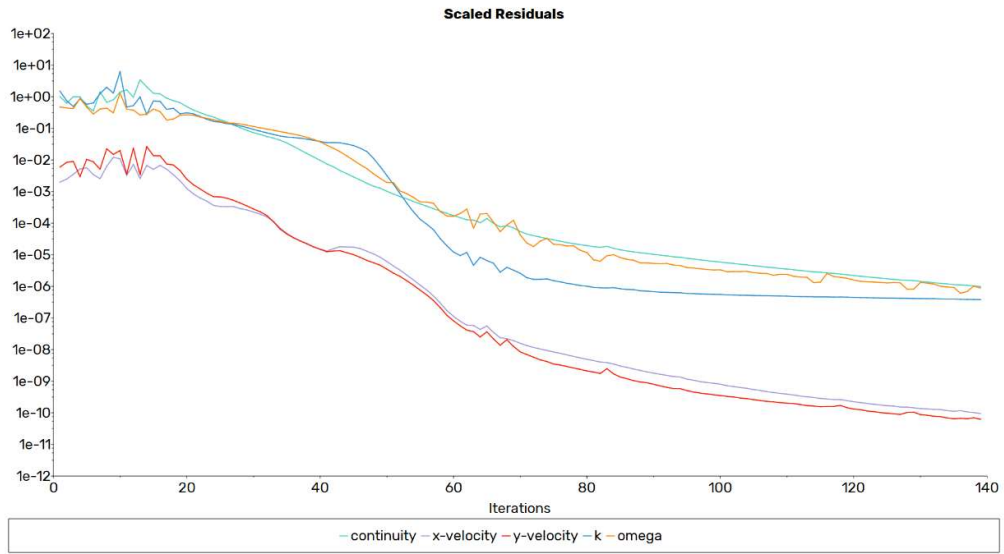


Figure D5: Scaled residuals plot for 0° angle of attack.

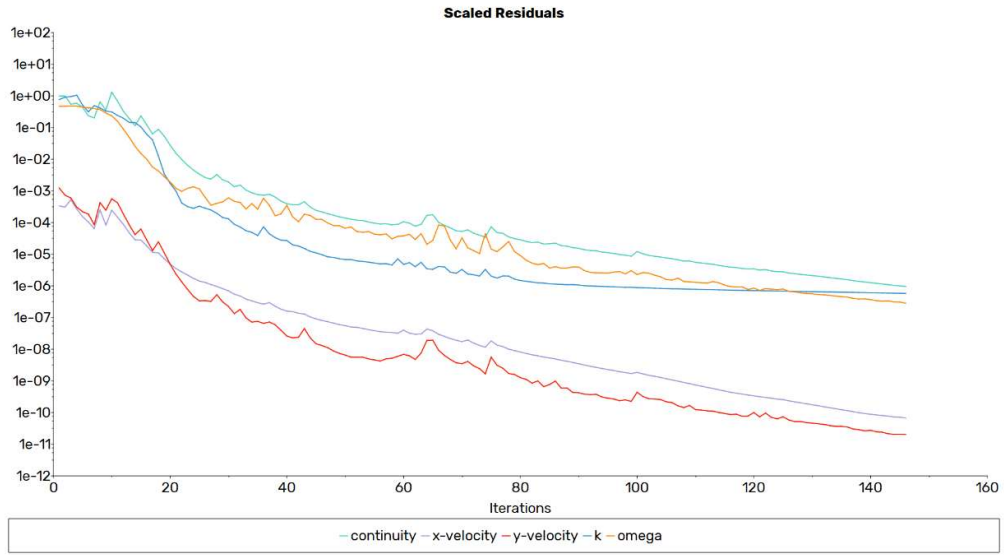


Figure D6: Scaled residuals plot for 2° angle of attack.

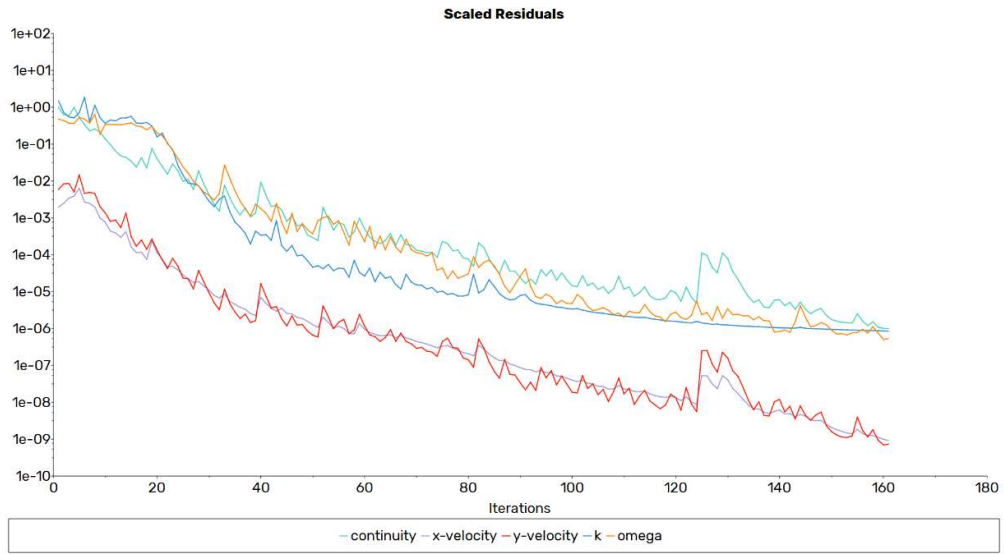


Figure D7: Scaled residuals plot for 4° angle of attack.

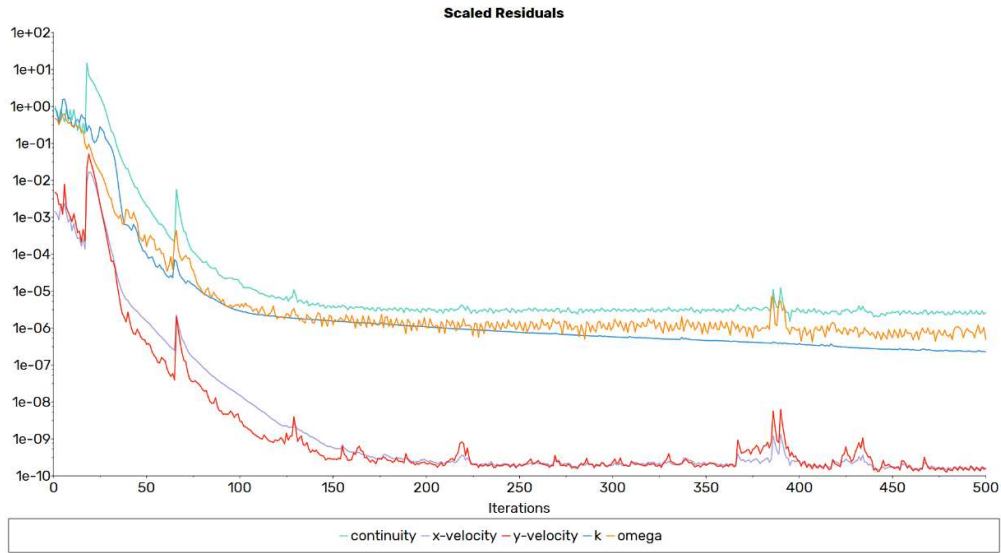


Figure D8: Scaled residuals plot for 6° angle of attack.

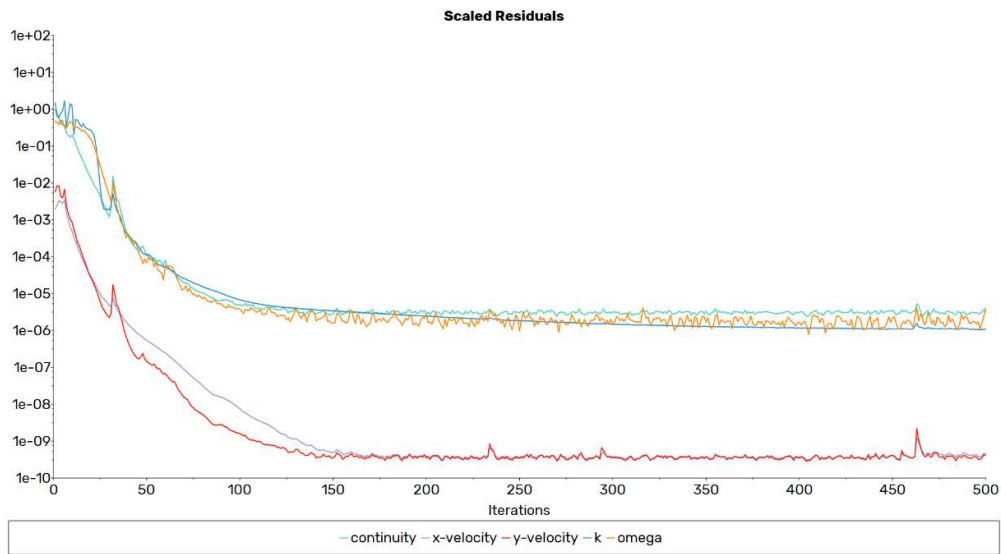


Figure D9: Scaled residuals plot for 8° angle of attack.

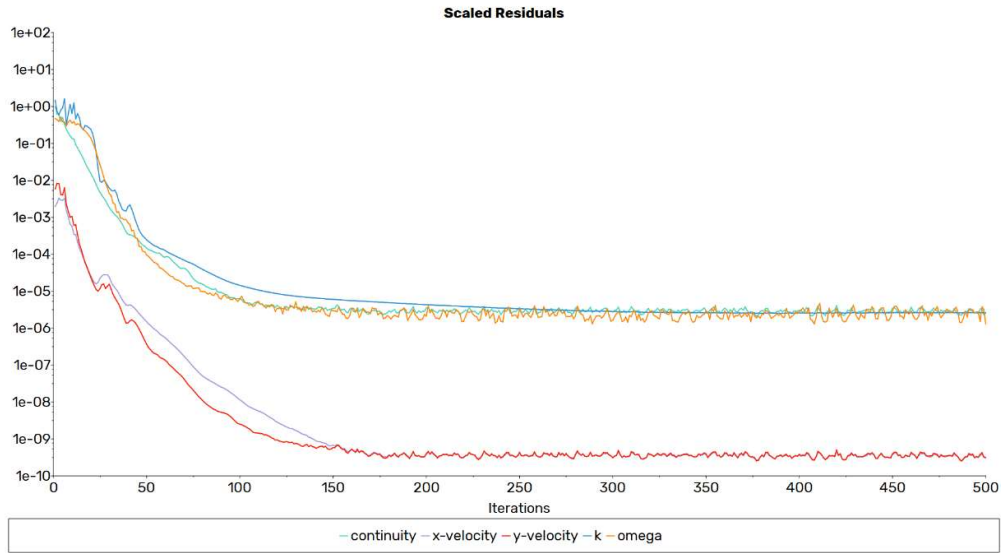


Figure D10: Scaled residuals plot for 10° angle of attack.

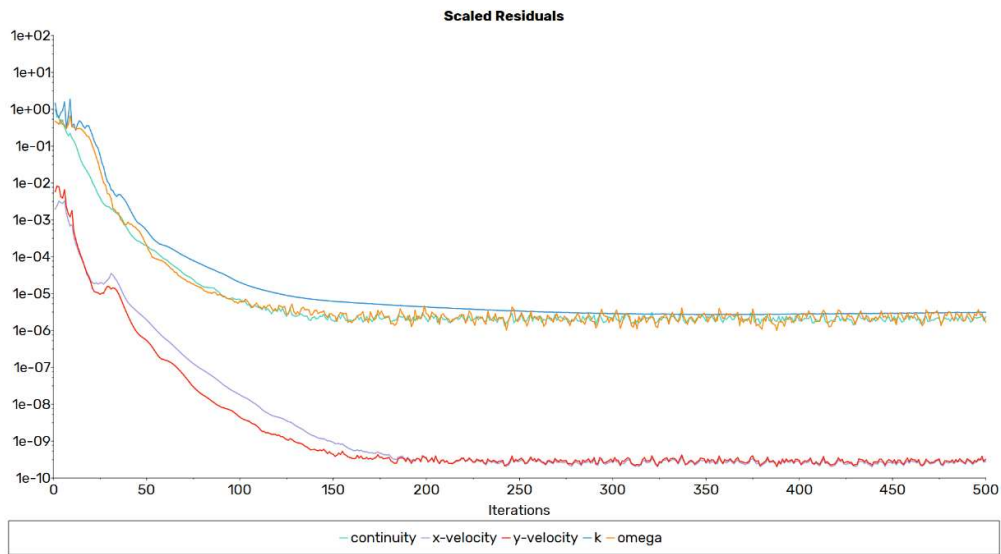


Figure D11: Scaled residuals plot for 12° angle of attack.

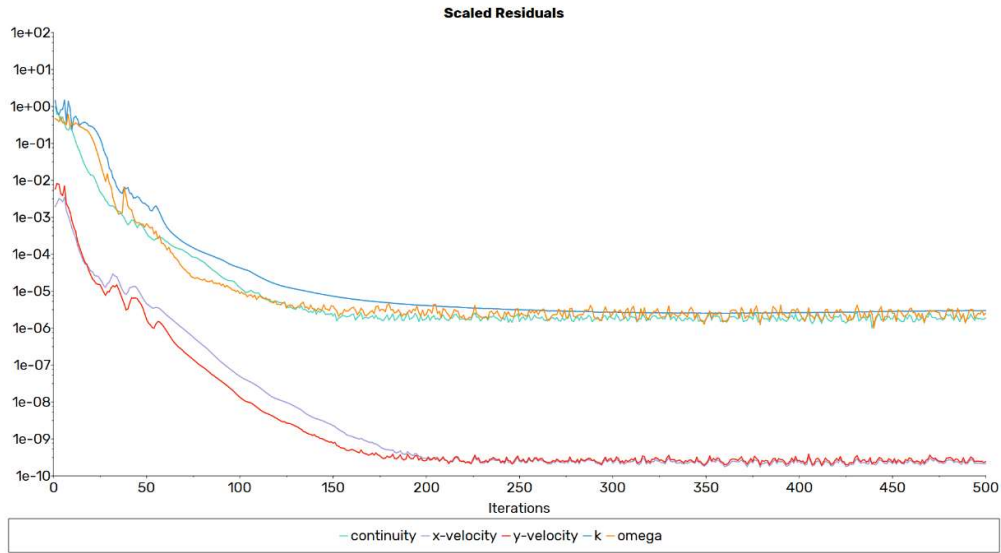


Figure D12: Scaled residuals plot for 13° angle of attack.

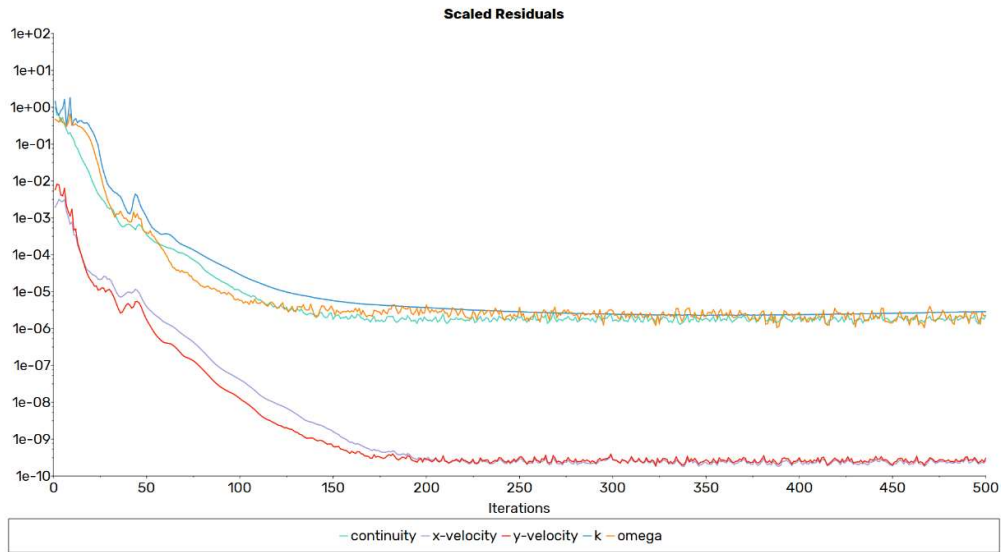


Figure D13: Scaled residuals plot for 14° angle of attack.

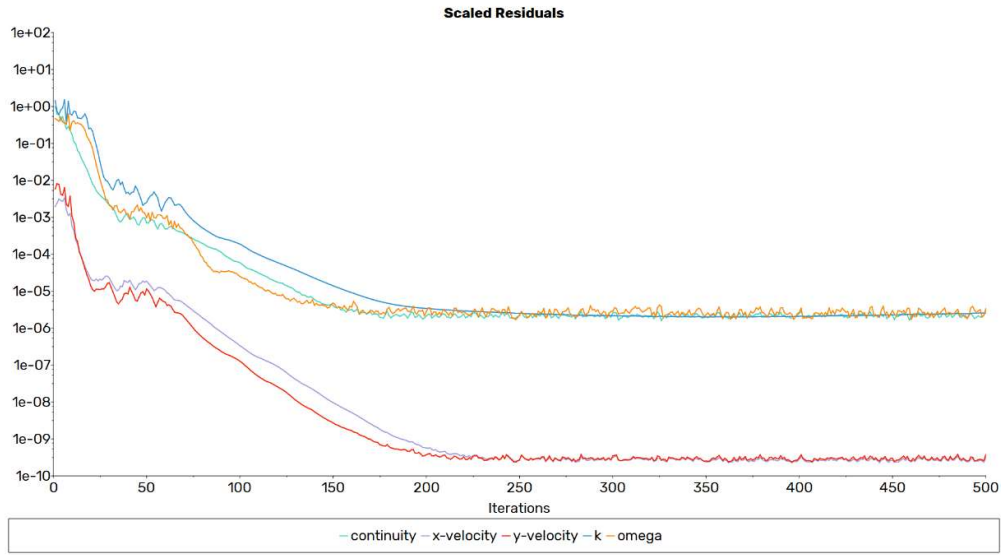


Figure D14: Scaled residuals plot for 15° angle of attack.

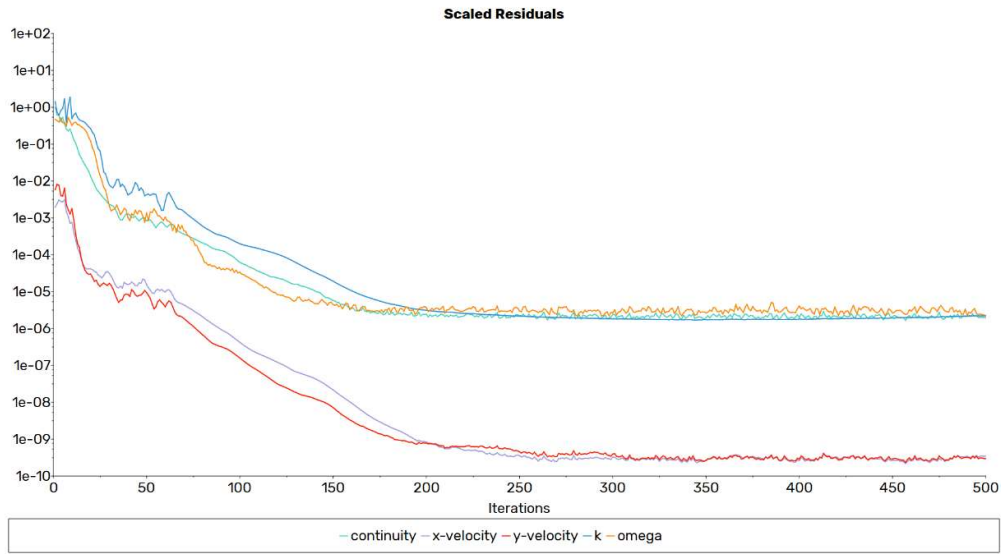


Figure D15: Scaled residuals plot for 16° angle of attack.

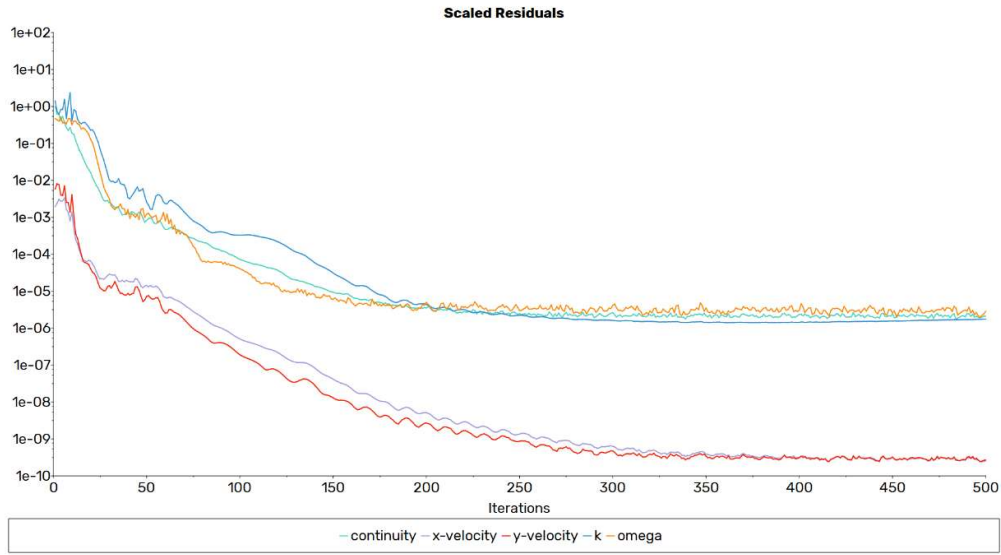


Figure D16: Scaled residuals plot for 17° angle of attack.

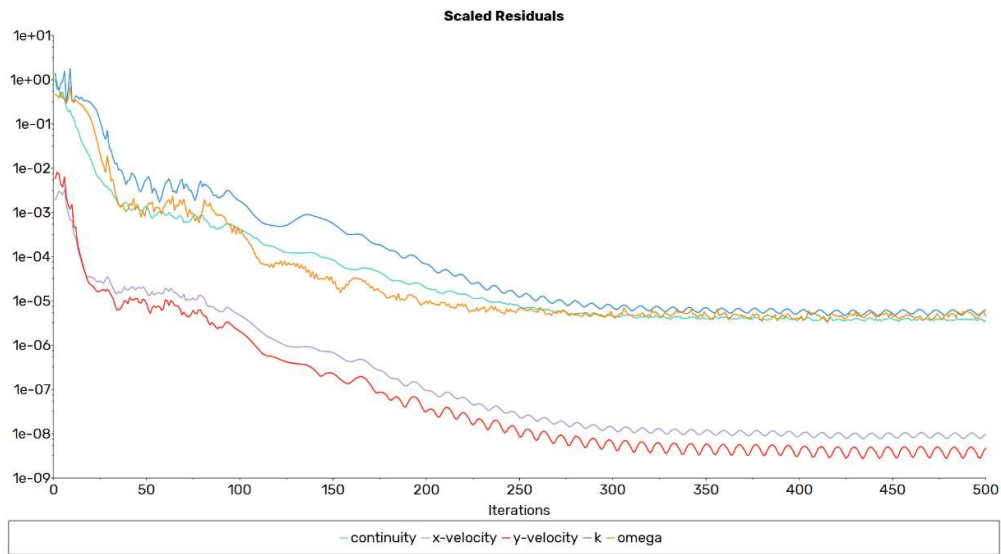


Figure D17: Scaled residuals plot for 18° angle of attack.

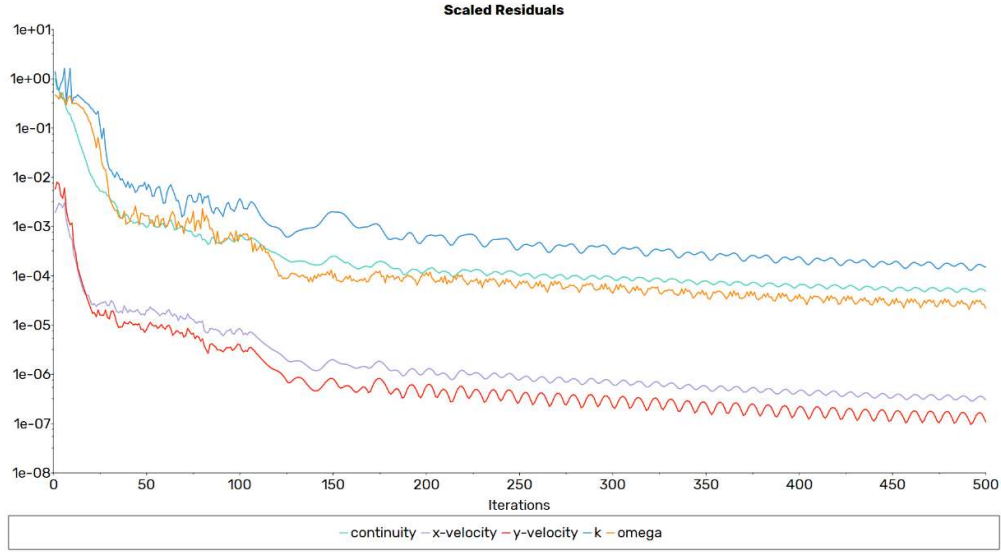


Figure D18: Scaled residuals plot for 19° angle of attack.

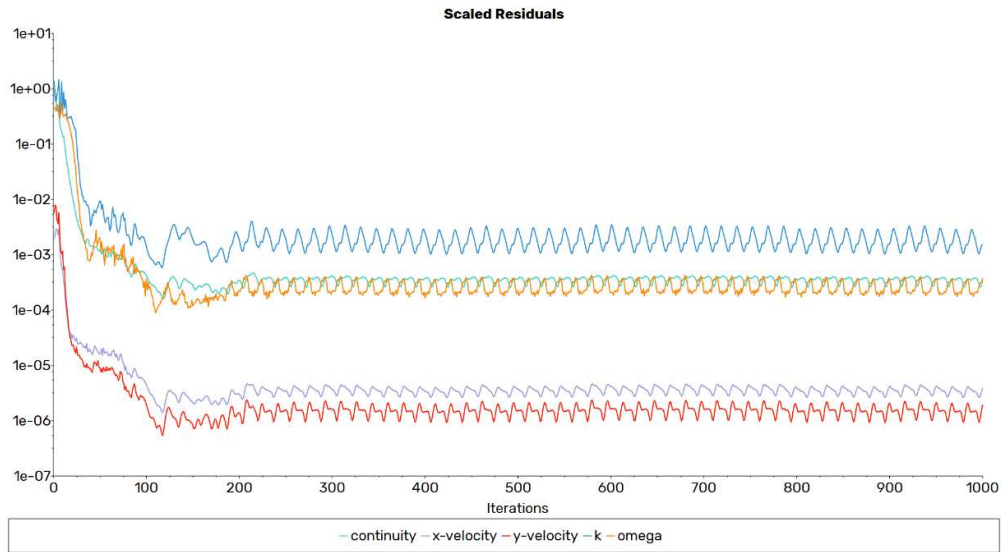


Figure D19: Scaled residuals plot for 20° angle of attack.

Appendix F — Wall y^+ Distribution

Figure F1 presents the wall y^+ distribution along the airfoil surface at 18° angle of attack — the case producing the maximum y^+ value of 1.24 across the full AoA sweep. The plot confirms y^+ remains below 1.24 across the entire airfoil surface, with the vast majority of the surface below 1.0, satisfying the boundary layer resolution requirements of the SST $k-\omega$ turbulence model.

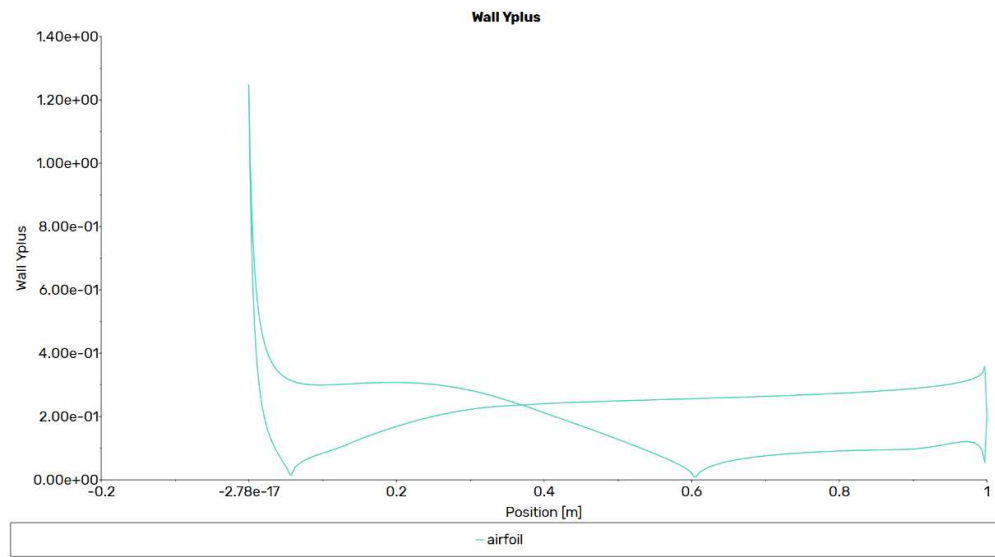


Figure F1: Wall y^+ distribution along the airfoil surface at 18° angle of attack